INCH-POUND
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SUPERSEDING
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w/INT. AMENDMENT 1
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DETAIL SPECIFICATION

CABLES, ELECTRIC, LOW SMOKE, HALOGEN-FREE, FOR SHIPBOARD USE GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers low smoke, halogen-free electric cable for Navy shipboard applications.

1.2 Classification. Cable constructions covered by this specification are either armored, unarmored, or unarmored with overall shielding; are classified as watertight, watertight (with circuit integrity), non-watertight, and non-watertight (with circuit integrity); and further classified for flexing and nonflexing service for power, lighting, control, communications, instrumentation, and electronic applications (see supplement). A part or identifying number (PIN) for cable is as specified in 1.3 and the applicable specification sheet (see 3.1).

1.3 Part or identifying number (PIN). PINs to be used for low smoke, halogen-free electric cable acquired to this specification are created as follows:

M	**24643**	**/XX**	**-**	**XX**	**A or U**	**O or D or N**
Prefix	Specification Number	Specification Sheet Number		Dash Number (See Specification Sheets)	A = Armored U = Unarmored	O = Overall shield D = Double overall shielded N = Not shielded

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6145

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-228 - Test Methods for Cables and Wire, Insulated

DEPARTMENT OF DEFENSE SPECIFICATIONS

(See supplement 1 for list of specification sheets.)

MIL-DTL-17 - Cables, Radio Frequency, Flexible and Semirigid General Specification for

MIL-PRF-572 - Cords, Yarns and Monofilaments Organic Synthetic Fiber

MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid

MIL-Y-1140 - Yarn, Cord, Sleeving, Cloth, and Tape-Glass

MIL-PRF-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5

MIL-DTL-16884 - Fuel, Naval Distillate

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-PRF-17672 - Hydraulic Fluid, Petroleum, Inhibited

MIL-PRF-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167

MIL-PRF-83282 - Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

MIL-PRF-87257 - Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2003-4 - Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@us.navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA WC 61 - Transfer Impedance Testing

ANSI/NEMA WC 53/ICEA T-27-581 - Standard Test Methods for Extruded Dielectric Power, Control, Instrumentation, and Portable Cables for Test

(Copies of these documents are available online at http://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM B3 - Standard Specification for Soft or Annealed Copper Wire

ASTM B8 - Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM B33 - Standard Specification for Tin-Coated Soft or Annealed Copper Wire for Electrical Purposes

ASTM B172 - Standard Specification for Rope-Lay-Stranded Copper Conductors Having Bunch-Stranded Members, for Electrical Conductors

ASTM B173 - Standard Specification for Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors

ASTM B174 - Standard Specification for Bunch-Stranded Copper Conductors for Electrical Conductors

ASTM B193 - Standard Test Method for Resistivity of Electrical Conductor Materials

ASTM B258 - Standard Specification for Standard Nominal Diameters and Cross-Sectional Areas of AWG Sizes of Solid Round Wires Used as Electrical Conductors

ASTM B286 - Standard Specification for Copper Conductors for Use in Hookup Wire for Electronic Equipment

ASTM B355 - Standard Specification for Nickel-Coated Soft or Annealed Copper Wire

ASTM D470 - Standard Test Methods for Crosslinked Insulations and Jackets for Wire and Cable

ASTM D512 - Standard Test Methods for Chloride Ion in Water

ASTM D770 - Standard Specification for Isopropyl Alcohol

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ASTM D1141	-	Standard Practice for the Preparation of Substitute Ocean Water
ASTM D1179	-	Standard Test Methods for Fluoride Ion in Water
ASTM D1246	-	Standard Test Method for Bromide Ion in Water
ASTM D1248	-	Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable
ASTM D2240	-	Standard Test Method for Rubber Property – Durometer Hardness
ASTM D3868	-	Standard Test Method for Fluoride Ions in Brackish Water, Seawater, and Brines
ASTM D3869	-	Standard Test Methods for Iodide and Bromide Ions in Brackish Water, Seawater, and Brines
ASTM D4327	-	Standard Test Method for Anions in Water by Suppressed Ion Chromatography
ASTM E230/E230M	-	Standard Specification for Temperature-Electromotive Force (EMF) Tables for Standardized Thermocouples
ASTM G21	-	Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

(Copies of these documents are available online at www.astm.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at www.asme.org.)

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

EIA 364-31 - TP-31F Humidity Test Procedure for Electrical Connectors and Sockets

(Copies of this document are available online at www.ecianow.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA T-28-562 - Test Method for Measurement of Hot Creep of Polymeric Insulations

(Copies of this document are available online at www.icea.net.)

MINISTRY OF DEFENCE STANDARDS (DEF STAN)

DEF STAN 07-247 - Selection of Materials on the Basis of their Fire Characteristics

(Copies of this document are available online from the Def Stan Index at https://www.dstan.mod.uk/StanMIS/Indexes/DefenceStandardSeries. To access the Def Stan Index, obtain a StanMIS account via the UK Defence Standardization StanMIS Public application online at https://www.dstan.mod.uk/StanMIS/Account/Login.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA WC 26/EEMAC 201 - Binational Wire and Cable Packaging Standard

(Copies of this document are available online at www.nema.org.)

UL SOLUTIONS

UL 1581 - Reference Standard for Electrical Wires, Cables, and Flexible Cords

UL 1685 - Vertical-Tray Fire-Propagation and Smoke-Release Test for Electrical and Optical-Fiber Cables

(Copies of these documents are available online at www.ul.com.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

ANSI/TIA 568.2 - Balanced Twisted-Pair Telecommunications Cabling and Components Standard

(Copies of this document are available online at www.tiaonline.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Cables furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.6).

3.3 Materials. Materials used in construction of those cables furnished under this specification shall be in accordance with the requirements shown in the applicable specification sheets and as follows. All non-metallic materials, including non-metallic film layers of foil/film tapes, shall contain no more than 0.2 percent halogen content. All cable materials shall be compatible in accordance with 4.8.5.

3.3.1 Conductor materials.

3.3.1.1 Bare (uncoated) copper. Unless otherwise specified in the applicable specification sheet, bare copper used for conductors (solid or stranded), drain wires (solid or stranded), and shield ends shall be in accordance with ASTM B3.

3.3.1.2 Tin-coated copper. Tin-coated copper used for conductors (solid or stranded), drain wires (solid or stranded), and shield ends shall be in accordance with ASTM B33.

3.3.1.3 Nickel-coated copper. Nickel-coated copper used for stranded conductors shall be in accordance with ASTM B355, class 2.

3.3.1.4 Thermocouple wire. Conductors for thermocouple cables shall be in accordance with ASTM E230/E230M.

3.3.2 Conductor construction.

3.3.2.1 Conductor stranding. The size and quantity of individual conductor strands and the total circular-mil area of the conductor shall be in accordance with ASTM B8, ASTM B172, ASTM B173, ASTM B174, or ASTM B286, as specified in the applicable specification sheet. Navy standard sized conductors shall be in accordance with table I.

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3.3.2.1.1 Concentric lay stranded. The length and direction of lay and the type and number of joints in concentric lay stranded conductors shall be in accordance with ASTM B8 or ASTM B286 as applicable.

3.3.2.1.2 Bunch stranded. The length and direction of lay and the type and number of joints in bunch stranded conductors shall be in accordance with ASTM B174.

3.3.2.1.3 Rope lay stranded. The length and direction of lay and the type and number of joints in rope lay stranded conductors shall be in accordance with ASTM B172 or ASTM B173 as applicable.

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TABLE I. Navy standard sizes.

Conductor size Navy standard	No. of strands (nom. [1/])	Strand dia. nom. [1/] (inch)	Cond. dia. nom. [1/] (inch)	Conductor cross-sectional area (circular mils)		Maximum conductor direct current (DC) resistance (ohms per 1,000 ft. at 77 °F [25 °C])		Weight, nom. [1/] (lbs. per 1,000 ft.)
				Nom. [1/]	Min.	Bare	Coated [2/]	
Concentric lay stranded								
400 (127)	127	0.057	0.742	413,600	405,400	0.0268	0.0273	1,300
Bunch lay stranded								
9 (90)	90	0.010	0.120	9,045	8,864	1.22	1.28	28
14 (140)	140	0.010	0.145	14,070	13,790	0.786	0.823	43
Rope lay stranded								
42 (209)	209	0.014	0.260	42,100	41,280	0.272	0.284	130
60 (304)	304	0.014	0.310	61,200	60,040	0.187	0.196	190
83 (418)	418	0.014	0.380	84,230	82,600	0.136	0.142	270
133 (684)	684	0.014	0.480	137,800	135,100	0.0830	0.0867	440
150 (760)	760	0.014	0.510	153,100	150,100	0.0747	0.0780	490
200 (988)	988	0.014	0.580	199,100	195,100	0.0575	0.0600	630
250 (1254)	1254	0.014	0.680	252,700	247,700	0.0453	0.0472	800
400 (2052)	2052	0.014	0.850	413,500	405,300	0.0277	0.0289	1,300
800 (4033)	4033	0.014	1.150	812,700	796,500	0.0141	0.0148	2,600

NOTES:

[1/] Values are for information only.

[2/] Copper coated with commercially pure tin.

3.3.3 Separators. Separators may be used at manufacturer's option where required to meet insulation removability requirements. Separators employed directly over conductors shall be applied to give not less than 100 percent coverage to the conductors. When used, separator shall be white (opaque).

3.3.4 Insulation.

3.3.4.1 Insulation type. Insulation type shall be as specified in the applicable specification sheet, except thermoset tape insulation shall be as specified in 3.3.4.2. Unless otherwise required in the specification sheet, insulation shall meet the applicable requirements below. Conductor insulation shall be readily removable by conventional wire-stripping devices without damage to the conductor, leaving no visible insulation residue when examined without magnification. Tapes used as insulation shall be smooth and free from wrinkles. The tape shall not split, crease, or tear when the insulated conductor is subjected to a 12-13X bend radius (see 4.7).

3.3.4.2 Thermoset tape. The thermoset tape shall be silicone rubber-coated glass fabric tape as specified in 3.3.4.2.1 or other equivalent thermoset material meeting the requirements of the applicable specification sheet.

3.3.4.2.1 Silicone rubber-coated glass fabric tape. The silicone rubber-coated glass fabric tape insulation shall, before application to a conductor, have a dielectric strength of not less than 475 volts per mil for tapes that are less than 8.8 mils thick and 575 volts per mil for tapes that are 8.9 mils or greater in thickness when verified using the test methodology specified in 4.8.5. Other physical and electrical properties shall be as specified in the specification sheets.

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3.3.4.3 Insulation resistance. The completed cable shall have the ability to transmit DC without exhibiting unacceptable current flow through insulation or jacketing material, either between conductors or between single conductor cables and any conducting substance that may be in contact with the surface of the cable during in-service use (see 4.9.5).

3.3.5 Glass braids. Glass braids for use on silicone rubber or thermoset extruded insulation shall have a thickness, coverage, and braid angle such that the completed cable shall meet all inspection and test requirements in accordance with MIL-Y-1140.

3.3.5.1 Crack resistance. The glass braid covering shall withstand bending and shall not puncture the overlaying covering in accordance with 4.8.30.

3.3.6 Covering. The covering shall consist of a layer or layers of a halogen-free material that provides high resistance to moisture, abrasion, and hydraulic fluids and provides a smooth surface for printed identification marking. The surface shall be smooth and free from irregularities and discontinuities.

3.3.6.1 Jackets. The material used for jackets over insulated conductors and groups of insulated conductors and for the cable jacket shall have the physical and electrical properties as required by the applicable specification sheet and as specified herein. The color of the cable jacket shall be black, white, or red, as specified (see 6.2). The use of red jackets requires written NAVSEA approval for the specific application.

3.3.7 Shields. The materials and constructions for shields of insulated conductors, groups of insulated conductors, and overall cable shall be as specified in the specification sheet. When AWG sizes are specified for wire shields, they shall be in accordance with ASTM B258.

3.3.8 Fillers. Unless otherwise specified in the applicable specification sheet, for fillers for cables of the flex/bending type intended for flexing applications, only fibrous fillers shall be used. For non-flexing types, solid or fibrous filler materials shall be acceptable. Water-blocking materials (3.3.9) and water-swelling fillers (3.3.9.1) are considered fillers for the purposes of halogen content, acid gas generation, smoke index, and toxicity index testing.

3.3.8.1 Fibrous fillers. Fibrous fillers shall be treated for flame or moisture resistance, or a combination of both, to meet the requirements for the particular type of cable.

3.3.8.2 Non-fibrous fillers (solids). Non-fibrous fillers shall consist of elastomeric material that is removable from insulation of conductors and insulating coverings over the shields without the use of chemicals or tools. Non-fibrous solid fillers in cables with swept electrical parameters include physical separation barriers placed between individual pairs to meet electrical performance requirements (spacer splines).

3.3.9 Water-blocking materials (watertight). Water-blocking materials used in cable and shield interstices shall be compatible with all other cable materials. The materials shall be clean and non-tacky to the touch and shall leave no residue on the installer’s hands. The materials shall be free-stripping from the cable components and shield by hand and shall not require the use of chemicals or other mechanical means of removal. The materials shall not interfere with any termination technique used with finished cable, shields, or components.

3.3.9.1 Water-swelling fillers (watertight). Water-swelling fillers used to water-block cable voids and shield interstices shall be compatible with all other cable materials. The fillers shall be clean and non-tacky to the touch. They shall be readily removable from the cable components and shields. The fillers shall not interfere with any termination technique used with finished cable, shields, or components.

3.3.10 Tapes. Tapes shall be of a material compatible with the other cable materials. For cable constructions not using water-swelling tapes, the tape material shall be halogen-free. Polyester tapes shall be in accordance with MIL-I-631. Tapes applied over shield braids for singles, twisted pairs, and triads shall be sealed or set to prevent unwinding freely but shall be easily removable for wire or shield termination.

3.3.10.1 Water-swelling tapes (watertight). Water-swelling tapes used to water-block cable voids and shield interstices shall be compatible with all other cable materials. The tapes shall be clean and non-tacky to the touch. They shall be readily removable from the cable components and shields. The tapes shall not interfere with any termination technique used with finished cable, shields, or components.

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3.3.11 Braids.

3.3.11.1 Braid identification. Colored braids used for conductor or group identification shall be in accordance with MIL-PRF-572.

3.3.11.2 Glass braids. Glass braids for use on silicone or thermoset extruded insulated conductors shall be composed of the appropriate size of either staple or continuous fiber in accordance with MIL-Y-1140.

3.3.12 Tie cords. Tie cords shall be of synthetic fiber with a diameter not greater than 0.065 inch and a breaking strength not less than 30 pounds.

3.3.13 Separators. Separators, when required, shall be of a material that is compatible with other cable materials.

3.3.14 Binders. Binders shall be of a material compatible with the other cable materials.

3.3.15 Armor wires. The armor wires shall consist of Alclad 5056 or 5154 aluminum alloy or equivalent with a diameter of 0.0126±0.0005 inch. They shall have a minimum tensile strength of 50,000 pounds per square inch (lb/in^2) and a minimum elongation (before application to the cable) of 2 percent in a 10-inch-length.

3.3.16 Fungus resistance. All nonmetallic materials shall be fungus inert and shall be certified to meet the requirements of ASTM G21 with a growth rating not greater than 1 (see 4.10).

3.3.17 Hazardous items and toxic materials. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions regarding this requirement shall be referred by the contracting activity to the appropriate department medical service who will act as an advisor to the contracting agency. Regardless of any other requirements, materials and parts containing asbestos, mercury or mercury compounds, or those giving off excessive toxic fumes (see 4.8.25) shall not be used.

3.3.18 Prohibited materials. The cables shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.3.19 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.20 Electrical moisture absorption. The maximum increase in capacitance from 1 to 14 days shall be 10 percent, and the maximum increase in capacitance from 7 to 14 days shall be 4 percent.

3.3.21 Temperature and humidity. After exposure to the temperature and humidity conditions of 4.8.33, the cable shall meet the attenuation requirements specified in the applicable specification sheet.

3.3.22 Stress resistance. The electrical characteristics of the cable shall not change when stressed in accordance with the conditions of 4.9.13.2.

3.3.23 Long term jacket immersion. After immersion in water for a period of 56 days, the cable shall exhibit the following when tested in accordance with 4.8.37: no increase in the overall shield resistance greater than +2 percent, no visible sign of water penetration into the overall cable shield, and no visible evidence of shield corrosion.

3.3.24 Reinforcement. Reinforcing binders or reinforcement between material layers shall be a cord of synthetic fibers having a minimum breaking strength of 15 pounds and shall be compatible with other cable components.

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3.4 Design and construction.

3.4.1 Fillers, binders, and reinforcement.

3.4.1.1 Fillers. Fillers may be used to provide firmness and roundness of completed cables and to provide watertightness when required by the specification sheet. Filler material shall be compatible with other cable components.

3.4.1.2 Binders. Binders may be used to provide firmness and roundness of completed cable and to provide watertightness when required by the specification sheet. Binder material shall be compatible with other cable components.

3.4.1.3 Reinforcement. When a reinforcement is specified in the applicable specification sheet, it shall be a cord of synthetic fibers applied as a braid or two reversed serves. The reinforcement shall be applied with approximately four picks of crossover per inch (see 3.3.24).

3.4.2 Shields.

3.4.2.1 Braided shields. When specified in the applicable specification sheet, braided shields shall be of the pushback type and shall be constructed to conform to the physical and electrical requirements specified in the applicable specification sheet. The shield shall be of a close-fitting braid construction and shall be free from irregularities and discontinuities. Individual wire strands may be spliced. Only one carrier shall be spliced at any one point. When the braided shield is cut, it shall be capable of sliding back not less than 4 inches on an exposed 24-inch-length of component wire or cable with the opposite end clamped.

3.4.2.1.1 Braided shield coverage. Braided wire shield shall have the angle of application and the percent coverage as required in the applicable specification sheet. The percent coverage and the angle of application shall be determined by the following formula:

Percent coverage = $100\ (2F-F^2)$

Where: $F = NPd/\sin(a)$

$\tan(a) = 2 \times 3.14159\ (D+2d)P/C$

a = acute angle of braid with axis of cable

d = diameter (inch) of individual braid wires

D = diameter (inch) of cable under braid

N = number of wires per carrier

C = number of carriers

P = picks per inch of cable length

NOTE: For non-water-blocked component pairs and triples, calculate D as follows:

Pairs: D = 1.637 × diameter of one insulated conductor in pair

Triple: D = 1.955 × diameter of one insulated conductor in triple

To this calculated value of D must be added any diameter build-up caused by wraps, coverings, tapes, or water-blocking materials applied over the twisted components prior to applying the shield.

3.4.2.1.2 Double shield. Double shields shall be constructed such that the angle of application, and the percent of coverage shall allow the completed cable to comply with the surface transfer impedance (see 4.9.11) and electromagnetic pulse (EMP) response (see 4.9.11.2) requirements of the applicable specification sheet. The percent of separator tape overlap between double shields shall be as required by the applicable specification sheet.

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3.4.2.2 Foil shields. Foil shields, when specified in the applicable specification sheet, shall be of the foil or film type: consisting of one or more layers of metallic foil, alternating with one or more layers of non-metallic film, and all layers bonded or laminated together. The non-metallic film shall be of material that is compatible with the temperature rating of the cable. Aluminum foil shall not be used in contact with uncoated copper drain wires or uncoated copper braided shields.

3.4.2.3 Shield insulation. Shield insulation shall consist of two tapes, sealed. Alternate shield insulation may consist of one tape, plus an extruded covering of a translucent material with a minimum average thickness of 0.003 inch. The standard identification code shall be applied by method 2 (see 3.4.8.2) on the inner tape and be legible through the outer layer or covering.

3.4.3 Watertightness. Where watertight cable construction is specified in the applicable specification sheet, voids within the cable construction shall be filled to limit the passage of water longitudinally through the cable.

3.4.4 Cabling. Unless otherwise specified in the applicable specification sheet, cable components shall be cabled together with a lay not greater than 24 times the pitch diameter of the layer. In multi-layer cables, the lay of each layer shall not exceed 24 times the pitch diameter of each individual layer. The cable lay direction shall be all left-hand or shall be alternating directions. Unless otherwise specified in the applicable specification sheet, the outer layer lay direction shall be left-hand.

3.4.5 Braid armor. The armor shall be applied in the form of a braid. The weave of the braid shall be of either the “one over - one under” or the “two over - two under” type. The weave shall be such that the wires of a carrier are laid closely together, flat, and parallel. The braid shall be applied with the maximum tension possible to prevent loosening and creeping but not to cause broken ends. Splices in the braid strands shall be infrequent, staggered, and inconspicuous so as not to increase the nominal diameter of the cable or result in rough spots. Broken ends shall be neatly tucked under the braid so they do not protrude. No more than 10 end breaks shall be allowed in a 50-foot section of cable (this does not include changes in carriers). When a full carrier is changed, the ends of the carrier shall also be neatly tucked under the braid so they do not protrude. No more than one carrier change shall be allowed within 10 feet of cable. If more than one carrier change is required, a new length of cable shall be created. The braid shall remain snug to the cable jacket and not spring away when cut. The selection of the number of wires per carrier and the number of carriers per braid shall be such as to produce a basket weave to give a minimum of 88 percent coverage and a braid angle within the limits as specified in table II. The number of wires per carrier shall be not greater than as specified in table III.

TABLE II. Armor braid angle.

Diameter over jacket (inches)	Braid angle (degrees)	
	Min.	Max.
0.100 - 0.600	30	60
0.601 - 1.000	35	60
1.001 - 1.500	40	70

TABLE III. Wires per carrier.

Diameter under braid (inches)	Max. number of ends/carrier	
	One over - one under	Two over - two under
0.100 - 0.400	8	5
0.401 - 0.800	12	10
0.801 - 1.500	16	10

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3.4.5.1 Coverage. The coverage shall be as determined by the following formula:

Percent coverage = $100\ (2F-F^2)$

Where: F = NPd/sin (a)

tan (a) = $2 \times 3.14159\ (D+2d)P/C$

a = acute angle of braid with axis of cable

d = diameter (inch) of individual braid wires

D = diameter (inch) of cable under braid

N = number of wires per carrier

C = number of carriers

P = picks per inch of cable length

3.4.6 Dimensional tolerances. Where minimum or maximum dimensions, or both, are specified, no minus or plus tolerances, respectively, shall be permitted. Where no minimum overall cable diameter is specified, the minimum permissible diameter shall be not less than 92½ percent of the specified maximum overall cable diameter. NOTE: Caution should be exercised when designing armored cables, since the use of 28 AWG aluminum armor results in an actual increase in overall diameter of as much as 0.060 inch.

3.4.6.1 Nominal dimensions. Where a dimension is specified as nominal, it shall be for reference purposes only and shall not be construed as a requirement.

3.4.6.2 Extruded and taped insulation wall thickness. For conductor insulation wall thickness specified as minimum average, the concentricity measured at any cross-section shall be not less than 90 percent.

3.4.6.3 Cable jacket thickness. In single and multi-conductor cables where cable jacket wall thickness is specified as a minimum average, the concentricity measured at any cross-section shall be not less than 80 percent.

3.4.7 Centering and concentricity.

3.4.7.1 Insulation. The insulation on the individual conductors shall be uniform in diameter throughout the conductor length. Concentricity at any cross-section shall be not less than 85 percent for specified minimum average wall thickness greater than 0.025 inch, nor less than 75 percent for specified minimum average wall thickness of 0.025 inch or less.

3.4.7.2 Cable jacket. The cable jacket shall be applied concentrically to the cable core in a manner to maintain circularity in the completed cable. Concentricity shall be not less than 75 percent.

3.4.8 Identification codes and methods. Individual conductors and groups of conductors shall be separately identified. The applicable identification code and method by which the code is applied shall be as specified in the specification sheet.

3.4.8.1 Identification codes.

3.4.8.1.1 Standard identification code. The standard identification code shall be in accordance with table IV.

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TABLE IV. Standard identification code.

Color, conductor, or group no.	Background or base color	First tracer color	Second tracer color
1	Black	----	----
2	White	----	----
3	Red	----	----
4	Green	----	----
5	Orange	----	----
6	Blue	----	----
7	White	Black	----
8	Red	Black	----
9	Green	Black	----
10	Orange	Black	----
11	Blue	Black	----
12	Black	White	----
13	Red	White	----
14	Green	White	----
15	Blue	White	----
16	Black	Red	----
17	White	Red	----
18	Orange	Red	----
19	Blue	Red	----
20	Red	Green	----
21	Orange	Green	----
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange
36	Orange	White	Blue
37	White	Red	Blue

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TABLE IV. Standard identification code – Continued.

Color, conductor, or group no.	Background or base color	First tracer color	Second tracer color
38	Brown	----	----
39	Brown	Black	----
40	Brown	White	----
41	Brown	Red	----
42	Brown	Green	----
43	Brown	Orange	----
44	Brown	Blue	----
45	White	Black	Blue
46	Red	White	Blue
47	Green	Orange	Red
48	Orange	Red	Blue
49	Blue	Red	Orange
50	Black	Orange	Red
51	White	Black	Orange
52	Red	Orange	Black
53	Green	Red	Blue
54	Orange	Black	Blue
55	Blue	Black	Orange
56	Black	Orange	Green
57	White	Orange	Green
58	Red	Orange	Green
59	Green	Black	Blue
60	Orange	Green	Blue
61	Blue	Green	Orange
62	Black	Red	Blue
63	White	Orange	Blue
64	Red	Black	Blue
65	Green	Orange	Blue
66	Orange	White	Red
67	Blue	White	Red
68	Black	Green	Blue
69	White	Green	Blue
70	Red	Green	Blue
71	Green	White	Red
72	Orange	Red	Black
73	Blue	Red	Black
74	Black	Orange	Blue

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TABLE IV. Standard identification code – Continued.

Color, conductor, or group no.	Background or base color	First tracer color	Second tracer color
75	Red	Orange	Blue
76	Green	Red	Black
77	Orange	White	Green
78	Blue	White	Green
79	Red	White	Orange
80	Green	White	Orange
81	Blue	Black	Green
82	Orange	White	----
83	Green	Red	----
84	Black	Green	----
85	White	Green	----
86	Blue	Green	----
87	Black	Orange	----
88	White	Orange	----
89	Red	Orange	----
90	Green	Orange	----
91	Blue	Orange	----
92	Black	Blue	----
93	White	Blue	----
94	Red	Blue	----
95	Green	Blue	----
96	Orange	Blue	----
97	Yellow	----	----
98	Yellow	Black	----
99	Yellow	White	----
100	Yellow	Red	----
101	Yellow	Green	----
102	Yellow	Orange	----
103	Yellow	Blue	----
104	Black	Yellow	----
105	White	Yellow	----
106	Red	Yellow	----
107	Green	Yellow	----
108	Orange	Yellow	----
109	Blue	Yellow	----
110	Black	Yellow	Red
111	White	Yellow	Red

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TABLE IV. Standard identification code – Continued.

Color, conductor, or group no.	Background or base color	First tracer color	Second tracer color
112	Green	Yellow	Red
113	Orange	Yellow	Red
114	Blue	Yellow	Red
115	Black	Yellow	White
116	Red	Yellow	White
117	Green	Yellow	White
118	Orange	Yellow	White
119	Blue	Yellow	White
120	Black	Yellow	Green
121	White	Yellow	Green
122	Red	Yellow	Green
123	Orange	Yellow	Green
124	Blue	Yellow	Green
125	Black	Yellow	Blue
126	White	Yellow	Blue
127	Red	Yellow	Blue

3.4.8.1.2 Telephone identification code. The conductor identification code for telephone cables shall be as follows:

Color or conductor no.	Color
1	Black
2	White
3	Red
4	Green
5	Orange
6	Blue

Color or conductor no.	Color
7	Brown
8	Gray
9	Yellow
10	Purple
11	Tan
12	Pink

The pairing of conductors for forming pairs shall be as follows:

a. No. 1 paired with nos. 2 through 12 for first 11 pairs
b. No. 2 paired with nos. 3 through 12 for next 10 pairs
c. No. 3 paired with nos. 4 through 12 for next 9 pairs
d. No. 4 paired with nos. 5 through 12 for next 8 pairs
e. No. 5 paired with nos. 6 through 12 for next 7 pairs
f. No. 6 paired with nos. 7 through 12 for next 6 pairs
g. No. 7 paired with nos. 8 through 12 for next 5 pairs
h. No. 8 paired with nos. 9 through 12 for next 4 pairs
i. No. 9 paired with nos. 10 through 12 for next 3 pairs

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j. No. 10 paired with nos. 11 through 12 for next 2 pairs

k. No. 11 paired with no. 12

3.4.8.1.3 Special identification code. The special identification code shall be in accordance with the following:

Color or conductor no.	Color
1	Black
2	White
3	Red
4	Green
5	Orange
6	Blue

Color or conductor no.	Color
7	Brown
8	Gray
9	Yellow
10	Purple
11	Tan
12	Pink

3.4.8.1.4 Letter identification code. The letter identification code shall consist of the letters A, B, C, and D printed in block type and with black, white, red, and green ink, respectively.

3.4.8.1.5 Twisted pair identification code. The twisted pair identification code shall consist of the numbers in sequence running from 1 through the number corresponding to the total number of twisted pairs in the cable. Both conductors in each pair shall be numbered the same, denoting the sequence number of the pair. Distinction between the two conductors is provided by different colored insulation. Conductors of a cable with a single pair need not be numbered.

3.4.8.1.6 Twisted triad identification code. The twisted triad identification code shall consist of the numbers in sequence running from 1 through the number corresponding to the total quantity of twisted triads in the cable. Three conductors shall be numbered the same, denoting the sequence number of the triad; distinction between the three conductors is provided by different colored insulation. Conductors of a cable with a single triad need not be numbered.

3.4.8.2 Identification methods.

3.4.8.2.1 Method 1. Identification method 1 shall be surface printing of both number and color designations. The legend shall be printed in contrasting color, preferably white ink on black or dark background or black ink on white or light background. The printing can be on the conductor insulation, provided the covering or jacket is transparent. The printing shall be on the covering or jacket if it is not transparent. The legend shall be repeated at intervals not exceeding 3 inches from the end of one legend to the beginning of the following legend, the legend shall be perpendicular to the longitudinal axis of the conductor, and alternate legends shall be inverted (for example, 10 ORANGE BLACK). The character type shall be block and shall have a height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.045 to 0.070	0.025
0.070 to 0.095	0.031
0.095 to 0.115	0.047
0.115 to 0.200	0.062
0.190 to 0.250	0.078
0.235 to 0.350	0.094
0.330 and larger	0.125

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3.4.8.2.2 Method 2. Identification method 2 shall be the use of translucent (opaque) polyester tapes that have been printed with both the number and the color designation. The legend shall be printed with black ink and shall be repeated at intervals not exceeding 3 inches from the end of one legend to the beginning of the following legend, and alternate legends shall be inverted. The character type shall be block and shall be approximately 3/32 inch high. Printed characters shall be perpendicular to the longitudinal axis of the tape prior to its application to the component.

3.4.8.2.3 Method 3. Identification method 3 shall be the use of solid base colors or solid base colors with tracers as required. The base color may be either the color of the insulation or the color of a coating or covering applied to the insulation. The tracers shall be approximately 1/32-inch-wide ink stripes of the required color applied helically with 1½±¼-inch lay. If two tracers are required, the second shall be half the width of the first.

3.4.8.2.4 Method 4. Identification method 4 shall be the use of colored braids. Tracers shall consist of the required colors applied by three adjacent carriers. Where two tracers are required, they shall be applied with reverse lay.

3.4.8.2.5 Method 5. Identification method 5 shall be the use of the printed letter on the outermost insulating tape or the printed letter on a polyester binder tape over the insulating tapes (see 3.4.8.1.4). The letters shall be approximately 3/16 inch high and shall have been printed at intervals not exceeding 3 inches prior to the application of the tape to the conductor. If the insulating tapes are white, no printing is required on the B (white) conductor. Printed letters may be perpendicular or parallel to the longitudinal axis of the tape prior to its application to the component.

3.4.8.2.6 Method 6. Identification method 6 shall consist of numerals printed in ink on the conductor insulation. For conductors having a jacket directly over the insulation, the numerals may be printed in ink on the jacket at the manufacturer's option. White ink shall be used for a red or black background; black ink shall be used for a white background. Numerals shall be perpendicular or parallel to the longitudinal axis of the conductor. Legends shall be alternately inverted and shall be repeated at intervals not greater than 1½ inches from the end of one legend to the beginning of the following legend.

a. When applied perpendicular to the longitudinal axis of the conductor, numeral character type shall be block and the height shall be proportional to the conductor's outside diameter (OD) as shown in method 1 (see 3.4.8.2.1). Each numeric legend shall be underlined.

b. When applied parallel to the longitudinal axis of the conductor, numeral width shall be proportional to the conductor's OD as shown in method 1 (see 3.4.8.2.1), and numeral height shall be approximately three times numeral width. Two-digit legends shall have only the bottom numeral underlined (see figure 1).

3.4.9 Manufacturer's identification tape. Unless otherwise indicated in the specification sheet, all cables shall contain a continuous, thin, moisture-resistant marker tape not less than 1/10 inch wide. The marker tape shall be placed directly under the cable's outermost tape or jacket. The tape shall be printed to show the following information at intervals not greater than 1 foot:

a. Name and location of manufacture
b. Year of manufacture
c. Specification number (MIL-DTL-24643)
d. Progressive serial number

The serial number is not necessarily a footage marker. A serial number shall not be repeated by a manufacture in any one year for any one type and size of cable.

3.4.10 Cable or surface marking. Unless otherwise specified in the applicable specification sheet, ink marking shall be used for overall cable jacket surface marking. The legend shall be printed in contrasting color: white ink on a black or red background or black ink on a white or light background. The legend shall consist of the manufacturer's name, the cable type and size designation when applicable, the part number, the year of manufacture, and the jacket type. The legend shall be repeated at intervals not greater than 1 foot from the end of one legend to the beginning of the following legend. The year of manufacture need not be in line with the balance of the legend.

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3.4.10.1 Character type. The character type shall be block, shall be perpendicular to the longitudinal axis of the cable, and shall have a minimum height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.125 to 0.200	0.047
0.200 to 0.285	0.063
0.285 to 0.350	0.078
0.350 to 0.500	0.094
0.500 and larger	0.125

3.4.10.2 Jacket material name abbreviations. Crosslinked polyolefin jackets shall be abbreviated XLPOLYO (for example, “Manufacturer’s name LS3SWU-19 M24643/36-05UN XLPOLYO 2007”, “Manufacturer’s name LS3SWU-19 M24643/36-05UN 2007 XLPOLYO”).

3.5 Electrical properties. When tested as specified in 4.9, the electrical properties of the completed cable shall be as specified in the applicable specification sheet.

3.6 Physical properties. When tested as specified in 4.8, the physical properties of the completed cable and cable components shall be as specified in the applicable specification sheet.

3.7 Repair of the cable jacket. Minor repair of the cable jacket shall be permitted. The materials and techniques used shall be such that the finished cable complies with all the requirements of this specification.

3.7.1 Conductor splices. Conductor splices in finished cable shall be removed prior to preparation for shipment.

3.8 Workmanship. The cable shall be a uniform and consistent product and shall be free of defects and surface contamination, which will adversely affect the serviceability of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified into three categories and are listed below. Inspections shall be conducted by the appropriate testing facility as specified in table V.

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

c. Qualification retention inspection (see 4.5).

4.1.1 Accredited independent testing laboratory (AL). An independent, third-party testing laboratory located in the continental United States and accredited to ISO/IEC 17025 (hereafter referred to as AL; see 6.8.1) for the specific tests shall be used to perform the flame propagation, gas flame (circuit integrity), smoke index, toxicity index, acid gas generation, halogen content, and watertightness tests noted in table V for initial qualification and retention of qualification testing. The cable manufacturer has the option of performing the flame propagation, gas flame (circuit integrity), and watertightness conformance tests at its in-house test facility if the cable material has not changed in composition, configuration, or supplier (see table V, [2]) and the cable manufacturer’s in-house test facility is certified as a test laboratory by an AL.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.6.

4.3 Qualification inspection. Qualification inspection shall consist of the inspections specified in table V and shall be in accordance with the applicable specification sheet.

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4.3.1 Sample selection. Cable group sample selection for qualification testing shall be in accordance with the information specified in table VI. When a manufacturer desires qualification of an individual type, size, or group, the selection of test specimens and scope of qualification will be subject to approval by NAVSEA. For specification sheets (see 3.1) having more than one type of insulation compound, each insulation type will require qualification.

4.3.2 Change approval. Changes in construction, formulation, materials, design, or manufacturing process of qualified products may require requalification. Unless otherwise specified by the qualifying activity, changes require written approval by NAVSEA.

TABLE V. Qualification and conformance inspection.

Inspection	Qualification	Qualification retention	Conformance	Requirement	Test method	Test facility 1/
Cable aging 500 °F (260 °C)	X			Spec sheet	4.8.4	MFR
Cable aging & compatibility	X			Spec sheet	4.8.5	MFR
Acid gas generation	X	X		Spec sheet	4.8.22	AL
Halogen content	X	X		Spec sheet	4.8.23	AL
Immersion	X	X		Spec sheet	4.8.26	MFR
Smoke index	X	X		Spec sheet	4.8.24	AL
Toxicity index	X	X		Spec sheet	4.8.25	AL
Weathering	X			Spec sheet	4.8.27	MFR
Fungus resistance	X			3.3.16	4.10	MFR
Gas flame (circuit integrity) 3/	X 5/	X 5/	X 2/ 4/	Spec sheet	4.8.13	AL
Accelerated service	X			Spec sheet	4.8.2	MFR
Cold working -4 °F (-20 °C)	X			Spec sheet	4.8.9	MFR
Armor	X			Spec sheet	4.8.36	MFR
Electrical moisture absorption	X			3.3.20 & spec sheet	4.8.32	MFR
Temperature/humidity	X			3.3.21 & spec sheet	4.8.33	MFR
Aging stability	X			Spec sheet	4.8.34	MFR
Dimensional stability	X			Spec sheet	4.8.35	MFR
Stress resistance	X			3.3.22 & spec sheet	4.9.13	MFR
Flame propagation 3/	X	X	X 2/	Spec sheet	4.8.12	AL
Long term jacket immersion	X			3.3.23	4.8.37	MFR
Surface transfer impedance 3/	X		X	Spec sheet	4.9.11	MFR

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TABLE V. Qualification and conformance inspection – Continued.

Inspection	Qualification	Qualification retention	Conformance	Requirement	Test method	Test facility [1/]
Basic electrical						
Conductor resistance	X		X	Spec sheet	4.9.4	MFR
DC resistance unbalance	X		X	Spec sheet	4.9.4	MFR
Voltage withstand	X		X	Spec sheet	4.9.8	MFR
Insulation resistance	X		X	Spec sheet	4.9.5	MFR
Conductor and shield continuity	X		X	Spec sheet	4.9.10	MFR
Jacket flaws	X		X	Spec sheet	4.9.9	MFR
Group A						
Visual and dimensional	X		X	3.3 through 3.8 & spec sheet	4.7	MFR
Concentricity & wall thickness	X		X	Spec sheet	4.7.1	MFR
Dimensions	X		X	Spec sheet	4.7.2	MFR
Capacitance	X		X	Spec sheet	4.9.2	MFR
Characteristic impedance	X		X	Spec sheet	4.9.3	MFR
Watertightness	X	X	X [2/]	Spec sheet	4.8.21	AL
Hydrostatic (open end)	X		X	Spec sheet	4.8.15	MFR
Abrasion resistance	X		X	Spec sheet	4.8.1	MFR
Crack resistance	X		X	Spec sheet	4.8.31 4.8.30	MFR
Swept electrical parameters	X		X	Spec sheet	4.9.12	MFR
Input impedance	X		X	Spec sheet	4.9.14	MFR
Insertion loss	X		X	Spec sheet	4.9.15	MFR
Group B						
Cold bending flexing cable	X		X	Spec sheet	4.8.8	MFR
Cold bending cable	X		X	Spec sheet	4.8.7	MFR
Thermoset proof	X		X	Spec sheet	4.8.31	MFR
Drip	X		X	Spec sheet	4.8.10	MFR
Twisting endurance	X		X	Spec sheet	4.8.20	MFR
Tear strength	X		X	Spec sheet	4.8.19	MFR
Physicals (unaged)	X		X	Spec sheet	4.8.18 and 4.8.29	MFR
Attenuation	X		X	Spec sheet	4.9.1	MFR
Bending endurance	X		X	Spec sheet	4.8.3	MFR
Durometer	X		X	Spec sheet	4.8.11	MFR

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TABLE V. Qualification and conformance inspection – Continued.

Inspection	Qualification	Qualification retention	Conformance	Requirement	Test method	Test facility 1/
Group C						
Physicals (aged)	X		X	Spec sheet	4.8.19 and 4.8.29	MFR
Permanence of print or stripe (conductor insulation)	X		X	Spec sheet	4.8.16	MFR
Permanence of printing (jacket)	X		X	Spec sheet	4.8.17	MFR
Heat distortion	X		X	Spec sheet	4.8.14	MFR
Cable sealant removability	X		X	Spec sheet	4.8.6	MFR
Shrinkage	X		X	Spec sheet	4.8.29	MFR
Mutual inductance	X		X	Spec sheet	4.9.6	MFR
Pulse response time	X		X	Spec sheet	4.9.7	MFR
Tensile & elongation	X		X	Spec sheet	4.8.28	MFR

NOTES:

1/ AL – Testing shall be conducted by an independent testing laboratory located in the continental United States and accredited to ISO/IEC 17025 for the specific test (see 4.1.1 and 6.8.1).

MFR – The manufacturer may conduct the testing in house without the requirement for the in-house test equipment, personnel, and procedures to be certified by an AL, or may contract an independent testing laboratory to conduct the testing. The independent testing laboratory does not need to be an AL for the specific test.

2/ The cable manufacturer has the option of performing the flame propagation, the gas flame (circuit integrity), and the watertightness conformance tests at its in-house test facility, if the cable material has not changed in composition, configuration, or supplier, and if the applicable test equipment, procedures, and personnel are certified as a test laboratory by an AL. If the cable manufacturer's in-house test facility is not certified as a test laboratory by an AL, then the flame propagation, gas flame (circuit integrity), and watertightness conformance testing shall also be performed by and at an AL. Contact NAVSEA 05S Command Standards via email at usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil for additional in-house testing requirements. For qualification and qualification retention, the flame propagation, gas flame (circuit integrity), and watertightness tests shall be performed by and at an AL.

3/ Group D (see 4.4.7).

4/ Gas flame (circuit integrity) conformance testing shall be performed for a 1-hour time duration.

5/ Gas flame (circuit integrity) qualification and qualification retention testing shall be performed for a 3-hour time duration.

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TABLE VI. Qualification and qualification retention samples and groups.

Group no.	Test samples		Types comprising the group 1/
	Additional cable types and sizes for flame propagation test only	All tests specified in the applicable specification sheet type and size	
1	LSDHOF-3	LSTHOF-9	All sizes 83 and smaller of LSSHOF, LSDHOF, LSTHOF, LSFHOF
2	None	LSTHOF-150	All sizes 133 and greater of LSSHOF, LSDHOF, LSTHOF, LSFHOF, plus LSSSF
3	LS2SJ-22 LSPBTMU-5	LSMDU-23 LSTPNW-1½	All sizes of LSMDU, LSMRI, LSPBTMU, LSMS, LSMU, LS2SJ, LS3SJ, LS4SJ, & LSTPNW, LSTPSJ
4	LSDCOP-1	LSTTOP-10	All sizes of LSMMOP, LSTTOP, LSDCOP, LSTCOP, & LSMHOF
5	LSDSGU-3 LSTTSU-1½	LSMSCU-7 LSTCTU-4 or LSTCJU-4	All sizes of LSMSCU, LSTCJU, LSTCTU, LSTTSU, 7 AWG and smaller of LSDSGU, LSTSGU, LSFSGU & LS7SGU, LSYSGU, LS2OW, LS3OW, LS4OW, LSSCF, LSFCF
6	LSDSGU-50	LSTSGU-150	All sizes greater than 7 AWG of LSSSGU, LSDSGU, LSTSGU, LSFSGU, LS6SGU & LS5KVTSGU, LSYSGU, LS2OW, LS3OW, LS4OW, LSSCF, LSFCF
7	LSTCTX-1 or LSTCKX-1, LSDPS-3	LSTCTX-7 or LSTCKX-7 LSPI-3	All sizes of LSTCTX, LSTCKX, LSTCJX, LSPI, LSDPS, LSTPS & LSFPS, LS7PS, LSTCJXN, LSTCKXN, LSTCTXN, LSDPSN, LSFPSN, LSTPSN, LS7PSN
8	LSTTRS-2	LSTTRS-8	All sizes of LSTTRS
9	LSDNW-3 LSMNW-7	LSTNW-150	All sizes of LSDNW, LSTNW, LSFNW, LSMNW, LS4NW, & LS8NW, & LSTCF
10	LS2CS-6 LS3U-3 LS2SU-3	LS1S50MU-16 LS2SU-10	All sizes of LS1S50MU, LS2AU, LS3SU, LS3U, LS1S75MU, LS1SMU, LS1SAU, LS3SF, LS2U, LS2CS, LS1SU, LS2SU, & LSMCOS
11	LS1SWU-2	LS2SWAU-10 LSECM	All sizes of LS2SWAU, LS1SWU, LS3SWU, LSECM, LS2WAU, LS1SMWU, LSSRW, LSDRW, LSTRW, LS2SWL, LS2UW, LS2WU, & LS2SWU
12	None	LSC5OSW	LSC5FSW, LSC5OSW, LSC5W
13	None	LSC5OS	LSC5FS, LSC5OS, LSC5, LSC5P, LSC5POS, LSC5POSR
14	None	LSPB2SDOSW	LSPB2SDOS, LSPB2SDOSW, LSPB2SD, LSPB2SDW
15	None	LS3C179DT	LS6C179DT, LS3C179DT, LSC264
16	None	LS5KVTEPSG	LS5KVTEPSG
17	None	LSTW-23	All sizes of LST, LSTW
18	None	LSE4EW-23	All sizes: LSE1E through LSE61E, LSE1EW through LSE61EW

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TABLE VI. Qualification and qualification retention samples and groups – Continued.

Group no.	Test samples		Types comprising the group 1/
	Additional cable types and sizes for flame propagation test only	All tests specified in the applicable specification sheet type and size	
19	None	LSE10U-26	All sizes: LSE6U through LSE60U, LSE7AU through LSE40AU, LSE3TTOP through LSE15TTOP, LSE7UW through LSE42UW, LSE7AUW through LSE40AUW
20	None	LSE7SPE-18	All sizes: LSE1SPE through LSE61SPE, LSE1SPEW through LSE61SPEW
21	None	LSE6OS-4	LSEC6FS, LSE6OS, LSE6OSW
22	None	LSE4CI-23	All sizes: LSE1CI through LSE91CI, LSE1CIW through LSE91CIW

NOTE:

1/ Includes all sizes and variations of the cable type unless specifically specified in the table.

4.4 Conformance inspection. Conformance inspection shall be performed on all completed cable in accordance with the procedure specified herein. This inspection shall consist of basic electrical tests plus groups A, B, C, and D inspections specified in table V and in accordance with the specification sheets.

4.4.1 Inspection lot. An inspection lot shall consist of the total number of units of product of any one type, size, and construction manufactured under essentially the same conditions.

4.4.2 Basic electrical tests. Basic electrical tests shall be in accordance with the applicable specification sheet and shall be performed on each length of completed cable. For electrical test purposes, length of completed cable shall be as defined in 6.8.5.

4.4.3 Sampling procedure. The required number of samples (see 6.8.3) for group A, B, C, and D inspections shall be selected at random from the inspection lot. Nonconforming starting and finishing ends of cable shall be removed by the manufacturer prior to selecting samples.

4.4.3.1 Sampling, accept/reject criteria. Material found unacceptable may be submitted for re-inspection only after all defective materials are removed or defects corrected. Material found acceptable during re-inspection can be re-identified and released as a separate lot.

4.4.4 Group A inspections. Group A inspections, as required by the specification sheet, shall be performed on specimens selected in accordance with 4.4.4.1.

4.4.4.1 Sampling for group A inspections. Specimens for group A inspections shall be selected from each lot in accordance with table VII.

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TABLE VII. Sampling for group A inspections.

Units of product in lot	Number of samples
1	1
2 through 10	2
11 through 30	3
31 through 90	7
91 through 210	15
211 and over	25

4.4.5 Group B inspections. Group B inspections, as required by the specification sheet, shall be performed on specimens selected in accordance with 4.4.5.1.

4.4.5.1 Sampling for group B tests. Specimens for group B tests shall be selected from each lot in accordance with table VIII.

TABLE VIII. Sampling for group B tests.

Units of product in lot	Number of samples
8 and under	1
9 through 30	2
31 through 90	3
91 through 210	4
211 and over	5

4.4.6 Group C inspections. Group C inspections, as required by the specification sheet, shall be performed on specimens (see 6.8.4) selected in accordance with 4.4.6.1.

4.4.6.1 Sampling for group C tests. Specimens for group C tests shall be selected in accordance with table IX.

TABLE IX. Sampling for group C tests.

Two month's production (units of product)	Number of samples
8 and under	1
9 through 30	2
31 through 90	3
91 through 210	4
211 and over	5

4.4.7 Group D inspections. Group D inspections, as required by the specification sheet, shall be performed on specimens selected in accordance with 4.4.7.1.

4.4.7.1 Sampling for group D tests. Samples for group D tests shall be selected from the first production run of each type and size of cable each year.

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4.4.7.2 Annual flame propagation conformance testing. Annual flame propagation testing shall be conducted in accordance with 4.8.12 by testing a sample selected from the first production run of each type and size of cable (see 4.4.7.1) each calendar year. The test results shall be recorded. Requirements for submittal of the test results will be provided.

4.4.7.3 Annual gas flame (circuit integrity) (1-hour) conformance testing. Annual gas flame (circuit integrity) testing shall be conducted in accordance with 4.8.13 by testing a sample selected from the first production run of each type and size of cable (see 4.4.7.1) each calendar year. The test results shall be recorded. Requirements for submittal of the test results will be provided.

4.4.7.4 Annual surface transfer impedance conformance testing. Annual surface transfer impedance testing shall be conducted in accordance with 4.9.11 by testing a sample from the first production run of each type and size of cable (see 4.4.7.1) each calendar year. The test results shall be recorded.

4.4.8 Certified test reports. Test reports, if required, shall be prepared as specified (see 6.2.1).

4.5 Qualification retention. Every 3 years after the initial qualification to this specification, the qualification retention inspections specified in table V shall be performed. Inspections shall be conducted on all materials for which a requirement is specified, and inspections shall be in accordance with the applicable test method and conducted by the appropriate testing facility as specified in table V.

4.5.1 Qualification retention sampling for flame propagation and gas flame (circuit integrity) testing. All cable samples selected shall have been manufactured in the current or previous year, as verified by the marker tape.

4.5.1.1 Sample selection. Cable group sample selection for qualification retention testing shall be in accordance with the information specified in table VI. The flame propagation test shall be performed on one cable type randomly selected from table X and one cable type randomly selected from table XI for a maximum of two cables of which the manufacturer is qualified to produce. If the manufacturer is not qualified in cable types from both tables, then only one cable type is selected for test. The gas flame (circuit integrity) test shall be performed on one additional cable type that the manufacturer is qualified to produce, randomly selected from table XI. The gas flame (circuit integrity) cable type shall be different than the cable type selected for flame propagation.

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TABLE X. Qualification retention samples and groups without circuit integrity.

Group no.	Cable types and sizes for flame propagation test only
1	LSSHOF-3, LSDHOF-3, LSTHOF-3
2	LSSSF-300
3	LSMDU-6, LSPBTMU-5, LSPBTMU-15, LSMU-14, LS2SJ-22, LS3SJ-22, LS2SJ-20, LSTPNW-1-1/2, LSTPNW-3, LSTPNW-5, LS1TPSJ-24, LS1TPSJ-22, LS1TPSJ-20
4	LSDCOP-1, LSDCOP-1-1/2, LSDCOP-2, LSMHOF-7, LSMHOF-10, LSMHOF-14, LSMMOP-5, LSTTOP-3, LSTTOP-5
8	LSTTRS-2, LSTTRS-4, LSTTRS-6
9	LSDNW-3, LSDNW-4, LSTNW-3, LSTNW-4, LSFNW-3, LSFNW-4, LSMNW-7, LSMNW-10, LS8NW-6, LS4NW-8, LSTCF-200
10	LSMCOS-2, LSMCOS-5, LSMCOS-6, LS1S50MU-16, LS2SU-3, LS2SU-7, LS3SU-3, LS3SU-7, LS3U-3, LS3U-7, LS1SMU-5, LS2U-10, LS2U-15, LS2U-19, LS2CS-6, LS2CS-18, LSLCFS-25
11	LS1SWU-12, LS2SWAU-3, LS2SWAU-7, LS2SWU-1, LS2SWU-3, LS3SWU-3, LS3SWU-7, LSSRW, LSDRW, LS2SWL-7, LS2UW-42
12	LSC5FS-4, LSC5FSW-4, LSC5OS-4, LSC5-4, LSC5W-4
13	LSC5P-4, LSC5POS-4, LSC5POSR-4
14	LSPB2SD-1, LSPB2SDW-1, LSPB2SDOS-1, LSPB2SDOSW-1
15	LS3C179DT, LS6C179DT, LSC264-4
16	LS5KVTEPSG-250

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TABLE XI. Qualification retention samples and groups with circuit integrity.

Group no.	Cable types and sizes for flame propagation and gas flame (circuit integrity) tests
5	LSDSGU-3, LSDSGU-4, LSTSGU-3, LSTSGU-4, LSFSGU-3, LSFSGU-4, LSMSCU-7, LSMSCU-10, LS7SGU-3, LS7SGU-4, LSTCJU-4, LSTCTU-4, LS6SGU-100, LS5KVTSGU-100, LSTTSU-1-1/2, LSTTSU-3, LSTTSU-5, LSYSGU-3, LSYSGU-4, LS2OW-3, LS2OW-4, LS3OW-3, LS3OW-4, LS4OW-3, LS4OW-4, LSSCF-3, LSSCF-4
6	LSSSGU-50, LSDSGU-50, LSTSGU-30, LSFSGU-50, LSSCF-26, LSFCF-42
7	LSTCJX-3, LSTCKX-1, LSTCTX-1, LSPI-3, LSDPS-3, LSDPS-4, LSTPS-3

4.6 Test conditions. Unless otherwise specified in the applicable specification sheet, the inspections specified in tables V and XII shall be accomplished at standard ambient conditions as follows:

a. Temperature: 73±4 ºF (23±2 °C)

b. Humidity: 50±40 percent RH

4.7 Visual and dimensional inspection. Cables shall be visually inspected to verify that the materials, design, construction, physical dimensions, marking, and workmanship are as specified in 3.3 through 3.8, and the specifications sheets. Concentricity, minimum wall, minimum average wall thickness, and dimensional inspections shall be made when applicable and as specified in 4.7.1 and 4.7.2.

4.7.1 Concentricity and wall thickness. The concentricity of the finished wire and cable jacket shall be determined by first locating and recording the minimum wall thickness measured on a cross-section of the finished wire or cable jacket. From this point on the outer rim of the finished wire or cable jacket (at which the minimum wall thickness was measured), three more reference points (90 degrees apart) on the outside rim of the finished wire or cable jacket shall be established. At each of these three reference points, the nearest member of the conductor or cable shall be selected, and the minimum wall thickness be measured. The average of the four readings shall be the measured minimum average wall thickness. (Note: In the case of cables with two or three components, two or three readings, respectively, shall be taken at the component high points; these readings shall be used to compute the minimum average). The percent concentricity shall be defined as 100 times the ratio of the measured minimum wall thickness to the measured minimum average wall thickness. Wall thickness measurements shall be made on cross-sectional insulation or jacket specimens removed from the finished wire or cable and viewed under magnification. The wall thickness shall be the shortest distance between the outer and inner rims of the wire insulation or cable jacket specimen.

4.7.2 Dimensions. Measurements shall be made on not less than a 12-inch-length of cable taken from the end of the sample unit. Inner components shall be made accessible by stripping and removing the outer components so as not to nick, cut, cold work, or otherwise damage the component to be measured. Four points for measurement shall be located 3 to 4 inches apart along the finished wire specimen or finished cable specimen lengths, as applicable. Measurements shall be made at each point at two perpendicular planes or as required to assure the minimum and maximum reading is attained at each point. A total of eight measurements shall be performed on each specimen. The minimum, maximum, and average value shall be recorded, as applicable. Measurements shall be made with a micrometer, caliper, or equal. A PI-tape may be used for measurements of diameters greater than 0.250 inch.

4.8 Test methods (physical).

4.8.1 Abrasion resistance. Abrasion resistance of insulation coverings and cable jackets shall be determined as specified in 4.8.1.1 through 4.8.1.4.

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4.8.1.1 Specimens. Specimens shall be as follows:

a. Insulation. Each specimen shall consist of a single insulated conductor (including insulation covering, if any) of not less than 22 inches in length. Specimens shall be removed from completed cable as follows:

(1) Cables containing four or fewer conductors - one specimen shall be removed from each insulated conductor.

(2) Cables containing between 5 and 20 different insulated conductors - a total of 4 specimens shall be removed.

(3) Cables containing 21 or more conductors - the number of specimens removed shall be equal to the square root (rounded to the nearest whole number) of the total number of conductors. Each specimen shall be removed from a different insulated conductor.

b. Cable jacket. One cable jacket test specimen shall be prepared by extruding material onto a 16 AWG (19/29) conductor. The material shall be manufactured using process conditions as close as possible to those used to produce the overall cable jacket and shall have an overall cable diameter of 0.119+0.001 inch.

4.8.1.2 Special apparatus. The apparatus shall include the following:

a. Abrading machine. A cylinder, which incorporates 2 abrading elements on its surface, that is motor driven to rotate about a horizontal axis at 17+3, - 2 revolutions per minute (rpm) and over which the specimen is draped, as shown on figures 2 and 3 and as described below:

(1) Each abrading element shall consist of a 5⁄16-inch square, high speed tool bit (Cleveland Twist Drill Company number 855 or equivalent) that has been ground on two adjacent, longitudinal sides to produce a single, sharp 90-degree longitudinal edge free of nicks. A medium grade grinding wheel (Norton Company number 39C6018VK or equivalent) shall be used for this grinding. Abrading elements shall be reground as required; an abrading element shall be not used in excess of 5 hours without regrinding. Whenever the perpendicular distance between either pair of opposite longitudinal sides of an abrading element becomes less than 0.3085 inch (as by repeated regrinding), then that abrading element shall be discarded and replaced.

(2) The cylinder shall be 8±⅛ inches in diameter and shall be rigidly fabricated from metal.

(3) Each of the two abrading elements shall be maintained in electrical contact with the metal cylinder by securing each in a notch cut into the cylinder surface, parallel to the cylinder axis; the two notches required shall be spaced 180±2 degrees apart around the circumference of the cylinder. These notches shall be cut such that the sharpened 90-degree longitudinal edge of each element shall be facing outward from the cylinder surface and such that the midpoint of a straight line drawn between the midpoint of the two longitudinal edges adjacent to the sharpened edge is tangential ±0.003 inch to the cylinder surface.

(4) An automatic counter shall be provided to totalize the number of times the test specimen is scraped by the abrading elements during the test.

(5) A 1-pound ±0.5 ounce-weight shall be provided for applying tension to the specimen.

b. Short circuit monitor. A voltage source of not less than 12 V (either DC or root mean square [RMS] alternating current [AC]) that can be applied between the specimen conductor and both abrading elements of the abrading machine as shown on figures 2 and 3. (The connection to the abrasion elements may be provided by a wiping contact on the cylinder.) A means (such as an electrical relay) shall be provided whereby an electrical connection between either abrading element and the specimen conductor shall automatically stop the rotation of the cylinder on the abrading machine (as by removing motor power).

4.8.1.3 Procedure. Specimens shall be wiped with a clean, dry cloth to remove any lubricant or dirt. Each specimen shall then be tested (either simultaneously or at one time) as follows. Each specimen shall be hung circumferentially over the cylinder of the abrading machine such that each specimen shall form an arc of not less than 170 degrees around the cylinder for the remainder of the test. One end of each specimen shall then be secured to a fixed surface. The abrading machine weight shall be attached to the opposite specimen end such that both this specimen end and the weight are freely suspended. The automatic counter shall be set initially to zero; the short circuit monitor shall be applied between both abrading elements on the abrading machine and the conductor of each specimen. The cylinder motor shall then be turned on, allowing the cylinder to rotate beneath each specimen from secured end to weighted end.

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4.8.1.4 Observation. Specimen failure shall be construed if electrical contact occurs between either abrading element and any specimen conductor (as evidenced by cessation of cylinder rotation) prior to completing the specified number of abrasive scrapes (see the applicable specification sheet).

4.8.2 Accelerated service. The ability of completed cable to withstand electrical current overload shall be determined as specified in 4.8.2.1 through 4.8.2.4.

4.8.2.1 Specimen. The specimen shall consist of a 15-foot ±2-inch-length of completed cable.

4.8.2.2 Special apparatus. The apparatus shall include the following, and separate voltage and current sources shall be used:

a. Voltage source. A three-phase, 60 hertz (Hz) adjustable voltage source, as shown on figure 4, for providing a separate, single-phase test voltage through a normal blow fuse or overload relay to each specimen conductor. This voltage source shall be in accordance with the following:

(1) Specimens rated for 1 kilovolt (kV). The test voltage shall be 450±10 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 ampere (A).

(2) Specimens rated for 5 kV. The test voltage shall be 4,160±100 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of ¼ A. The secondary winding voltage rating of any constituent voltage step up transformers shall be not less than the required test voltage. Each phase of the voltage source shall supply the specimen with a current that shall be not less than twice the fuse or overload relay rating.

b. Current source. A three-phase, 60 Hz adjustable voltage source shall be used with three current transformers, as shown on figure 4, to provide three adjustable, single-phase test currents as specified. The secondary winding of each current transformer shall be isolated from electrical ground. Bus bar conducting test current without generating significant amounts of heat shall be used to convey the test currents from the current transformers to the specimen conductors.

c. Electrical metering. Instrumentation shall be provided, as shown on figure 4, such that the test voltage and current applied to each specimen conductor shall be measured with an error of not more than ±2 percent.

d. Heating chamber. A heating chamber shall be provided for supporting the requirements as specified in 4.8.2.3. This chamber shall have interior finishes and dimensions that discourage the reflection of radiant heat (generated by the test currents) onto the specimen.

e. Thermocouple devices. Four thermocouples and an associated temperature-indicating device shall be provided for use, and temperatures shall be as specified in 4.8.2.3. Temperatures shall be measured with an error of not greater than ±1.8 °F (±1 °C).

4.8.2.3 Procedure. The specimen shall be formed into a single, horizontal loop within the heating chamber to permit free circulation of convection air currents. Air temperature within the chamber shall be maintained at 86±3.6 °F (30±2 °C) during the test (except in the immediate vicinity of the specimen) and means shall be provided to prevent forced air drafts from directly striking the specimen. The four thermocouples shall be placed on the top of the specimen, near its midpoint, and in firm contact with its jacket. These thermocouples shall be placed longitudinally along the specimen jacket at 2±1/16-inch intervals. On unarmored specimens, firm jacket contact can be assured by the use of a length of string with a weight on each end as this can be hung over the jacket after forming a single wrapping around the jacket and over the thermocouple bead. On armored specimens, firm jacket contact can be assured by opening a small space between the jacket and an armor carrier (as with an ice pick), and inserting the thermocouple bead within this space, afterwards pulling the armor carrier longitudinally to press the thermocouple bead tightly against the jacket. The jacket shall not be cut and tapes shall not be used to hold the thermocouple beads since these may cause premature specimen failure due to elevated jacket temperatures (see 4.8.2.4). Two thermocouples shall be used to measure ambient temperature within the chamber. The test voltages and currents shall be simultaneously applied to the specimen as shown on figure 4. If the specimen is armored, an additional test voltage shall be applied by grounding the armor, as shown on figure 4. This shall effectively apply an additional voltage that is approximately equal to the test voltage divided by the square root of 3 between the armor and the specimen conductors. The test currents and voltages shall remain continuously applied to the specimen for a period of not less than 7 hours, unless specimen failure (see 4.8.2.4) occurs prior to the end of this period; then the test shall be terminated prematurely. The following notes shall apply for specimens with more than three conductors:

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a. Unarmored four-conductor specimens. The green insulated conductor shall be isolated from all test voltages and currents.

b. Armored four-conductor specimens. The green insulated conductor shall be electrically connected to the armor during the test.

c. Six-conductor specimens (armored or unarmored). The radially opposed conductors within the core of the specimen shall be electrically connected, at both specimen ends, to effectively form three conductors during the test.

d. Seven-conductor specimens (armored or unarmored). The radially opposed conductors within the core of the specimen shall be electrically connected, at both specimen ends, to effectively form three conductors during the test. In unarmored specimens, the central core conductor shall be isolated from all test voltages and currents. In armored specimens, the central core conductor shall be electrically connected to the armor during the test.

4.8.2.4 Observation. Any of the following shall constitute specimen failure, if occurring prior to the expiration of the 7-hour minimum test period:

a. The blowing of any fuse or the tripping of any overload relay.

b. Any thermocouple indication of the jacket temperature exceeding 275 °F (135 °C).

c. Jacket sagging on an unarmored specimen.

d. Jacket exudation through the armor of an armored specimen. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)

4.8.3 Bending endurance. The ability of completed cable to withstand repeated smooth and continuous, reversing bending motion while subjected to a specified temperature shall be determined.

4.8.3.1 Specimen. The specimen shall consist of a 30±6-inch-length of completed cable. Specimen conductor ends shall be electrically interconnected such that a single series electrical circuit is formed by the conductors. The two end conductors of this series circuit shall be connected as specified in 4.8.3.3.

4.8.3.2 Special apparatus. The apparatus shall include the following:

a. Flexing machine. A device for suspending the specimen vertically and for automatically bending the upper portion of the specimen alternately left and right by 90±5 degrees from vertical over rollers as shown on figure 5 and as further described below.

(1) The specimen shall be suspended and bent by means of a rigid, motor-driven arm, that initially shall be vertical, shall be horizontally pivoted about its lower end, and shall incorporate a clamp at its upper end for securing the upper end of the specimen jacket. The distance between the upper end of the specimen jacket and the nearest edge of this clamp shall be not less than 1 inch. When its motor is turned on, the arm shall move at the rate of not more than 14, nor less than 12 cycles per minute, where a cycle is 180±10 degrees of clockwise arm travel plus 180±10 degrees of counterclockwise arm travel (360±20 degrees total travel). The specimen clamp shall be positioned such that when the arm is vertical (straight up), the longitudinal axis of the specimen shall intersect the pivoting axis. The lower edge of the clamp shall be 8.5±0.5 inches above a horizontal line drawn across the tops of the upper rollers.

(2) The specimen shall be bent over two solid steel cylindrical rollers (the upper rollers as shown on figure 5) of the diameter specified. Unless otherwise specified in the applicable specification sheet: for specimens over 1.0 inch in measured diameter, the maximum roller diameter shall be four times the maximum specified overall cable diameter; for specimens with a diameter of 0.750 to 1.000 inch, the maximum roller diameter shall be 3.5 times the maximum specified overall cable diameter; and for specimens with a measured diameter less than 0.750 inch, the maximum roller diameter shall be three times the maximum specified overall cable diameter. Both rollers shall have a smooth surface finish on their curving surfaces, shall be pivoted about their axes, and capable of free rotation at all times. They shall be positioned such that their axes are horizontal and parallel and share a common horizontal centerline. The tops of the rollers shall be below the horizontal centerline of the flexing machine arm pivoting axis by a distance equal to one half of the maximum specified overall cable diameter ±10 percent. The rollers shall be located equidistantly on opposite sides of the vertical centerline passing through the flexing machine arm-pivoting axis. They shall be spaced apart such that their nearest edges are ⅛±½ inch from the specimen passing between them when the flexing machine arm is vertical.

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(3) Two cylindrical rollers (the lower rollers as shown on figure 5), of not less than ⅜ inch diameter, shall be provided to restrain lateral motion of the specimen. Both rollers shall be pivoted about their axes and capable of free rotation at all times. They shall be positioned such that their axes are horizontal centerline. The axes of these rollers shall be 7±1 inches below the axes of the rollers over which the specimen is bent (the upper rollers as shown on figure 5). The lower rollers shall be located equidistantly on opposite sides of the vertical centerline passing through the flexing machine arm pivoting axis. They shall be spaced apart such that their nearest edges are ⅛±1/32 inch from the specimen passing between them when the flexing machine arm is vertical.

(4) A weight shall be provided for applying tension to the specimen. This weight shall be attached to the specimen by means of a second clamp. The lower end of the specimen jacket shall be a distance of not less than 1 inch below the nearest edge of this clamp. The weight shall be chosen as follows:

Specimen tension produced by the clamp plus the weight in pounds = $10(3.14159)d^2 \pm (3.14159)d^2/2$

Where: d = specified maximum overall cable diameter, in inches.

The clamp and weight shall be attached such that the specimen tension is applied vertically downward along the specimen axis.

(5) Both the specimen clamp on the flexing machine arm and the specimen clamp supporting the tensioning weight shall apply radial compression to the specimen such that neither the specimen nor any of its internal components will slip in the clamping area during the test.

(6) An automatic counter shall be provided to total the number of bending cycles performed on the specimen during a test.

(7) The specimen shall not come into contact with any piece of machinery at any time during the test, except for the upper and lower rollers, the two specimen clamps, and the circuit continuity monitor.

b. Circuit continuity monitor. The circuit continuity monitor is a means for continuously monitoring the electrical continuity of the series through connected specimen conductors, which shall automatically stop the flexing machine arm (as by removing motor power) when a specimen conductor breaks.

c. Test chambers. Test chambers are chambers for maintaining the specified air temperature within ±3.6 °F (2 °C) for accommodating the flexing machine with the specimen attached and for supporting the requirements as specified in 4.8.3.3. The chamber air temperature shall be measured in the immediate vicinity of the upper roller of the flexing machine. The chamber shall incorporate a viewing port for observing the specimen under test.

d. Voltage withstand test apparatus. The voltage withstand test apparatus shall be as specified in 4.9.8.2.

4.8.3.3 Procedure. The flexing machine shall be installed in the test chamber and the specimen installed in the flexing machine as shown on figure 5. The circuit continuity monitor shall be connected between the two end conductors of the single series circuit within the specimen. The test chamber doors shall then be closed and the air temperature within the chamber shall be brought to the specified value within ±3.6 °F (2 °C). After the air temperature within the chamber has been maintained at this value for a period of not less than 1 hour, the automatic counter shall be set at zero and the bending endurance test begun by turning on the motor driven flexing machine arm. The specified number of bending cycles shall be performed. At all times during the test, the air temperature within the chamber shall be maintained as specified within ±3.6 °F (2 °C), and the test chamber doors shall remain closed.

4.8.3.4 Observation. Either of the following shall constitute specimen failure, except the specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest using different specimen from the same length of cable:

a. Stoppage of the flexing machine arm caused by the circuit continuity monitor (indicating conductor breakage), prior to the completion of the specified number of bending cycles. (Loss of conductor continuity from loss of connectors shall not be construed as failure. If this situation occurs, reconnect connectors and continue testing.)

b. Rupture of the specimen jacket prior to the completion of the number of specified bending cycles.

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4.8.3.5 Further procedure. If the specimen shows no failure, it shall be allowed to return to room temperature and shall be subjected to the voltage withstand test as specified in 4.9.8, using the specified voltages. Following this, that portion of the specimen that had been repeatedly bent over the upper rollers shall be dissected. Each of the constituent components shall be visually inspected for deterioration (see 4.8.3.6). Fillers are not considered components for this test.

4.8.3.6 Further observation. Either of the following shall constitute specimen failure, except the specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest using different specimen from the same length of cable:

a. Specimen fails the voltage-withstand test (see 4.9.8).

b. Specimen upon dissection and inspection exhibits visible distortion or cracking of any specimen component, including strand breakage on any conductor, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service. Fillers are not considered components for this test.

4.8.4 Cable aging (500 °F [260 °C]). This test shall be conducted to determine the ability of armored cable with silicone rubber insulation and jacketing to withstand accelerated heat aging.

4.8.4.1 Specimen. The specimen shall consist of a 48±1 inch-length of completed cable.

4.8.4.2 Special apparatus. The apparatus shall include the following:

a. A forced fresh air circulating oven shall be used for supporting the requirements as specified in 4.8.4.3. The oven air temperature shall be measured in the immediate vicinity of the specimen.

b. Tensile strength and elongation test apparatus shall be as specified in 4.8.31.

4.8.4.3 Procedure. The specimen shall be bent at the approximate middle of its length to form a symmetrical U shape, such that its inside diameter (ID) shall be six times the maximum overall cable diameter ±¼ inch, as specified in the applicable specification sheet. The specimen shall be secured in this position using heat-resistant twine and shall be freely suspended vertically such that the specimen ends face downwards within the oven. The air temperature within the oven shall be raised in a uniform manner from room temperature to a temperature of 500±18 °F (260±10 °C) over a period of not greater than 3 hours and not less than 2 hours. The oven air temperature shall be maintained at 500±18 °F (260±10 °C) for a period of not less than 24 hours and shall then be returned to room temperature. Provision shall be made to catch, for later weighing, any material that may fall away from the specimen ends during this test. Following the test, the air within the oven shall be allowed to return to room temperature, and the collected material that fell away from the specimen shall be weighed. The specimen shall then be removed from the oven.

4.8.4.4 Observation. Any of the following shall constitute specimen failure:

a. The falling away of any material in excess of 0.0353 ounce (1 gram) from the specimen ends during the test.

b. Jacket exudation through the armor of an armored cable. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)

c. Jacket shrink-back in excess of ⅛ inch from either end of the specimen.

4.8.4.5 Further procedure. If the specimen shows no failure, its jacket shall be subjected to the tensile and elongation tests as specified in 4.8.31.

4.8.4.6 Further observation. Specimen failure shall be construed if either the jacket tensile strength or elongation is less than that specified in the applicable specification sheet.

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4.8.5 Cable aging and compatibility (257 °F [125 °C]). This test shall be performed to detect any significant degradation due to component incompatibility or prolonged over heating of completed cable containing thermosetting polyolefin or silicone rubber and glass tape insulation. The dielectric strength values specified in 3.3.4.2.1 shall be met after the tapes have been conditioned for 24 hours at 73.4 °F (23 °C) and 96 percent relative humidity and tested while maintaining temperature and humidity conditions of the short time dielectric breakdown test in accordance with 4.6. Silicone rubber glass tape shall be held from unwinding from conductors when the conductors are exposed and subjected to normal handling during splicing and terminating. This test permits either of two methods (the current overload method or the hot air oven method) to be used for heat aging.

4.8.5.1 Specimen. Specimens for the cable aging and compatibility test shall be as follows:

a. For the current overload method, the specimen shall consist of a 40±4-foot-length of completed cable.

b. For the hot air method, two specimens of completed cable shall be required. Specimen number one shall have a length of 30±3 feet. Specimen number two shall have a length of not less than the sum of 12 inches plus 20 times the maximum overall diameter of the cable specified in the applicable specification sheet. Both specimens shall be removed from the same length of cable.

4.8.5.2 Special apparatus. The apparatus shall include the following:

a. The oven (required for either heat aging method) shall support the requirements as specified in 4.8.5.3, as applicable. The oven air temperature shall be measured in the immediate vicinity of the specimen.

b. The mandrel (required for either heat aging method) shall be rigid, smooth cylinder or partial cylinder, with a continuous curved surface of not less than 180 degrees, of the length and construction for the specimen bending as specified in 4.8.5.5. The mandrel diameter shall be approximately 12 times the maximum overall cable diameter specified in the applicable specification sheet.

c. The voltage withstand test apparatus (required for either heat aging method) shall be as specified in 4.9.8.3.

d. The insulation resistance test apparatus (required for either heat aging method) shall be as specified in 4.9.5.3.

e. A thermocouple device (required for the current overload method if sub-method 4.8.5.3.a(1) is used) and an associated temperature-indicating device shall be provided for the use and temperatures specified in 4.8.5.3.a. Temperatures shall be measured with an error of not greater than ±1.8 °F (1 °C).

f. The current source (required for current overload method only) shall be an adjustable source of either DC or 60 Hz single-phase AC for maintaining specimen conductors at an elevated temperature as specified in 4.8.5.3.a. A means shall be provided for measuring the current produced by this source with an error of not greater than ±5 percent.

g. A resistance meter (required for the current overload method if sub-method 4.8.5.3.a is used) shall be provided for measuring the resistance of a single specimen conductor at room temperature and at elevated temperatures, as specified in 4.8.5.3.a. The meter shall exhibit a measurement error of not more than ±1 percent.

4.8.5.3 Procedure. Select and perform either of the two heat aging methods, the current overload method or the hot air method, specified in a or b below, respectively. The oven air temperature (in both methods), conductor temperature, and current (in the current overload method only) shall be recorded at intervals of not greater than 15 minutes during the first hour after attaining the heat aging temperature, at intervals of not greater than 1 hour for the following 5 hours and not less than twice daily thereafter for the duration of the heat aging. The interval between the consecutive temperature recordings shall not exceed 17 hours at any time during heat aging.

a. Prior to heat aging, the current overload method specimen shall be subjected to the insulation resistance and voltage withstand test in accordance with 4.9.5 and 4.9.8, respectively. The voltage withstand test shall use the voltages specified in the applicable specification sheet. Following these, specimen conductor ends shall be electrically interconnected such that a single series electrical circuit is formed in the conductors. Provision shall then be made to measure specimen conductor temperature, using either of the following sub-methods.

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(1) The smallest practicable “V” shaped cut shall be made along the specimen at a point approximately midway between the specimen ends, and the thermocouple shall be placed beneath the resulting triangular tab. The thermocouple shall be positioned such that it is in firm contact with any single specimen conductor layer. (For specimens with only a single layer of conductors, any specimen conductor may be used.) The thermocouple shall be used with its associated temperature-indicating device to measure specimen conductor temperature during heat aging.

(2) The resistance meter shall be connected to measure the resistance between the ends of any single specimen conductor in the innermost conductor layer. (For specimens with only a single layer of conductors, any specimen conductor may be used.) The meter leads shall be positioned on the conductor extremities such that all conductor interconnections are excluded from resistance measurements. The conductor temperature shall then be determined by using the following formula:

Conductor temperature, in °F = (454.1 + t) (R/r) 454.1

Where: R = Measured conductor resistance, in ohms, during heat aging.

t = Measured room temperature, in °F, to which the specimen is exposed prior to heat aging. Temperature t shall be measured in the immediate vicinity of the specimen; measurement accuracy shall be within ±0.9 °F (0.5 °C).

r = Measured conductor resistance, in ohms, when the specimen is exposed to room temperature. It shall be held constant, within ±0.9 °F (0.5 °C), for a period of not less than 2 hours prior to this measurement.

The specimen shall next be placed into the oven such that a length of not less than the sum of 12 inches plus 20 times the maximum overall cable diameter specified in the applicable specification sheet remains straight and horizontal and such that the remainder of the specimen is formed into the largest practicable horizontal coil (to permit free circulation of convection air currents). The current source shall be connected between the two end conductors of the single series circuit within the specimen. Unless otherwise specified in the applicable specification sheet, the specimen shall then be subjected to heat aging, the air temperature within the oven shall be raised to 122±5.4 °F (50±3 °C), and the current source adjusted to produce a conductor temperature of 257±9 °F (125±5 °C). These temperatures shall be maintained for a continuous period of not less than 400 hours. The current source shall not be shut off during heat aging, except for momentary shut off for taking resistance measurements when sub-method a(2) herein is used. Following heat aging, the specimen shall be allowed to cool throughout to room temperature. The conductor temperature shall then be recorded. A recording that differs from room temperature by more than 9 °F (5 °C) shall be considered to indicate failure of the conductor measurement means. If any such failure should occur, the heat aging test shall be considered to be invalid and shall be repeated using a different specimen removed from the same cable. (A possible advance indication of such a failure and the need to retest may be the need to make prominent adjustments of conductor current in order to maintain indicated conductor temperature during heat aging.) Following heat aging, the specimen shall be subjected a second time to the insulation resistance and voltage-withstand tests as specified in 4.9.5 and 4.9.8, respectively. A sample, of length not less than 12 inches plus 20 times the maximum overall cable diameter specified in the applicable specification sheet, shall then be removed from that portion of the specimen that had been kept straight during the heat aging. This sample shall be subjected to the bending procedure as specified in 4.8.5.5. A portion of the remainder of the specimen shall be subjected to the cable filler removability test as specified in 4.8.7.

b. Prior to heat aging, hot air oven method specimen one shall be subjected to the insulation resistance and voltage withstand tests as specified in 4.9.5 and 4.9.8, respectively. The voltage withstand test shall use the voltages specified in the applicable specification sheet. Both specimens shall then be placed within the oven; specimen number one shall be formed into the largest practicable horizontal coil within the oven and specimen number two shall be positioned such that it is both straight and horizontal. Both specimens shall be simultaneously subjected to heat aging. Unless otherwise specified in the applicable specification sheet, the air temperature within the oven shall be raised to 257±9 °F (125±5 °C). This temperature shall be maintained for a continuous period of not less than 400 hours. Following heat aging, both specimens shall be allowed to cool to room temperature. Specimen one shall then be subjected a second time to the insulation resistance and voltage withstand tests as specified in 4.9.5 and 4.9.8, respectively. A portion of specimen one shall next be subjected to the cable filler removability test as specified in 4.8.7. Specimen two shall be subjected to the bending procedure as specified in 4.8.5.5.

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4.8.5.4 Observation. Any of the following shall constitute specimen failure:

a. The falling away of any material from either end of any specimen during heat aging.

b. Jacket sagging on any unarmored specimen.

c. Jacket exudation through the armor of any armored specimen. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)

d. Specimen (specimen one if the hot air oven method is used) fails either of the insulation resistance tests (see 4.9.5).

e. Specimen (specimen one if the hot air oven method is used) fails either of the voltage withstand tests (see 4.9.8).

f. Specimen (specimen one if the hot air oven method is used) fails the cable filler removability tests (see 4.8.7).

4.8.5.5 Further procedure. The mandrel shall be secured to a fixed surface; one end of the specimen shall be secured such that it is both in contact with the curved surface of the mandrel and perpendicular to the mandrel center of the curvature axis. The opposite end of the specimen shall be bent circumferentially around the curved mandrel surface through an arc of not less than 170 degrees. Bending shall be accomplished at the rate of approximately 20 degrees per second. The specimen shall then be secured to maintain its bent shape (as with a piece of twine, stretched taut and tied between each end of the specimen) and shall be removed from the mandrel. Not less than 2/3 of the bent portion of the specimen shall then be dissected; each of its constituent components shall be visually inspected for deterioration (see 4.8.5.6).

4.8.5.6 Further observation. Specimen failure shall be construed if the specimen, upon dissection and inspection, exhibits visible deterioration of such a nature or extent as to impair the performance of the cable in service. Deterioration sufficient to constitute specimen failure shall include, but is not limited to, distortion or cracking of any constituent component, hardening of filler material, or any discoloration indicating material incompatibility.

4.8.6 Cable sealant removability. This test shall determine whether or not filler materials can be readily removed from the constituent components of cable. This test shall not apply to filler material used to fill voids between either conductor strands or braid strands.

4.8.6.1 Specimen. The specimen shall be a 2-foot ±2-inch-length of completed cable.

4.8.6.2 Procedure. Overall specimen components (such as armors, jackets, overall binders, or shields) shall be removed from one end of the specimen for a distance of not less than 10 inches, thereby exposing the cable core. Using fingers only, sealant material shall then be removed from all exposed insulated conductors or from all groupings of insulated conductors that have a common covering (such as insulated conductor pairs or triads, which have an overall shield or binder) for a distance of not less than 8 inches. Tools, cloths, or solvents shall not be used to assist with sealant removal. The common covering and sealant material shall then be removed from not less than one grouping of insulated conductors, on specimens incorporating such groupings, for a distance of not less than 5 inches. Fingers only shall then be used for sealant removal. If occasional particles of sealant material remain on any constituent component of the cable core, then the removal of these particles shall be attempted by means of light brushing with fingers or a dry cloth.

4.8.6.3 Observation. Any of the following shall constitute specimen failure:

a. Sealant material that is nonflexible.

b. Sealant material that adheres to the finger or to any component of the cable core. The adherence of occasional small particles of sealant material to the cable core is acceptable unless there are small particles that cannot be removed from the core components by light brushing with fingers or a dry cloth.

c. Sealant material that leaves a residue on cable core components.

d. Sealant material that cannot be removed in less than 5 minutes.

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4.8.7 Cold bending cable. This test shall determine the ability of completed cable, which is not intended to be flexed during use, to withstand bending at reduced temperature, such as might be encountered during shipboard construction or repair.

4.8.7.1 Specimen. The specimen shall consist of a piece of completed cable that shall have a length of not less than the sum of 12 inches plus 1.6 times the mandrel diameter specified in the applicable specification sheet.

4.8.7.2 Special apparatus. The apparatus shall include the following:

a. The refrigeration chamber shall support the requirements as specified in 4.8.8.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

b. The mandrel shall be a rigid, smooth cylinder or partial cylinder, with a continuous, curved surface of not less than 180 degrees, of the length and construction for the specimen bending of 4.8.7.3. Unless otherwise specified in the applicable specification sheet, the mandrel diameter shall be a minimum of 11 and a maximum of 13 times the maximum specified overall cable diameter. The mandrel shall be provided with a clip for affixing one specimen end (see 4.8.7.3).

4.8.7.3 Procedure. The specimen shall be straightened. If it does not remain straight of itself, it shall be held straight by securing it (as with twine to a straight wooden bar). Unless otherwise specified in the applicable specification sheet, the straightened specimen (including wooden bar if applicable) shall be placed within the refrigeration chamber and subjected to an air temperature of minus 4±4 °F (minus 20±2 °C) for a continuous period of not less than 6 hours. Prior to any change of chamber temperature, the specimen shall be removed from the chamber (following detachment from the wooden bar, if applicable) and one specimen end shall be inserted into a clip that has been affixed to (or adjacent to) the mandrel, such that the specimen is both in contact with the curved surface of the mandrel and also perpendicular to the mandrel center of curvature axis. The opposite end of the specimen shall then be bent circumferentially around the curved mandrel surface through an arc of not less than 170 degrees. This bending shall be accomplished at a rate of not less than 20 degrees per second. The period of time between the removal from the refrigeration chamber and the completion of specimen bending shall be not greater than 40 seconds. Care shall be taken to avoid heating the specimen, as by excessive handling, during removal from the refrigeration chamber and during bending. Following bending, the specimen shall be secured to maintain its bent shape (as with a piece of twine stretched taut and tied between each end of the specimen) and shall then be removed from the mandrel. Not less than two thirds of the bent portion of the specimen shall then be dissected and each of its constituent components visually inspected for deterioration (see 4.8.7.4).

4.8.7.4 Observation. Specimen failure shall be construed if the specimen, either during bending or upon dissection and inspection, exhibits distortion or cracking of any specimen component or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

4.8.8 Cold bending flexing cable. This test shall determine the ability of completed cable to withstand bending at reduced temperature, such as might be encountered during shipboard construction or repair.

4.8.8.1 Specimens. The required specimens shall be as follows:

a. The cable specimen shall consist of a completed cable, which shall have a length of not less than the sum of 12 inches plus 60 times the maximum overall cable diameter specified in the applicable specification sheet.

b. Insulated conductor specimens shall consist of not fewer than two different insulated conductors removed from the completed cable. Each insulated conductor shall have a length of not less than the sum of 12 inches plus the sum of 30 times the maximum measured insulated conductor diameter of the specimen conductors removed from the completed cable.

4.8.8.2 Special apparatus. The apparatus shall include the following:

a. A refrigeration chamber for supporting the requirements as specified in 4.8.8.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

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b. Two mandrels, as specified in (1) and (2), shall be provided and shall be mounted within the test chamber such that the axis of each is horizontal and each supports the requirements as specified in 4.8.8.3. Provision shall be made whereby each mandrel can be rotated about its axis at the rate specified in 4.8.8.3, by an operator located outside the chamber (as by crank, the shaft of which passes through the chamber wall and mandrel axes to which the mandrels are affixed, or by remotely controlled and appropriately geared motor, which is located within the chamber) and whereby each mandrel can be locked in a desired rotational position.

(1) Cable mandrel shall be a rigid, smooth cylinder and shall have a diameter of not less than 2.5 and not more than 3 times the minimum overall cable diameter specified in the applicable specification sheet. The mandrel shall be of the length and construction for the specimen wrapping as specified in 4.8.8.3.

(2) Insulated conductor mandrel shall be a rigid, smooth cylinder and shall have a diameter of not less than 85 percent and not greater than that of the maximum measured insulated conductor diameter of the specimen conductors removed from the completed cable and that shall be of the length and construction for the specimen wrapping as specified in 4.8.8.3.

c. A magnifying glass that shall provide a magnification of not less than three diameters.

4.8.8.3 Procedure. One end of the cable specimen shall be secured to the curved surface of the cable mandrel, and one end of each insulated conductor specimen shall be secured to the curved surface of the insulated conductor mandrel, such that all specimens are vertically suspended without obstruction and that free space exists on each mandrel to accommodate the specimen wrapping to follow. A weight shall be attached to the opposite end of each specimen such that the suspended portion of each specimen shall remain vertical during wrapping. The refrigeration chamber doors shall then be closed, and the specimen shall be subjected to an air temperature of minus 40±4 °F (minus 40±2 °C) for a period of not less than 20 hours. Prior to any change of chamber temperature, the mandrels shall be rotated from outside the chamber at a rate of 2.5±0.5 revolutions per minute, such that the specimens are wrapped around the mandrels for not fewer than five closely spaced turns. The mandrels shall then be locked to maintain the specimens in their wrapped position. The chamber doors shall be opened, and the specimen shall be secured to the mandrels to prevent the loosening or unwrapping. The mandrels, with specimens attached, shall then be removed from the chamber and the entire wrapped portion of each specimen shall be examined using the magnifying glass.

4.8.8.4 Observation. Specimen failure shall be construed if the specimen, upon visual inspection, exhibits distortion or cracking of any specimen component or exhibits any other visible deterioration of such a nature or extent to impair the performance of the cable in service.

4.8.9 Cold working (minus 4±4 °F [minus 20±2 °C]). This test shall determine the ability of completed cable, which is not intended to be flexed during use, to withstand a traveling bend at a reduced temperature, such as might be encountered during shipboard construction or repair.

4.8.9.1 Specimen. The specimen shall consist of a 20±1-foot length of completed cable.

4.8.9.2 Special apparatus. The apparatus shall include the following:

a. A refrigeration chamber for supporting the requirements as specified in 4.8.9.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

b. Bending apparatus shall consist of two identical sheaves, as shown on figures 6 and 7. The sheaves are attached to a fixed surface such that they are pivoted about their axles and at all times their axles are both horizontal and parallel, the corresponding points on both sheaves are located in the same vertical plane, their bending surfaces are separated by not more than four times the maximum overall cable diameter specified in the applicable specification sheet, and they support the requirements as specified in 4.8.9.3. Each sheave shall have a bending surface width of not more than three times the maximum overall cable diameter specified in the applicable specification sheet and a bending surface diameter of not more than 13 nor less than 11 times the maximum overall cable diameter specified in the applicable specification sheet.

c. Two pieces of rope or twine, each of which shall have a length of not less than 25 feet and shall support the requirements as specified in 4.8.9.3.

4.8.9.3 Procedure. One end of one piece of rope or twine shall be firmly attached to either end of the specimen and one end of the second piece of rope or twine shall be firmly attached to the opposite specimen end. Attachment may be either with a cable grip or by directly tying the rope or twine to the specimen. The specimen with the rope

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or twine attached shall be placed within the refrigeration chamber and shall be subjected to a temperature of minus 4±4 °F (minus 20±2 °C) for a period of not less than 6 hours. Prior to any change of chamber temperature, the specimen with rope or twine attached shall be removed from the chamber, and one rope or twine shall be wound through the bending apparatus into an S shape as shown on figure 6 not more than 20 seconds after removing from the chamber. The specimen shall be subjected to cold working as follows. The specimen shall be drawn completely through the bending apparatus at a rate of 35±5 feet per minute by means of pulling on the rope or twine that initially passes through the apparatus. Immediately thereafter, the specimen shall be drawn completely through the bending apparatus in the opposite direction, at the same rate, by means of pulling on the opposite end of the rope or twine. At all times during cold working, both ropes, or twine, shall be directed such that the bending surface of each sheave is in contact with either the specimen or the rope or twine for an arc of not less than 170 degrees. Following cold working, the specimen shall be dissected for a distance of not less than 3 feet, and each of the constituent components shall be visually inspected for deterioration (see 4.8.9.4).

4.8.9.4 Observation. Specimen failure shall be construed if the specimen, either during cold working or upon dissection and inspection, exhibits distortion or cracking of any specimen component or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

4.8.10 Drip. This test shall determine whether or not fillers will exude and drip from the cable ends when cables containing these fillers are subjected to high temperatures. Excessive filler dripping may produce deleterious effects, such as the coating or bridging of underlying electrical contacts.

4.8.10.1 Specimen. For all single conductor cables, and all multi-conductor cables that have conductors that are smaller than Navy size 50 (3 AWG), the specimen shall consist of an 18±0.5-inch-length of completed cable. For multi-conductor cables that have conductors that are Navy size 50 (3 AWG) and larger, the manufacturer has the option of either of two acceptable specimens:

a. 18+½-inch-length of completed cable or

b. 24+½-inch-length of completed cable, with 6 inches of cable jacket, shield(s) if present, and binder(s), removed from one end to expose the insulated conductor core. From this 6-inch-exposed end, remove all non-fibrous (solid) filler(s) if present and cable core waterblock filling material, both from all outer cable core valleys and from all interior voids, leaving only 6 inches of completely exposed insulated conductors. Interior void removal of 6 inches of solid filler(s) or waterblock filling material shall be done to the practical extent possible without damaging the insulated conductors.

4.8.10.2 Special apparatus. The apparatus shall include a heating chamber for supporting the requirements as specified in 4.8.10.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

4.8.10.3 Procedure. The specimen shall be straightened and shall be suspended from one end within the heating chamber, such that the specimen hangs freely downward. For 24-inch specimens, the exposed insulated conductors shall be at the bottom end of the specimen. The temperature within the chamber shall then be raised to the value specified in the applicable specification sheet within +1.8 °F (1 °C). This chamber temperature shall be maintained for a continuous period of not less than 18 hours. Provision shall be made to collect any material that may fall away from the lower specimen end during this test. Following the test period, the collected material that fell away from the specimen shall be weighed.

4.8.10.4 Observation. For 18-inch specimens, failure shall be construed if any material falls away from the lower specimen end during the test. In the event of failure of an 18-inch specimen multi-conductor cable that has conductors that are Navy size 50 (3 AWG) and larger, the test may be repeated using a 24-inch specimen (see 4.8.10.1.b). For 24-inch specimens with 6 inches of exposed insulated conductors, failure shall be construed if the weight of collected material that falls away from the lower specimen end during the test exceeds the following maximum amounts:

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Sum of specified cross sections of all specimen conductors (see 3.1) (circular mils)	Collected material that has fallen away (ounce, maximum)
Less than 100,001	0.00
100,001 to 200,000	0.0265
200,001 to 500,000	0.035
500,001 to 800,000	0.0441
800,001 or greater	0.0529

4.8.11 Durometer hardness. This test shall be conducted to determine whether or not jacketing material exhibits sufficient hardness (see 4.8.11.4) for use in cable construction.

4.8.11.1 Specimen. The specimen shall consist of a block of material of the dimensions specified in ASTM D2240, which shall have been cured in effectively the same manner as when used in cable manufacture.

4.8.11.2 Special apparatus. The apparatus shall include a type A or D durometer, as specified in the applicable specification sheet, which shall be in accordance with ASTM D2240.

4.8.11.3 Procedure. The specimen shall be tested in accordance with ASTM D2240.

4.8.11.4 Observation. Specimen failure shall be construed if the specimen exhibits a durometer hardness other than that specified in the applicable specification sheet.

4.8.12 Flame propagation. The method given below shall be used to determine the relative ability of grouped cable to resist the propagation of fire along its vertical length. Two cable type specimens shall be used per each flame propagation test.

4.8.12.1 Specimen. The specimen shall consist of a length of completed cable sufficient to comply with the tray fill requirement of the specified flame propagation test procedure.

4.8.12.2 Procedure. The test shall be performed according to the UL 1685 Flame Exposure Test, without the smoke requirements.

4.8.12.3 Observation. Specimen failure shall be construed if the cable char height exceeds the requirement of the UL 1685 flame test procedure without the smoke requirements. Both cable samples shall pass the test criteria in order to achieve a successful overall test.

4.8.13 Gas flame (circuit integrity). The method given below shall be used to determine the ability of completed cable type specimen to maintain electrical circuit integrity in the presence of flame. The 3-hour test shall be performed for qualification and qualification retention testing and the 1-hour test shall be performed for conformance testing.

4.8.13.1 Specimen. One cable type specimen consisting of a 48±6-inch-length of the completed cable type shall be tested. The jacket and blocking materials shall be removed for a distance of 3 inches on each end of the cable type sample. Sufficient insulation shall be removed from one end of each conductor for making electrical connections as follows:

a. Specimens with one conductor shall be tested with single-phase voltage. The specimen shall be covered with an aluminum or copper braid shield providing no less than 50 percent coverage. The braid covering shall be attached to ground.

b. Specimens with two conductors shall be tested using two phases of the three-phase source.

c. Specimens with three conductors shall be tested with each conductor connected to a separate phase.

d. Specimens with four conductors shall have physically opposite conductors connected together in pairs to form two effective electrical circuits.

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e. Specimens with six conductors shall have physically opposite conductors connected together in pairs to form three effective electrical circuits.

f. Specimens with seven or more conductors shall have conductors connected together to form three effective electrical conductors. Connections shall be made such that physically adjacent conductors shall be members of different effective electrical conductors to the maximum practicable extent and such that the number of conductors constituting each effective electrical conductor does not differ by more than one.

4.8.13.2 Special apparatus. The apparatus shall include the following:

a. A metered and adjustable source of natural gas or other flammable gas and an adjustable source of air able to support the requirements as specified on figure 10. The gas and air support shall be made available as a mixture through a single conduit.

b. A ribbon burner, which accepts gas and air from the gas and air supply via a single, 0.5-inch nominal pipe set vertically into its underside, thoroughly blends the gas and air within an internal mixing chamber, and allows the blended gas and air mixture to exit for burning via a multiplicity of holes in its top, shall be fabricated in accordance with figures 8 and 9. The mixing chamber, fabricated from number 16 United States Standard Gage (USSG) steel, shall be horizontally centered within the burner and shall be airtight except for the specified holes on its two sides and through the inside of the ½-inch nominal pipe. The burner exterior shall be fabricated from number 14 USSG steel and shall be airtight except for the specified holes on its top and the specified holes in the mixing chamber.

c. A three-phase, 60 Hz voltage source, as shown on figure 10, shall provide a separate, single-phase test voltage through a normal blow fuse or overload relay to each specimen conductor. This voltage source shall be in accordance with the following:

(1) Specimens rated for 1 kV which incorporate conductors with cross-sections of less than 14,000 circular mils and all specimens rated for less than 1 kV: the test voltage shall be 120±5 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 A.

(2) Specimens rated for 1 kV which incorporate conductors with cross-sections of 14,000 circular mils or more: the test voltage shall be 450±10 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 A.

(3) Specimens rated for 2 kV: the test voltage shall be 2,000±50 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of ½ A.

(4) Specimens rated for 5 kV: the test voltage shall be 4,160±100 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of ¼ A.

The secondary winding voltage rating of any constituent voltage step up transformer shall be not less than the required test voltage. Each phase of the voltage source shall have the capacity to supply the specimen with a current which shall be not less than twice the fuse and overload relay rating.

d. A chamber enclosing the horizontally mounted burner and the suspended specimen (see 4.8.13.3): this enclosure shall incorporate an exhaust fan at its top and shall incorporate apertures for admitting air along its sides adjacent to its base. The exhaust fan shall be operated to produce no more suction than is necessary to carry off smoke and gases. The enclosure shall be designed to both minimize horizontal air drafts to the greatest practicable extent and to allow free circulation of convection air currents generated during the gas flame (circuit integrity) test.

4.8.13.3 Procedure.

a. The burner shall be mounted within the chamber, such that the holes in its top are in common horizontal plane. The specimen shall be suspended within the enclosure, such that it is parallel to the burner, horizontally centered directly above the burner and spaced 1⅞±⅛ inches vertically above the top of the burner surface. Specimen suspension shall be accomplished in a manner (such as by wire loops) that shall not significantly shield the specimen from flame and that shall prevent any portion of the specimen from dropping closer than ¾ inch to the top surface of the burner during specimen burning. The gas air supply shall then be attached to the burner, and the gas shall be adjusted to provide a flow rate of not less than that calculated by the following formula:

Gas flow rate, in cubic feet per hour = 1,387 (T+273)/(PN)

Where: T = Measured ambient temperature in the vicinity of the burner, in ° C. Measurement accuracy shall be within ±0.5 °C.

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P = Local atmospheric pressure, in lb/in^2 absolute. Measurement accuracy shall be within ±2 percent.

N = Gross heating value, in BTU/ft^3, of the gas at an atmospheric pressure of 14.69 lb/in^2 absolute and a temperature of 73 °F (23 °C); this value shall be accurate within ±2 percent.

b. The gas exiting the burner shall then be ignited and the air flow to the burner adjusted to produce the smallest blue flame possible. The short circuit monitor test voltages shall then be applied to the specimen, as shown on figure 10. If the specimen is armored, an additional test voltage shall be applied by grounding the armor as shown on figure 10. (This will effectively apply an additional voltage that is approximately equal to the test voltage divided by the square root of three between the armor and the specimen conductors.) The flame and test voltages shall remain continuously applied to the specimen for a period of not less than that specified in the applicable specification sheet, unless specimen failure (see 4.8.13.4) occurs prior to the end of this period, in which case the test may be terminated.

4.8.13.4 Observation. Specimen failure shall be construed if any fuse blows or if any overload relay trips prior to completion of the test period specified in the applicable specification sheet.

4.8.14 Heat distortion. This test shall determine the ability of insulation and jacket material to resist physical deformation when subjected to an elevated temperature.

4.8.14.1 Specimen. Each specimen shall be removed from completed cable in accordance with a and b(1) or b(2), as noted below, whichever is indicated. The number of specimens that shall be required shall be in accordance with b(3), b(4), or b(5), as noted below.

a. Cable jacket. Each specimen shall consist of a piece of jacketing that has been removed from a finished cable and cut to form a rectangular strip. This strip shall be not less than ⅞ inch long nor less than 9/16±1/16 inch wide and shall be ground and buffed to a thickness of 0.050±0.010 inch with a grinding apparatus (see 4.8.14.2.c). The thickness of the specimen shall be made as uniform as practicable.

b. Insulation.

(1) For cables that incorporate conductors with cross-sections of more than 190,000 circular mils, each specimen shall consist of a piece of insulation that has been removed from a conductor and that has been cut to form a rectangular strip. This strip shall be not less than ⅞ inch long nor less than 9/16±1/16 inch wide and shall be ground and buffed to a thickness of 0.050±0.010 inch with a grinding apparatus (see 4.8.14.2.c). The thickness of the specimen shall be made as uniform as practicable.

(2) For cables that incorporate conductors with cross-sections of 190,000 circular mils or less, each specimen shall consist of a single insulated conductor of a length not less than ⅞ inch. NOTE: In order to prevent misleading results, the conductor should be removed and replaced with a solid copper rod of similar diameter.

(3) For cables containing four or fewer conductors, one specimen shall be prepared from each insulated conductor.

(4) For cables containing more than four but fewer than 21 conductors, one specimen shall be prepared from each of four different insulated conductors (a total of four specimens).

(5) For cables containing 21 or more conductors, the number of specimens prepared shall be equal to the square root (rounded if necessary, to the nearest whole number) of the total number of conductors. Each specimen shall be prepared from a different insulated conductor.

4.8.14.2 Special apparatus. The apparatus shall include the following:

a. A self-standing thickness gauge, for supporting the requirements as specified in 4.8.14.3, which shall exhibit a measurement error of not more than ±0.001 inch. This gauge shall incorporate a flat horizontal plate upon which the entire specimen shall be laid flat during measurement and shall incorporate a foot that rests upon the top of the specimen during measurement. This foot shall be attached to a plunger that moves freely in the vertical direction and that is attached to a dial or other indicating device from which the specimen thickness may be read. The foot shall present a horizontal disk of 3/8±1/64 inch diameter to the specimen, which shall bear down on the specimen with a force of 85±4 grams and shall be loaded with weights (see 4.8.14.3) to present additional force to the specimen.

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b. A micrometer (required for insulated conductor specimens only) with flat, parallel measurement surfaces on both spindle and anvil, for the specimen thickness measurement as specified in 4.18.14.3.b(1), which shall exhibit a measurement error of not more than ±0.001 inch.

c. A motor driven grinding wheel and a motor driven buffing wheel, or their equal, for the specimen preparation as specified in 4.8.14.1.a and b(1). Guides shall be provided to assure that the specimen is pulled tangentially to the surface of each wheel during grinding and buffing.

d. An oven for accommodating the gauge with the specimen inserted and for supporting the requirements as specified in 4.8.15.3. The oven air temperature shall be measured in the immediate vicinity of the specimen.

4.8.14.3 Procedure. The initial insulation thickness of each specimen (T in the formula below) shall be determined (see a(1) and b(1) below), and the oven shall be preheated to 250±2 °F (121±1 °C). This oven temperature shall be maintained for the remainder of the test. The gauge foot shall then be loaded with weights (see a(2) and b(2) below) and the gauge placed within the oven. Not less than 1 hour later, one specimen shall be selected and placed within the oven. Not less than 1 hour after placing the specimen within the oven, the specimen shall be placed beneath the gauge foot such that the gauge indicates the specimen thickness. Not less than 1 hour after placing the specimen beneath the gauge foot, the final specimen thickness or diameter (as appropriate) shall be read from the gauge and the final insulation thickness (t in the formula below) determined (see a(3) and b(3) below). In an identical manner, the remaining specimens shall also be tested. The percentage heat distortion of the insulation shall then be calculated by using the following formula:

Percentage heat distortion = Average value of 100 (T-t)/T for all tested specimens.

Where: T = Initial insulation thickness of the specimen, in thousandths of an inch (see a and b below).

t = Final insulation thickness of the specimen, in thousandths of an inch (see a and b below).

The following additional details apply:

a. Rectangular strip specimens:

(1) The initial insulation thickness (T) shall be the initial specimen thickness, as measured with the gauge, prior to loading the gauge with weights and prior to inserting either the gauge or the specimen into the oven.

(2) The gauge foot shall be loaded with 4.41±0.22 pounds (2,000±100 grams).

(3) The final insulation thickness (t) shall be the final specimen thickness, measured as specified.

b. Insulated conductor specimen:

(1) T shall be defined as one half of the difference between the measured overall specimen diameter and the measured specimen conductor diameter. Measurements shall be made using the micrometer. The overall specimen diameter shall be measured in the same radial direction as will be measured when t (see b(3) below) is determined.

(2) The gauge foot shall be loaded in accordance with the below table:

Conductor cross-section of specimens (circular mils)	**Weight**
2,700-3,999	0.88±0.044 pound (400±20 grams)
4,000-21,999	1.10±0.55 pounds (500±25 grams)
22,000-94,999	1.65±0.77 pounds (750±35 grams)
95,000-189,999	2.20±0.11 pounds (1,000±50 grams)

(3) t shall be defined as one half of the difference between the final specimen diameter, measured as specified, and the previously measured specified conductor diameter.

4.8.14.4 Observation. Specimen failure shall be construed if the percentage heat distortion is greater than that specified in the applicable specification sheet.

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4.8.15 Hydrostatic (open end). This test shall determine the ability of completed cable, which is intended to pass through watertight bulkheads, to prevent the longitudinal flow of high-pressure water when properly installed in a bulkhead penetrator.

4.8.15.1 Specimen. The specimen shall consist of a 60±2-inch-length of completed cable. Armored cable shall have the armor removed to expose the cable outer jacket. The specimen shall be cut to length using a scissors-action cable cutter. Saws shall not be used.

4.8.15.2 Special apparatus. The apparatus shall include a tank that can be filled with water, internally pressurized to the value specified in the applicable specification sheet and used for applying this water pressure to one specimen end, as specified in 4.8.15.3. An appropriate stuffing tube shall be welded into the tank wall to provide a means for sealing the one specimen end within the tank. An appropriate packing assembly shall be available to effect this sealing (see 4.8.15.3). A means shall be provided whereby the pressure within the tank can be measured within an accuracy of ±5 percent.

4.8.15.3 Procedure. One end of the specimen shall be installed through the tank stuffing tube and shall be sealed in place by using a cable packing assembly. The opposite specimen end shall remain freely exposed to air for the remainder of the test. The tank shall then be closed and filled with water such that the specimen end within the tank remains directly and entirely exposed to water for the remainder of the test and positioned such that no portion of the specimen is at a greater elevation than where the specimen enters the tank through the stuffing tube. The water shall then be pressurized to a value of not less than that specified in the applicable specification sheet. This pressure shall be maintained for a continuous period of not less than that specified in the applicable specification sheet, unless specimen failure (see 4.8.15.4.a) occurs prior to the end of this period, in which case the test may be terminated prematurely. During the test period, water leaking from the specimen end open to free air shall be collected into an initially dry container. Following the test period, the volume of this collected water shall be measured, the specimen shall be removed from the tank, and the end that had been exposed to water pressure shall be examined.

4.8.15.4 Observation. Either of the following shall constitute specimen failure:

a. Any water leakage from the free end of the specimen in excess of that specified in the applicable specification sheet at any time during or at the conclusion of the test.

b. Slippage of more than ¼ inch, with respect to the specimen jacket, of any specimen component at the end of the specimen that had been exposed to water pressure.

4.8.16 Permanence of printing (conductor insulation). This test shall determine the ability of printed information on insulated conductors to remain legible in the presence of repeated abrasion.

4.8.16.1 Specimen. Each specimen shall consist of a single insulated conductor that shall have sufficient length for use in the test as specified in 4.8.16.3. Specimens shall be removed from the completed cable as follows:

a. Cables containing four or fewer conductors: one specimen shall be removed from each insulated conductor.

b. Cables containing more than 4 but fewer than 21 conductors: 1 specimen shall be removed from each of four different insulated conductors (a total of four specimens).

c. Cables containing 21 or more conductors: the number of specimens removed shall be equal to the square root (rounded, if necessary, to the nearest whole number) or the total number of conductors. Each specimen shall be removed from a different insulated conductor.

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4.8.16.2 Special apparatus. The apparatus shall include an abrading machine, which shall secure a specimen horizontally between two fixed supports and abrade the specimen printed identification by means of a motor driven, reciprocating tape consisting of number 50-2/20 unbleached cotton braid. One possible arrangement for the abrading machine is shown on figure 11. The braided cotton tape shall lie flat on the specimen surface, shall remain perpendicular to the specimen axis, and shall abrade an unchanging area of the specimen periphery, over an arc of not less than 135 degrees, at all times during the test. One end of the tape shall be reciprocated at a rate of 28±3 cycles per minute, such that the tape is drawn across the specimen for a distance of 10±1 inches in each direction (20 inches minimum total excursion during cycle). The opposite end of the tape shall be maintained under tension by a freely suspended ½±1/64-pound weight. Each side of the tape shall be used to perform not more than 500 cycles; the tape shall then be discarded and replaced. The abrading machine shall incorporate an automatic counter to total the number of abrasion cycles to which the specimen is subjected during the test.

4.8.16.3 Procedure. Each specimen shall be wiped with a clean, dry cloth to remove any lubricant or dirt. One specimen shall then be selected and mounted between the two fixed supports of the abrading machine. Mounting shall be accomplished such that the specimen shall be as taut as practicable and such that the braided cotton tape shall repeatedly abrade the same portion of the specimen printed identification at all times during the test. The automatic counter shall be set initially to zero. The abrading machine motor shall then be turned on. The braided cotton tape shall be allowed to reciprocate and abrade the specimen printed identification either until the printed identification is no longer legible in its abraded region or until not fewer than the number of abrasive cycles specified in the applicable specification sheet have been completed, whichever occurs first. The number of abrasive cycles completed shall then be noted, and the remaining specimens then tested in an identical manner.

4.8.16.4 Observation. Specimen failure shall be construed if the average value of the number of abrasive cycles completed on each specimen is less than that specified in the applicable specification sheet.

4.8.17 Permanence of printing (jacket). This test shall determine the ability of printed information of jacketing material to remain legible in the presence of repeated abrasion.

4.8.17.1 Specimen. The specimen shall consist of a piece of completed cable that shall have sufficient length for use in the test as specified in 4.8.17.3.

4.8.17.2 Special apparatus. The apparatus shall include an abrading machine that shall support and secure the specimen such that it is straight and horizontal and that shall abrade the specimen printed identification by means of a motor-driven, transversely reciprocating steel pin. The steel pin shall have a diameter of 0.025±0.001 inch where it abrades the specimen and shall have a surface roughness of not less than 2 microinches, as specified in ASME B46.1. (A selected sewing needle may satisfy these requirements.) One possible arrangement for the abrading machine is shown on figure 12. The steel pin shall be horizontal and perpendicular to the specimen axis, shall ride along the top of the specimen, and shall be weighted to bear down on the specimen with a force of 1±1/16 pound at all times during the test. The pin shall be reciprocated at a rate of 60±2 cycles per minute such that the pin is drawn along the specimen for a distance of ⅜±1/16 inches in each direction (¾±⅛-inch minimum total excursion) during each cycle. The abrading machine shall incorporate an automatic counter to total the number of times that the specimen is abraded by the steel pin during the test.

4.8.17.3 Procedure. The specimen shall be wiped with a clean, dry cloth to remove any lubricant or dirt and shall be secured in the abrading machine with the specimen printed identification facing upwards, where it is to be abraded by the steel pin. The automatic counter shall be set initially to zero. The abrading machine motor shall be then turned on, allowing the steel pin to reciprocate and abrade the specimen printed identification either until printed identification is no longer legible in the abraded region, or until not fewer than 125 abrasive cycles have been completed, whichever occurs first. The number of abrasive cycles completed shall then be noted. This test shall next be repeated four more times (five times total), subjecting a fresh portion of the specimen printed identification to abrasion each time.

4.8.17.4 Observation. Specimen failure shall be construed if the average value of the number of abrasive cycles completed during each of the five tests is less than 125.

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4.8.18 Physical tests (aged) on insulation and jacket. This test shall determine by means of tension measurements whether various insulating and jacketing materials have been properly processed. This test makes provision for making tension measurements both before and after these materials have been artificially aged.

4.8.18.1 Specimen. Each specimen shall consist of a single piece of insulation or jacketing, which shall have sufficient length for use in the test as specified in 4.8.29. Specimens shall be removed from completed cable.

4.8.18.2 Special apparatus. The apparatus shall consist of a forced fresh air circulating oven for supporting the requirements as specified in 4.8.18.4.a. The oven air temperature shall be measured in the immediate vicinity of the specimens.

4.8.18.3 Procedure. Specimens shall be maintained at ambient temperature for a period of not less than 30 minutes immediately prior to accelerated aging. Accelerated aging procedures shall be performed as follows, using one specimen. Each specimen shall be freely suspended vertically, secured by one end, within the oven. The air temperature within the oven shall then be raised to the value specified in table XII and shall be maintained at this value for a continuous period of not less than that specified in table XII. Each specimen shall then be removed from the oven and tested as specified in 4.8.29.

TABLE XII. Details for accelerated aging procedures.

Specimen material	Procedure	Temperature (°F)	Minimum period (hours)
Insulation specimens			
Thermoset	Air oven	329±3.6	4
Polyolefin	Air oven	212±4	48
Jacketing specimens			
Silicon rubber	Air oven	500±4	24
Cross linked-polyolefin	Air oven	329±3.6	4

4.8.18.4 Observation. Any of the following shall constitute specimen failure:

a. A specimen tensile strength that is less than that specified in the applicable specification sheet.

b. A specimen elongation that is less than that specified in the applicable specification sheet.

4.8.19 Tear strength. This test shall determine the ability of elastomeric jacketing material to withstand the propagation of a cut passing through a portion of its length.

4.8.19.1 Specimen. This test shall consist of a piece of jacketing material, which shall have been cut to the dimensions specified in ASTM D470. Specimens shall be removed from the completed cable. A minimum of six specimens shall be required.

4.8.19.2 Special apparatus. The apparatus shall include the following:

a. A micrometer with flat, parallel measurement surfaces on both spindle and anvil that shall be used for the specimen measurements as specified in 4.8.19.3 and exhibit a measurement error of not more than ±0.001 inch.

b. A motor driven tensile machine that shall be for applying increasing tension to the specimen and that shall automatically indicate within ±1 percent the maximum tension experienced by the specimen prior to tearing. This tensile machine shall incorporate two parallel and opposing jaws, between which the specimen can be secured, as specified in 4.8.19.3. The two jaws shall increase their separation at the uniform rate of 20±2 inches per minute by means of the tensile machine motor.

4.8.19.3 Procedure. The specimen shall be tested in accordance with ASTM D470.

4.8.19.4 Observation. Specimen failure shall be construed if the average value of the measured tear strengths of all specimens is less than that specified in the applicable specification sheet.

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4.8.20 Twisting endurance. This test shall determine the ability of completed cable, which is intended to be flexed during use, to withstand repeated and reversing torsional motion while subjected to a specified temperature.

4.8.20.1 Specimen. The specimen shall consist of a piece of completed cable that shall have a length of not less than the sum of 18 inches plus 10 times the maximum overall cable diameter specified in the applicable specification sheet, except that no specimen shall have a length of less than 24 inches. Specimen conductor ends shall be electrically interconnected such that a single series electrical circuit is formed by the conductors. The two end-conductors of this series circuit shall be connected as specified in 4.8.20.3.

4.8.20.2 Special apparatus. The apparatus shall include the following:

a. A twisting machine for suspending the specimen vertically and for automatically twisting the upper specimen end alternately clockwise and counterclockwise by 180±10 degrees with respect to the lower specimen end as described below:

(1) The specimen shall be suspended and repeatedly twisted from one extremity by means of a clamp, which shall be fastened to a motor-driven, rotating and reversing mechanism (such as a pinion connected to a rack, which in turn is made to reciprocate by an arm from a motor). The upper specimen end shall be twisted about its axis at a rate of not more than 14, nor less than 12, cycles per minute. Each cycle shall produce 180±10 degrees of clockwise axial specimen rotation plus 180±10 degrees of counterclockwise axial specimen rotation (360±20 degrees total rotation). The upper end of the specimen jacket shall be a distance of not less than 1 inch above the nearest edge of the clamp.

(2) A weight shall be provided for applying tension to the specimen. This weight shall be attached to the lower specimen extremity by means of a second clamp, which shall be constrained from rotating but that shall be free to move in the vertical direction at all times during the test. The weight shall be chosen such that the:

Tension from the second clamp plus the weight in pounds = $10(3.14159)d^2 \pm (3.14159)d^2/2$

Where: d = Maximum overall cable diameter, in inches as specified in the applicable specification sheet

This second clamp shall be separated from the rotating clamp by a distance that shall not exceed the sum of 6 inches plus 10 times the maximum overall cable diameter specified in the applicable specification sheet. The second clamp and weight shall be constructed such that tension is applied vertically downward along the specimen axis. The lower end of the specimen jacket shall be a distance of not less than 1 inch below the nearest edge of a clamp.

(3) Both the rotating clamp and the tensioning weight clamp shall apply radial compression to the specimen such that neither the specimen nor any of its internal components will slip in the clamping area.

(4) An automatic counter shall be provided to total the number of twisting cycles performed on the specimen during a test.

(5) The specimen shall not come into contact with any piece of machinery, except for the rotating clamp, the tensioning weight clamp, and the circuit continuity monitor (4.8.20.2.b) at any time during the test.

b. Circuit continuity monitor (a means for continuously monitoring the electrical continuity of the series connected specimen conductors) shall automatically stop the motion of the rotary clamp by removing motor power when a specimen conductor breaks.

c. A test chamber for maintaining the air temperature specified in the applicable specification sheet within ±3.6 °F (2 °C), for accommodating the twisting machine with specimen attached and for supporting the requirements specified in 4.8.20.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen, approximately midway between the two specimen clamps of the twisting machine. The test chamber shall have a viewing port for observing the specimen under test.

d. The voltage withstand test apparatus shall be as specified in 4.9.8.2.

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4.8.20.3 Procedure. The twisting machine shall be installed in the test chamber, and the specimen shall be installed in the twisting machine. The circuit continuity monitor shall be connected between the two end conductors of the single series circuit within the specimen. The test chamber doors shall then be closed, and the air temperature within the chamber shall be brought to the value specified in the applicable specification sheet within ±3.6 °F (2 °C). After the air temperature within the chamber has been maintained at this value for a period of not less than 1 hour, the automatic counter shall be set initially to zero and the twisting endurance test begun by turning on the motor-driven, rotating clamp. The number of twisting cycles specified in the applicable specification sheet shall be performed, unless specimen failure (see 4.8.20.4) occurs prior to the completion of this number of cycles. The test may then be terminated prematurely. At all times during the test, the temperature within the chamber shall be maintained as specified in the applicable specification sheet within ±3.6 °F (2 °C), and the test chamber doors shall remain closed. Following the test, the specimen shall be removed from the twisting machine and the test chamber.

4.8.20.4 Observation. Either of the following shall constitute specimen failure, except that specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest using a different specimen from the same length of cable:

a. Rotary clamp stoppage caused by the circuit continuity monitor (indication specimen conductor breakage), prior to completion of the number twisting cycles specified in the applicable specification sheet.

b. Rupture of the specimen jacket prior to completion of the number of bending cycles specified in the applicable specification sheet. Specimens exhibiting no failure shall be tested further, as specified in 4.8.20.5.

4.8.20.5 Further procedure (may not be required, see 4.8.20.4). The specimen shall be allowed to return to room temperature and shall be subjected to the voltage withstand test as specified in 4.8.8, using the voltages specified in the applicable specification sheet. Following this, the portion of the specimen that had been between the specimen clamps of the twisting machine shall be dissected, and each of its constituent components visually inspected for deterioration (see 4.8.20.6). Fillers shall not be considered components for this test.

4.8.20.6 Further observation. Either of the following shall constitute specimen failure, except that specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure but shall require a complete retest of twisting endurance using a different specimen from the same length of cable.

a. Specimen fails the voltage withstand test (see 4.8.8).

b. Specimen, upon dissection and inspection, exhibits visible distortion or cracking of any specimen component, including strand breakage on any conductor, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service. Fillers shall not be considered components for this test.

4.8.21 Watertightness. This test shall determine the ability of completed cable, which is intended to pass through watertight bulkheads, to prevent the longitudinal flow of low-pressure water. Cables manufactured with the types of watertight fillers specified in 3.3.9.1 shall be tested for watertightness in accordance with 4.8.21 with seawater in accordance with ASTM D1141 and shall meet the requirements for allowable water-leakage in accordance with table XIII.

4.8.21.1 Specimen. The specimen shall consist of a 60±2-inch-length of completed cable. Armored cable shall have the armor removed to expose the outer cable jacket. The specimen shall be cut to length using a scissors-action cable cutter. Saws shall not be used.

4.8.21.2 Special apparatus. The apparatus shall include the following:

a. A source of pressurized water, which shall be provided at a regulated pressure of 27±1 lbs/in^2 and which shall be for use with the terminal fitting (see 4.8.21.2.b) as specified in 4.8.21.3. If tapes or yarns are used within the cable specimen, then ocean water (mixed in accordance with ASTM D1141) shall be used. Otherwise, tap water shall be used.

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b. A metal terminal fitting that applies the source of pressurized water to one end of the specimen, which supports the requirements as specified in 4.8.21.3 and which shall be fabricated as specified in the following. Figure 13 shows one possible arrangement for the terminal fitting. The fitting shall admit the specimen end for the distance specified in 4.8.21.3, and shall have an ID, where it fits over the specimen, of not greater than the measured overall specimen diameter plus 0.5 inch. The fitting shall have a means for introducing the source of pressurized water to the specimen end and a means for bleeding off any air that might be trapped between the specimen end and the source of pressurized water. The fitting shall also have an aperture for introducing a hardening sealant (see 4.8.21.2.c below), to produce a pressure-tight bond between the fitting and the specimen jacket. A plug (such as a thick wrapping of rubber tape around the specimen) shall be provided at the place where the specimen enters the fitting to prevent sealant loss and to approximately center the specimen within the fitting.

c. A liquid hardening sealant that shall be used to produce a pressure-tight bond between the fitting and the specimen jacket when applied as specified (see 4.8.21.3) and allowed to harden. The sealant shall be at a temperature of not greater than 212 °F (100 °C) when poured into the terminal fitting and shall not expand in volume while hardening. An appropriate metal alloy is recommended as a sealant since it can be remelted and reused. Any metal alloy used shall have a melting point of not greater than 190 °F (88 °C).

4.8.21.3 Procedure. The specimen shall be secured in a vertical position, the terminal fitting (including the plug) shall be fitted over its upper end, and the hardening sealant shall be poured through the appropriate terminal fitting aperture to bond the fitting to the specimen. The specimen shall be centered within the terminal fitting, such that not more than a 6-inch-length of specimen jacket is exposed to the sealant, the end face of the specimen is located not more than 1 inch above the sealant surface and contains no sealant, and the aperture through which the sealant was introduced is made pressure-tight. The sealant shall be allowed to harden, following which the terminal fitting plug shall be removed or cut away so that it exerts no radial compression on the specimen. The specimen shall then be positioned such that it is straight, and no portion of the specimen is at a greater elevation than the specimen end within the terminal fitting. The fitting shall then be connected to the source of pressurized water, and excess air within the fitting shall be bled off until it can be certain that the entire enclosed specimen end face is exposed to the pressurized water for the remainder of the test. The source of pressurized water shall be slowly raised to 27.5±0.5 lb/in^2 over a period of 3±0.5 minutes and remain connected to the fitting for a continuous period of 6±0.25 hours, unless specimen failure (see 4.8.21.4) occurs prior to the end of this period, in which case the test may be terminated prematurely. During this test period, water (if any) leaking from the specimen end opposite the terminal fitting shall be collected into an initially dry container. Following the test period, the volume of this collected water shall be measured.

4.8.21.4 Observation. Specimen failure shall be construed if the collected water volume exceeds that shown in table XIII or exceeds that specified in the applicable specification sheet.

TABLE XIII. Limits for water leakage.

Sum of specified cross-sections of all specimen conductors (see 3.1) (circular mils)	**Collected water volume (cubic inches, max.)**
Less than 9,001	4
9,001 to 15,000	5
15,001 to 25,000	6
25,001 to 50,000	8
50,001 to 100,000	9
100,001 to 200,000	10
200,001 to 500,000	11
500,001 to 800,000	12
800,001 or greater	13

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4.8.22 Acid gas generation. The method given below shall be used to determine the total emission of any strong soluble acids (pH less than 3). When present in the cable sample, jacket, filler, and insulation (extruded or taped), material shall be individually tested and the acid gas generation shall be determined following the analyses of the chemical composition of all ingredients that are present.

4.8.22.1 Specimen. The specimen shall consist of a weighed sample, normally 0.25 to 0.50 gram, of each of the jacket, fillers, and insulation materials, when present in the finished cable type. For extruded insulation, the required mass of material removed from a 16 AWG or smaller insulated wire shall be used. Any overlaying braid, covering, or jacket shall be removed from the insulated wire prior to obtaining the required extruded material sample.

4.8.22.2 Procedure. The required apparatus is shown on figure 14. The specimen sample shall be placed in a silica boat that is put into the center of a silica tube with a length of 19.69 to 23.62 inches (50 to 60 cm) and internal diameter of 0.79 to 0.87 inch (20 to 22 mm). Each material shall be tested individually. The silica tube shall be placed in the tube furnace. An air supply, derived from a blower or compressed air cylinder, at the rate of 1 liter per minute ±5 percent, shall be passed through the silica tube and then through four absorber flasks each containing 1.18 inches (30 mm) of deionized water. The furnace heating shall be commenced, and the temperature of the tube and sample shall be raised to 1,472±18 °F (800±10 °C) over a period of approximately 40 minutes and then held at temperature for a further 20 minutes. During the heating period, any acid gases produced will be carried over into the absorber flasks. On completion of the heating cycle, the acid content of the fluids in the absorber flasks shall be titrated against 0.1 N sodium hydroxide solution using the color red as an indicator. The total titrate indicates the total soluble acids. 0.0338 fluid ounces (1.0 mL) of 0.1 N sodium hydroxide solution is equivalent to 1.287E-4 ounces (3.65 mg) of acid expressed as “acid equivalent relative to hydrochloric acid.”

4.8.22.3 Observation. Specimen failure shall be construed if the acid equivalent percentage, by weight of the sample, is greater than that specified in the applicable specification sheet.

4.8.23 Halogen content. The methods given below shall be used to determine the halogen content of the sample cable type. When present in the cable sample, jacket, fillers, and insulation (extruded or taped), material shall be individually tested and the halogen content shall be determined by X-ray fluorescence or analytically after performing an inspection and analyses of the chemical composition of all ingredients used.

4.8.23.1 X-ray fluorescence. An X-ray fluorescence instrument shall be used to determine the amount of all four halogens in the jacket, fillers, and insulation materials individually. The sample shall not be combusted prior to making the measurement. The X-ray fluorescence instrument used shall have sufficient measurement detection limits for each of the four halogens.

4.8.23.2 Ion chromatograph. Approximately 0.0088 to 0.017 ounce (0.25 to 0.5 gram) of jacket, filler, and insulation material shall be combusted individually using the same procedure in the acid gas generation test (see 4.8.22). Upon completion of the heating cycle, the amount of all four halogens in the absorber solution shall be determined in accordance with ASTM D4327.

4.8.23.3 Ion selective electrode. Approximately 0.0088 to 0.017 ounce (0.25 to 0.5 gram) of jacket, filler, and insulation material shall be combusted using the same procedure used in the acid gas generation test (see 4.8.22). Upon completion of the heating cycle, the amount of all four halogens shall be determined using the following methods. Fluoride content shall be determined in accordance with ASTM D1179 or ASTM D3868. Chloride content shall be determined in accordance with ASTM D512. Bromide content shall be determined in accordance with ASTM D1246. Iodide content shall be determined in accordance with method C of ASTM D3869.

4.8.23.4 Observation. Specimen failure shall be construed if the halogen content is greater than that specified in the applicable specification sheet.

4.8.24 Smoke index. The smoke index of the cable sample, using the methods described in 4.8.24.1, 4.8.24.2, and 4.8.24.3 below, shall be determined as a calculated measure of the relative quantity of smoke generated that may affect visibility in and around the vicinity of a fire. When present in the cable sample, jacket, filler, and insulation (extruded or taped), material shall be individually tested. Three specimens of jacket, filler, and insulation material shall each be tested and an average value for each material shall be determined.

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4.8.24.1 Specimen.

a. Jacket. Each specimen shall consist of a sufficient number of 3-inch (75-mm) long strips cut from the cable jacket to completely cover the face area of the sample holder. To prevent excessive buckling and distortion of the specimen during test, a wire mesh, manufactured from either a stainless steel wire 0.04 inch (1.0 mm) in diameter with a spacing of 0.5 inch (13 mm) and a square mesh configuration or a 40-mil stainless steel wire 2-mesh screen (two openings per inch), shall be placed inside and across the face of the sample holder. The specimen shall be fabricated by placing the specimen holder (with wire mesh) test face down onto a flat surface and positioning each 3-inch (76 mm) length in the holder in a parallel arrangement so that when the holder is in the test position the strips will be vertical. An insulating block 0.4 inch (10 mm) thick completely wrapped in aluminum foil shall be placed on top of the strips followed by a tension spring and locking pin.

b. Insulation.

(1) Extruded. The test specimen shall be a 3.28-foot (1-meter) length of any finished AWG wire.

(2) Taped. The test specimen shall be tape of sufficient length to provide a minimum of 9 square inches of material. In the event tapes of different thicknesses of the same material are used in the construction of tape insulation walls, the thickest tape is to be used for the test specimen.

c. Fillers and sealants.

(1) Non-fibrous. The test specimen shall be 3 by 3 inches of a 0.070±0.010-inch-thick block of the same material used in the cable.

(2) Fibrous. The test specimen shall consist of as many 2.95-inch (75-mm) lengths of 0.070±0.010-inch diameter strands necessary to fill the 3- by 3-inch test specimen holder one layer deep and as closely spaced as possible. The test specimens shall be prepared as described in 4.8.24.1.a.

4.8.24.2 Procedure. The smoke index shall be measured as specified in DEF STAN 07-247 (with the DS 02-711 test method), using the specimens as described in 4.8.24.1 and with the following procedure modifications:

a. Only the use of propane gas shall be allowed.

b. The radiometer shall be used to calibrate the heat flux at the initial start of each test period.

c. The test chamber shall be calibrated using a National Institute of Standards and Technology or equivalent Cellulose Standard Reference Material at the initial start of each test period.

d. The chamber shall be inspected to assure air tightness. Seals and gaskets shall be replaced as needed.

e. The test shall be run using a minimum of three specimens. If test values are below the limits specified in the applicable specification sheet, the test is complete. If one test value deviates from the average value by more than 20 percent, two additional specimens shall be tested. The high and low test values shall be discarded, and the remaining three averaged.

4.8.24.3 Observation. Specimen failure shall be construed if any smoke index is greater than that specified in the applicable specification sheet.

4.8.25 Toxicity index. The toxicity index of the cable sample, using the methods described in 4.8.25.1, 4.8.25.2, and 4.8.25.3 below, shall be derived by chemical analysis and calculation, as a calculated measure of the health risk posed to personnel in and around the vicinity of a fire. When present in the cable sample, jacket, filler, and insulation (extruded or taped), material shall be individually tested. A minimum of two specimens of the sample cable type jacket, filler, and insulation material, when present in the cable sample, shall be tested and an average value for each material shall be determined.

4.8.25.1 Specimen. For extruded insulation materials, the sample shall be the mass of insulation on a 3.28-foot (1-meter) length of any finished AWG wire. For tape used as conductor insulation, the sample shall be 0.105 ounces (3 grams) of the tape material. For jacket and filler materials, the sample shall be 3.527 ounces (100 grams) of material.

4.8.25.2 Procedure. The toxicity index shall be determined as specified in DEF STAN 07-247 (with the DS 02-713 test method), using the specimens as described in 4.8.25.1.

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4.8.25.3 Observation. Specimen failure shall be construed if any toxicity index is greater than that specified in the applicable specification sheet.

4.8.26 Immersion tests. Specimens of the cable jacket material (see 4.8.18.1) shall be immersed in the fluids shown in table XIV for 24 hours (except for MIL-PRF-17331, which shall be immersed for 18 hours) at the temperatures specified. The specimens shall then be removed, blotted to remove excess fluid, and suspended in air at room temperature for not less than 3½ hours or more than 4½ hours. Each specimen shall be tested as specified in 4.8.28. In the event a listed fluid has been deleted or rendered inactive for new design, and the fluid specification designates a replacement fluid (or fluids), the replacement fluid may be used instead of the original specified fluid. If more than one replacement fluid is allowed, any one of said replacement fluids may be substituted for the original specified fluid, at the manufacturer's option.

TABLE XIV. Immersion test temperatures.

Fluid	Test temperature (°F [°C])
Fuel oil, MIL-DTL-16884	208 to 212 (98 to 100)
Turbine fuel, JP-5, MIL-DTL-5624	118 to 122 (48 to 50)
Cleaner, isopropyl alcohol, ASTM D770	68 to 77 (20 to 25)
Hydraulic fluid, MIL-PRF-5606 1/	118 to 122 (48 to 50)
Hydraulic fluid, MIL-PRF-83282 1/	118 to 122 (48 to 50)
Hydraulic fluid, MIL-PRF-87257 1/	118 to 122 (48 to 50)
Hydraulic fluid, MIL-PRF-17672	118 to 122 (48 to 50)
Lubricating oil, MIL-PRF-23699	208 to 212 (98 to 100)
Coolant, Monsanto Coolanol 25 or equivalent	68 to 77 (20 to 25)
Lubricating oil, 2190 TEP, MIL-PRF-17331	248 to 252 (120 to 122)

NOTE:

1/ Only one of these three hydraulic fluids need be used for immersion testing. MIL-PRF-83282 and MIL-PRF-87257 are acceptable alternative fluids for use in place of MIL-PRF-5606, at the manufacturer's option.

4.8.26.1 Observation. Any of the following shall construe specimen failure:

a. A specimen tensile strength which is less than that specified in the applicable specification sheet.

b. A specimen elongation which is less than that specified in the applicable specification sheet.

4.8.27 Weathering. Jacket material shall be subjected to 720 hours of the xenon-arc lamp weathering resistance test in accordance with the xenon-arc test of UL 1581.

4.8.27.1 Observation. Specimen failure shall be construed if the tensile strength and elongation retention is less than 75 percent or if the specimen surface exhibits signs of cracking when examined using 3 times magnification.

4.8.28 Tensile strength and elongation. Both aged and unaged specimens of the insulation and jacket material shall be tested in accordance with methods 3021 and 3031 of FED-STD-228. Unless otherwise specified in the applicable specification sheet, there shall be 1-inch bench marks, 1-inch jaw separation, and a rate of jaw travel of 10 inches per minute. For thermoset insulation meeting the criteria of ASTM D1248 for types II, III, and IV insulation, the rate of jaw travel shall be 2 inches per minute. The thickness of the specimen shall be measured using any standard micrometer.

4.8.28.1 Observation. Specimen failure shall be construed if the tensile strength and elongation retention is less than that specified herein or in the applicable specification sheet.

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4.8.29 Shrinkage. A 12-inch specimen of cable shall be cut so that all components are flush at both ends. Unless otherwise specified in the applicable specification sheet, the specimen shall then be aged at 277±5 °F (136±3 °C) for 6 hours in an air circulating oven. At the end of this period, the specimen shall be removed from the oven and allowed to cool to room temperature. Shrinkage of the jacket shall then be measured to determine the total distance the jacket has receded from both ends of the conductor.

4.8.29.1 Observation. Specimen failure shall be construed if the total shrinkage of the jacket is greater than 0.25 inch.

4.8.30 Crack resistance. The glass braid covering shall withstand bending, cracking, and puncturing of the overlay covering in accordance with 4.8.30.1 through 4.8.30.6.

4.8.30.1 Specimen. Each specimen shall consist of a single insulated conductor (including insulation covering) of a length not less than that specified in table XV. Specimens shall be removed from the completed cable as follows:

a. Cables containing four or fewer conductors: one conductor shall be removed from each insulated conductor.

b. Cables containing more than 4 but fewer than 21 different insulated conductors; a total of 4 specimens shall be removed. Each specimen shall be removed from a different insulated conductor.

c. Cables containing 21 or more conductors: the number of specimens removed shall be equal to the square foot (rounded to the nearest whole number) of the total number of conductors. Each specimen shall be removed from a different insulated conductor.

An additional specimen, which shall consist of a piece of insulated conductor (including insulation covering) of not less than 10 feet in length, may be required (see 4.8.30.5).

TABLE XV. Specimen and mandrel dimensions for crack resistance test.

Conductor size (AWG)	Min. specimen length (inches)	Mandrel diameter (inches)
7 or larger	24	1½±1/16
9	18	1±1/16
10	12	½±1/32
14	12	½±1/32
16	8	⅓±1/64
18	8	⅓±1/64
Less than 18	6	¼±1/64

4.8.30.2 Special apparatus. The apparatus shall include the following:

a. The mandrel shall be a rigid, smooth cylinder of the diameter specified in table XV and the length and construction for the specimen wrapping as specified in 4.8.30.3.

b. A violet aqueous solution, of a quantity and concentration for staining as specified in 4.8.30.3 or 4.8.30.5, as applicable.

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4.8.30.3 Procedure. Each specimen shall be tightly hand-wrapped for not fewer than three circumferential and continuous, 360-degree turns around the mandrel. Following this, each specimen shall be removed from the mandrel and continuous 360-degree turns shall be made in the opposite lateral direction. Each specimen shall next be removed from the mandrel and straightened to the extent that the specimen is nowhere bent along a radius of less than 12 times the maximum measured specimen diameter. Each straightened specimen shall then be submerged, except for the extreme ends (which shall remain exposed to the air), in the violet aqueous solution for a period of not less than 30 minutes. The specimen shall then be removed from the violet aqueous solution and visually inspected for defects (see 4.8.30.4).

4.8.30.4 Observation. Specimen failure shall be construed if the insulation covering of any specimen exhibits visible cracking or peeling that allows penetration of the violet aqueous solution. If the insulation covering of any specimen exhibits an average of more than one pinhole, which allows penetration of the violet aqueous solution for each 12 inches of length, then the further procedure as specified in 4.8.30.5 shall be performed.

4.8.30.5 Further procedure (may not be required, see 4.8.30.4). An additional specimen, which shall consist of a piece of insulated conductor (including insulation covering) of not less than 10 feet in length, shall be prepared. This entire specimen, except for the extreme ends (which shall be exposed to air), shall be submerged in violet aqueous solution for a period of not less than 30 minutes. To facilitate submersion, the specimen may be coiled to the extent that it is nowhere bent along a radius of less than 12 times the maximum measured specimen diameter. The specimen shall then be removed from the violet aqueous solution and visually inspected for defects (see 4.8.30.6).

4.8.30.6 Further observation. Specimen failure shall be construed if the insulation covering of the specimen exhibits an average of more than one pinhole, which allows penetration of the violet aqueous solution for each 12 inches of length.

4.8.31 Thermoset proof. The hot creep test shall be in accordance with ICEA T-28-562. The test shall be performed as specified for thermoset insulations. The procedure for jacket testing shall be modified to run at 392 °F (200 °C). Jacket percent of elongation shall be as specified in the applicable specification sheet.

4.8.31.1 Observation. Specimen failure shall be construed if the insulation or jacket hot creep elongation percent is more than that specified in the applicable specification sheet.

4.8.32 Electrical moisture absorption. The test shall be performed in accordance with ICEA T-27-581/ANSI/NEMA WC 53, Accelerated Water Absorption Test, Electrical Method at 60 Hz (EM-60), with the following requirements. The test shall be performed on a 14 AWG wire. The extruded insulation thickness shall be 0.045±0.004 inch. The test shall be performed in water at 167+1.8 °F (75+1 °C). The dissipation factor, stability factor, and dielectric constant portions of the test method are not applicable for this requirement (see also 3.3.20).

4.8.32.1 Observation. Specimen failure shall be construed if the percent increase in capacitance exceeds the values specified in 3.3.20.

4.8.33 Temperature/humidity. When specified in the applicable specification sheet, the cable shall be tested in accordance with EIA 364-31, humidity test procedure, method IV, with a cold temperature step. After completion of the testing, the cable shall be allowed to return to 68 °F (20 °C).

4.8.33.1 Observation. Specimen failure shall be construed if the attenuation requirements of the applicable specification sheet are not met.

4.8.34 Aging stability. When specified in the applicable specification sheet, the inspection shall be in accordance with MIL-DTL-17.

4.8.34.1 Observation. Specimen failure shall be construed if the overall cable jacket or print legend displays any of the defects specified for this test in MIL-DTL-17.

4.8.35 Dimensional stability. When specified in the applicable specification sheet, the inspection shall be in accordance with MIL-DTL-17.

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4.8.35.1 Observation. Specimen failure shall be construed if the dimensional differences exceed those specified in the applicable specification sheet.

4.8.36 Armor. Cable armor performance shall be determined as specified in 4.8.36.1 through 4.8.36.4.

4.8.36.1 Specimen. Each specimen shall consist of a single piece of armor wire not less than 6 inches in length, which has been removed from a spool of armor wire prior to cable armoring. Each specimen shall be removed from a different spool. Ten specimens shall be required.

4.8.36.2 Special apparatus. The apparatus shall include the following:

a. Mandrel. A ¼±1/64-inch diameter steel cylinder, of length and construction for the springiness test as specified in 4.8.36.3.

b. Scale. A spring or other type of scale, the measurement error of which shall not exceed ±5 percent for use in the springiness and toughness tests specified in 4.8.36.3.

c. Special vise. A special vise, with two parallel edges of 0.030±0.001-inch radius between which a specimen may be clamped to perform the toughness test as specified in 4.8.36.3.

4.8.36.3 Procedure. The two following tests shall be performed using no fewer than five specimens. The specimen shall be tested one at a time for each test. The same specimen shall not be used for more than one test.

a. Springiness. One specimen shall be selected and secured by one end to the mandrel such that it is both in contact with the curved surface of the mandrel and perpendicular to the mandrel axis. The opposite end of the specimen shall be attached to the scale, and the specimen shall be wrapped circumferentially around the mandrel, by handling the scale only, for not fewer than four complete turns, while maintaining a specimen tension of not less than 3.5 ounces (100 grams) or more than 4.4 ounces (125 grams). Care should be taken to pull the scale in such a direction that it accurately indicates the specimen tension. The specimen shall then be released from the scale and allowed to uncoil. Any specimen that is kinked at any place other than the attachment points at either end shall be considered an invalid specimen; the test shall be repeated using a different specimen. This test shall be successfully repeated four more times (five times total), using a different specimen each time.

b. Toughness. One specimen shall be selected and secured by one end between the rounded, horizontal, and parallel vise edges, such that not less than 2 inches of the specimen extends vertically above the vise. The scale shall then be attached to the free end of the specimen and a specimen tension of 3.527±0.352 ounces (100±10 grams) shall be established by handling the scale only. The specimen shall be bent 90±15 degrees alternately in both directions perpendicular to the vise edges while maintaining the specified vise tension. Not fewer than ten 90-degree bends shall be made, unless the specimen breaks. The bending rate shall not exceed fifty 90-degree bends per minute. This test shall be repeated four more times (five times total), using different specimens each time.

4.8.36.4 Observation. Either of the following shall constitute specimen failure:

a. Springiness. The outer diameter of the coil formed by any of the specimens around the mandrel, as measured at a middle portion of the specimen, exceeds ⅜ inch.

b. Toughness. Any specimen withstands fewer than ten 90-degree bends without breaking.

4.8.37 Long term jacket immersion. The cable shall meet the requirements of 3.3.23 when tested as specified in 4.8.37.1 through 4.8.37.4.

4.8.37.1 Specimen. The specimen shall consist of three 10±0.50-foot lengths of finished cable.

4.8.37.2 Special apparatus. The apparatus shall include a water tank of a size sufficient to contain the specimen, which shall be used as specified in 4.8.37.3 below. The tank shall contain tap water at 68±9 °F (20±5 °C).

4.8.37.3 Procedure. One inch of jacket material from both ends of each specimen length shall be removed. The exposed shield at both ends of each length shall be visually examined (see 4.8.37.4) and shall show no sign of moisture or corrosion.

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a. Specimen lengths shall be coiled or U-shaped (if coiled, the inside coil diameter shall be no smaller than 24 times the nominal overall cable diameter). The ends of each length shall be brought together, crossed over each other, and tied together and shall be sufficiently separated from each other to allow for DC resistance test connections. The specimen lengths shall then be held at 68±9 °F (20±5 °C) for 24 hours prior to the test. The specimen lengths shall remain in the selected shape throughout the 56-day immersion period, including during all DC resistance testing.

b. The initial DC resistance measurement and immersion in the water tank shall be made within 15 minutes after removal from being conditioned. Upon removal from conditioning, the cable shield of each specimen length shall be tested for DC resistance, and the value recorded. The cable temperature at the time of the initial DC resistance test measurement shall be recorded. The measured DC resistance shall be corrected to a 68 °F (20 °C) ambient temperature reference value.

c. The three specimen lengths shall be submerged in the water tank, except for the extreme ends, which shall remain exposed to air. A minimum of 8 feet of each specimen length shall be immersed. The water temperature shall be maintained at 68±9 °F (20±5 °C) for the entire 56-day immersion period. It is permissible for the immersed lengths to be in contact with each other, the tank walls, and the bottom. The three specimen lengths shall be immersed in water for eight cycles of seven days for a total of 56 days. After each 7-day period of immersion, the shield DC resistance of each specimen length shall be measured and recorded.

d. The 7-day cycle DC resistance tests may be conducted on the lengths while immersed, or each length may be removed from the tank for a maximum of 10 minutes for the tests with the following exception: after the DC resistance test at the 56-day mark, the sample specimens are not returned to the tank). The water temperature at the time of each 7-day DC resistance test measurement shall be recorded. The measured DC resistance shall be corrected to a 68 °F (20 °C) ambient temperature reference value.

e. All measured DC resistance values shall be corrected to 68 °F (20 °C) ambient temperature reference value by means of the following formula:

DCR20 = DCRT/(1 + [0.00375 * (T-20)])

Where: DCRT = DC resistance measured value at temperature T

T = Temperature of specimen or water at the time of testing

0.00375 = Arbitrary value for composite shield temperature coefficient of resistance

f. If the shield DC resistance of any specimen length increases by greater than 2 percent from the original initial measured value of that length before the end of the 56-day immersion period, the test shall be terminated. At the end of the 56-day immersion period, the specimen lengths shall be removed from the water, and the last DC resistance test cycle shall be performed within 10 minutes of said removal. If the shield DC resistance of all three specimen lengths has not increased by greater than 2 percent from the original initial measured values after the last shield resistance measurements at the end of the 56-day immersion period, the specimen lengths shall be straightened and the cable jacket surface shall be towel dried. Hot air drying shall not be permitted. The jacket of each specimen length shall then be completely removed. Each specimen length's outer braided shield (and optional binder tape, if applied) under the jacket shall be visually examined (see 4.8.37.4) to verify if there is any sign of moisture or shield corrosion.

4.8.37.4 Observation. Specimen failure shall be construed if any of the three lengths exhibit the following:

a. An increase in shield DC resistance greater than 2 percent from the original initial measured value of that length during or at the end of the 56-day immersion period.

b. The outer braided shield (or optional binder tape, if applied) under the jacket shows any sign of moisture or shield corrosion.

4.9 Test methods (electrical).

4.9.1 Attenuation. This test shall determine the ability of completed cable, which is intended for the transmission of data, voice, or control information, to propagate a signal at a specific frequency without causing an unacceptable reduction of signal amplitude.

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4.9.1.1 Specimen. The specimen shall consist of a piece of completed cable, which shall have sufficient length to exhibit an electrical attenuation of not less than 3 decibels (dB) at the test frequency specified in the applicable specification sheet. Individual transmission lines within this specimen (where each single insulated conductor plus surrounding shield or each insulated conductor pair plus surrounding shields is considered to be a transmission line) shall be selected for testing as follows:

a. Specimens containing four or fewer transmission lines: every transmission line shall be tested.

b. Specimens containing more than 4 but fewer than 21 transmission lines: 4 different transmission lines shall be tested. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

c. Specimens containing 21 or more transmission lines: the number of transmission lines tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of transmission lines. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

An electrical connector, which shall be designed for use at the test frequency specified in the applicable specification sheet and which shall be used with the specimen, shall be attached to each end of each transmission line selected for testing.

4.9.1.2 Special apparatus. The apparatus shall be used at the test frequency specified in the applicable specification sheet, shall be chosen such that all electrical interconnections (see 4.9.1.3) are of the same nominal characteristic impedance throughout (except that connectors and specimens in which a single conductor plus surrounding shield is tested may be of a different impedance than this apparatus, if necessary), and shall include the following:

a. A sinusoidal signal generator, which shall produce an unchanging signal of the test frequency (specified in the applicable specification sheet within a frequency accuracy of 0.25 percent). The signal generator may be amplitude-modulated in a constant and unchanging manner if the detector (see 4.9.1.2.b) requires an amplitude-modulated signal for proper operation. The signal generator shall generate power to produce a reasonable indication on the detector when used as specified in 4.9.1.3; however, the signal generator shall not generate so much power that the specimen or any piece of test apparatus will consequently experience a significant temperature rise.

b. A crystal diode, bolometer, or other detector, with its associated indicating device, which shall be used as specified in 4.9.1.3.

c. A calibrated variable attenuator, which shall produce a maximum attenuation greater than that exhibited by the specimen and shall resolve attenuation within an accuracy of ±0.25 dB.

d. Two fixed attenuators, each of which shall exhibit an attenuation of not less than 10 dB.

e. Impedance matching transformers (required only when testing a conductor pair plus surrounding shield).

f. Two winding transformers: one winding of each transformer shall have an impedance equal to the nominal impedance of the fixed attenuators, the second winding of each transformer shall be centertapped and shall have an impedance equal to the impedance of the specimen specified in the applicable specification sheet. When testing a single conductor plus surrounding shield, impedance matching transformers may be used, if necessary, to match the nominal unbalanced impedance of each fixed attenuator (see 4.9.1.2.d) to the unbalanced impedance of the specimen specified in the applicable specification sheet, as shown on figure 15.

g. An adapter connector (may not be required, see 4.9.1.3) for joining the two fixed attenuators directly together. This adapter connector shall exhibit an attenuation that shall be not greater than 2 percent of the specimen attenuation. If desired, the adapter connector may be fabricated in the same manner and of the same components as the specimen, except that the shortest practicable length of completed cable shall be used to join the two connectors.

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4.9.1.3 Procedure. The test apparatus shall be electrically interconnected as shown on figure 15 (when the transmission line to be tested is a single insulated conductor plus surrounding shield) or as shown on figure 16 (when the transmission line to be tested is an insulated conductor pair plus surrounding shield), as applicable; one of the selected transmission lines shall then be connected where shown. Following this, the calibrated variable attenuator shall be adjusted to a low value and the signal generator adjusted to produce a test signal of sufficient amplitude to produce a reasonable indication on the detector. The setting on the calibrated variable attenuator and the indication on the detector shall then be noted. Next, the transmission line shall be disconnected from the test apparatus, and the two fixed attenuators shall be connected together either directly, or, if necessary, by means of an adapter connector. The signal generator shall not be readjusted. The calibrated variable attenuator shall then be readjusted until the detector produces the same indication as was noted when the transmission line was connected to the test apparatus. The new setting on the calibrated variable attenuator shall then be noted, and the transmission line attenuation shall be calculated by using the following formula:

Attenuation, in dB per 100 feet = $100 (A_2 - A_1)/L$

Where: A_1 = Initial setting of the calibrated variable attenuator (when the transmission line is connected to the test apparatus), in dB.

A_2 = Final setting of the calibrated variable attenuator (following transmission line removal from the test apparatus), in dB.

L = Measured specimen length, in feet; measurement accuracy shall be within ±1 percent.

In an identical manner, the remaining selected transmission lines shall also be tested.

4.9.1.4 Observation. Specimen failure shall be construed if any transmission line exhibits an attenuation that is greater than that specified in the applicable specification sheet.

4.9.2 Capacitance. This test shall determine the capacitive characteristics of completed cable.

4.9.2.1 Component capacitance. This test shall determine the capacitance between conductors and between conductors and shields.

4.9.2.1.1 Specimen. The specimen shall consist of a 122±0.5-inch-length of completed cable, which shall have overall components (such as armors, jackets, or overall binders) removed from each end for a distance of 2±0.5 inches, thereby exposing the insulated conductors (and shields, if applicable) at each end of the cable core. Each exposed shield (if any) at each cable end shall be unwrapped from the end of its enclosed insulated conductor(s) for a distance of 1±⅛ inch; the unwrapped portion of each shield shall then be tightly twisted to form a pigtail for electrical connection. The insulation shall next be stripped from each end of each insulated conductor for a distance of 0.5±0.125 inch. If the characteristic impedance test or mutual inductance test is not required by the applicable specification sheet, the insulation may be stripped for the specified distance from all of the insulated conductors at one specimen end only.

4.9.2.1.2 Special apparatus. The apparatus shall include an instrument for measuring capacitance (such as a capacitance bridge, Q-meter, or other instrument) that shall be for supporting the requirements as specified in 4.9.2.1.3. This apparatus shall have a two-terminal measurement port, which shall use a sinusoidal wave of the frequency specified in the applicable specification sheet for making measurements and shall make measurements within an accuracy of ±1 percent.

4.9.2.1.3 Procedure. The appropriate capacitance test method shall be selected and performed from the following:

a. For specimens incorporating any number of individually shielded, insulated conductor transmission lines, one transmission line shall be selected for measurement, and each end shall be positioned such that its conductor and shield shall not come into mutual electrical contact, nor into electrical contact with any other conductor or shield that may be within the specimen. One end of this selected transmission line shall then be appropriately connected electrically to the capacitance-measuring instrument, and its capacitance shall be measured. In an identical manner, the capacitance of every other transmission line within the specimen shall also be measured. The capacitance per foot of each transmission line shall then be calculated by using the following formula:

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Capacitance in picofarads per foot = C/10

Where: C = Measured capacitance of the transmission line, in picofarads.

b. For specimens incorporating any number of individually shielded, insulated conductor pair transmission lines, one transmission line shall be selected for measurement, and each end shall be positioned such that neither its conductors nor its shield come into mutual electrical contact or into electrical contact with any other conductor or shield that may be within the specimen. Three capacitances shall then be measured at one end of this transmission line, where one conductor of this line has been designated as conductor number one and the other conductor of this line has been designated as conductor number two as follows:

(1) Conductor number one and the shield shall both be electrically connected to one terminal of the capacitance-measuring instrument, and conductor number two shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_a.

(2) Conductor number two and the shield shall both be electrically connected to one terminal of the capacitance-measuring instrument, and conductor number one shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_b.

(3) Both conductor number one and conductor number two shall then be electrically connected to one terminal of the capacitance-measuring instrument, and the shield shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_c.

The mutual capacitance per foot and, if specified in the applicable specification sheet, the capacitance unbalance (sometimes called the coefficient of asymmetry) shall then be calculated by using the following approximate formulas:

Mutual capacitance, in picofarads per foot = $(C_a + C_b)/20 - C_c/40$

Capacitance unbalance, in percent = $10[C_a - C_b]/C_m$

Where: C_a, C_b, and C_c are as defined in (1), (2), and (3) respectively, in picofarads.

C_m = Calculated mutual capacitance, in picofarads per foot.

In an identical manner, the mutual capacitance and, if specified in the applicable specification sheet, the capacitance unbalance of every other transmission line shall also be measured and calculated.

c. For specimens incorporating multiple, unshielded, insulated conductor pair transmission lines, the mutual capacitance, and, if specified in the applicable specification sheet, the capacitance unbalance of every transmission line shall be determined using the same method as used for specimens incorporating individually shielded, insulated conductor pair transmission lines, with the following change: The overall specimen braid (if any) and all specimen conductors except the conductor pair under test shall be connected together electrically and considered to be the shield of the transmission line under test.

d. For specimens incorporating any number of individually shielded, insulated conductor triad transmission lines, one transmission line shall be selected for measurement, and each end shall be positioned such that none of its conductors, nor its shield, come into mutual electrical contact, nor into electrical contact with any other conductor or shield that may be within the specimen. Four capacitances shall then be measured at one end of this transmission line as follows, where one conductor of this line has been designated as conductor number one, another conductor of this line has been designated as conductor number two, and the remaining conductor of this line has been designated as conductor number three as follows:

(1) Conductors number two and three and the shield shall be electrically connected to one terminal of the capacitance-measuring instrument, and conductor number one shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_a.

(2) Conductors number one and three and the shield shall next be electrically connected to one terminal of the capacitance-measuring instrument, and conductor number two shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_b.

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(3) Conductors number one and two and the shield shall next be electrically connected to one terminal of the capacitance-measuring instrument and conductor number three shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_c.

(4) Conductors number one, two, and three shall then be electrically connected to one terminal of the capacitance-measuring instrument, and the shield shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_d.

The mutual capacitance per foot and, if specified in the applicable specification sheet, the capacitance unbalance (sometimes called the coefficient of asymmetry) shall then be calculated by using the following approximate formulas:

$$Cm = [(C_a + C_b + C_c)/40] - [C_d/120]$$

The capacitance unbalance or coefficient of asymmetry (K) of a shielded triad, expressed in percent, shall be determined by the formula:

$$K_1 = 10[C_a - C_b]/C_m$$

$$K_2 = 10[C_a - C_c]/C_m$$

$$K_3 = 10[C_b - C_c]/C_m$$

Where: K_1 = The percent of capacitance unbalance of conductor No. 1 in relation to conductor No. 2.

K_2 = The percent of capacitance unbalance of conductor No. 2 in relation to conductor No. 3.

K_3 = The percent of capacitance unbalance of conductor No. 3 in relation to conductor No. 1.

NOTE: In the formula, the relative positions of C_a, C_b, and C_c may be transposed as necessary to avoid negative values.

In an identical manner, the mutual capacitance and, if specified in the applicable specification sheet, the capacitance unbalance of every other transmission line shall also be measured and calculated.

e. For specimens incorporating multiple, unshielded, insulated conductor triad transmission lines, the mutual capacitance and, if specified in the applicable specification sheet, the capacitance unbalance of every transmission line, shall be determined using the same method as used in 4.9.2.1.3.d, with the following change: the overall specimen braid (if any) and all specimen conductors except the conductor triad under test shall be connected together electrically and considered to be the shield of the transmission line under test.

4.9.2.1.4 Observation. Specimen failure shall be construed if the specimen exhibits a mutual capacitance or unbalance that is greater than that specified in the applicable specification sheet.

4.9.2.1.5 Characteristic impedance and mutual inductance. If the characteristic impedance or mutual inductance test is specified in the applicable specification sheet, then the specimen shall be retained for use in that test.

4.9.2.2 Pair to ground capacitance unbalance. This test shall determine the capacitance unbalance to ground of the conductors of any one pair against all other pairs (and overall cable shields, if present).

4.9.2.2.1 Specimen and procedure. The specimen preparation and test procedure shall be in accordance with ANSI/TIA 568.2.

4.9.2.2.2 Observation. Specimen failure shall be construed if the specimen exhibits a pair to ground capacitance unbalance that is greater than that specified in the applicable specification sheet.

4.9.3 Characteristic impedance. This test, when required in the applicable specification sheet and using MIL-DTL-17, shall determine the characteristic impedance encountered by signals propagating along completed cable. The mutual capacitance (see 4.9.2) shall be calculated for each transmission line within the completed cable prior to performing the characteristic impedance test. This is necessary since values of mutual capacitance are used to calculate values of characteristic impedance.

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4.9.3.1 Specimen. The specimen shall be that which was previously used to determine capacitance characteristics (see 4.9.2.1.1).

4.9.3.2 Special apparatus. The apparatus shall include an instrument for measuring inductance (such as an inductance bridge, Q-meter, or other instrument) that shall be capable of supporting the requirements as specified in 4.9.8.3. This apparatus shall have a two-terminal measurement port, which shall use sinusoidal wave of the frequency specified in the applicable specification sheet for making measurements. Measurements shall be made within an accuracy of ±1 percent.

4.9.3.3 Procedure. The appropriate characteristic impedance test shall be selected and performed from the following:

a. Specimens incorporating any number of individually shielded, insulated conductor transmission lines shall be laid straight. At one specimen end, the exposed conductor of each transmission line shall be electrically connected to the pigtail of its associated shield by tightly twisting both together. At the opposite specimen end, one transmission line shall be selected for measurement, which shall then be connected to the inductance measuring instrument. The inductance of this transmission line shall be measured. In an identical manner, the inductance of every other transmission line within the specimen shall also be measured. The characteristic impedance of each transmission line shall then be calculated by using the following formula:

Characteristic impedance, in ohms = $(10^5 L/C_m)^{1/2}$

Where: C_m = Mutual capacitance, in picofarads per foot, as calculated in 4.9.2.1.3.

L = Measured inductance, in microhenries/10 feet.

b. Specimens incorporating any number of individually shielded or unshielded insulated conductor pair transmission lines shall be laid straight. At one specimen end, the two exposed insulated conductors of each transmission line shall be laid adjacent to each other (to minimize superfluous inductance) and electrically connected by tightly twisting both together. At the opposite specimen end, one transmission line shall be selected for measurement and its two conductors connected electrically to the inductance-measuring instrument. Its shield (if any) shall remain disconnected. The inductance of this transmission line shall be measured. In an identical manner, the inductance of every other transmission line within the specimen shall also be measured. The characteristic impedance of each insulated conductor pair shall then be calculated by using the formula shown in 4.9.3.3.a.

4.9.3.4 Observation. Specimen failure shall be construed if the specimen exhibits a characteristic impedance other than that specified in the applicable specification sheet.

4.9.4 Conductor resistance and DC resistance unbalance. These tests shall determine the conductor resistance characteristic of completed cable.

4.9.4.1 Conductor resistance. This test shall determine whether conductors within the completed cable are fabricated from the specified material.

4.9.4.1.1 Procedure. Conductor resistance shall be tested in accordance with ASTM B193 and shall be corrected to 77 °F (25 °C).

4.9.4.2 DC resistance unbalance. When specified in the applicable specification sheet, this test shall determine the DC resistance unbalance between conductors within individual twisted pairs in completed cable.

4.9.4.2.1 Procedure. DC resistance unbalance shall be tested in accordance with ANSI/TIA 568.2.

4.9.4.3 Observation. Specimen failure shall be construed if any specimen conductor exhibits a conductor resistance or DC resistance unbalance greater than that specified in the applicable specification sheet.

4.9.5 Insulation resistance. Insulation resistance shall meet the requirements of 3.3.4.3 when tested as specified in 4.9.5.1 through 4.9.5.5. The voltage withstand test (see 4.9.8) shall be performed prior to the insulation resistance test.

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4.9.5.1 Selection of insulated conductors. The conductor insulation resistance test shall be performed on all conductors that are smaller than 18 AWG (less than 0.0403 inch in diameter if solid wire); for conductors that are 18 AWG or greater (0.0403 inch in diameter if solid wire), the conductor insulation resistance test shall be performed on conductors selected as follows:

No. of insulated conductors in cable	No. of insulated conductors selected for test
1 through 4	All
5 through 20	4
21 through 30	5
31 through 42	6
43 through 56	7
57 through 72	8
73 through 91	9
92 through 110	10
111 through 132	11
133 through 156	12

4.9.5.2 Specimen. The specimen shall be that which was previously used to perform the voltage withstand test (see 4.9.8.1).

4.9.5.3 Special apparatus. The apparatus shall include the following:

a. A megohmmeter, megohm bridge, or other standard instrument that uses a DC test voltage of not less than 200 V and not greater than 1/2 of the rated voltage of the specimen (test) cable, which shall support the requirements as specified in 4.9.5.4, and make measurements within an accuracy of ±10 percent.

b. A water tank (may not be required [see 4.9.5.4.b and e]) of a size to contain the specimen, which shall be used as specified in 4.9.5.4.b and e.

4.9.5.4 Procedure. The specimen shall be maintained at a constant ambient temperature, within ±5.4 °F (±3 °C), for a period of not less than 5 hours immediately prior to testing. For testing in water, the temperature of the water surrounding the specimen shall be measured after the specimen has been submerged for not less than 1 hour. This temperature shall be noted and shall be used to calculate F (see formula below) when measurements are made in accordance with a, b, c, or d. The megohmmeter shall then be electrically connected to specimen as specified (see a, b, c, d, and e). For each connection, the test voltage from the megohmmeter shall be continuously applied to the specimen for a period of at least 1 minute, but for no longer than a period of 1 minute + 15 seconds; the resistance reading of the megohmmeter shall be noted immediately following the measurement. Care shall be taken during each measurement to assure that any electrical jumper leads connecting the specimen to the megohmmeter shall be spaced sufficiently far from each other and from any common conducting surface so that measurement accuracy shall not be impaired.

a. Specimens incorporating one or more individually shielded insulated conductor transmission lines: one transmission line shall be selected for measurement. One conductor of this transmission line shall be electrically connected to one terminal of the megohmmeter, and the shield and all other conductors (if any) of this transmission line shall be electrically connected to the other megohmmeter terminal. The resistance measurement shall then be performed as specified in 4.9.5.4. In an identical manner, the resistance between each remaining conductor in this transmission line (if any) and the short-circuited combination of all other conductors (if any), plus the shield in this transmission line shall be measured. Additionally, in an identical manner, resistance measurements shall be made for all remaining transmission lines within the specimen. (Additional measurements may be required; see c and d.)

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b. Specimens incorporating multiple insulated conductors without individual shields: one insulated conductor shall be selected and electrically connected to one terminal of the megohmmeter. All other specimen conductors, including the overall shield (if any) shall be electrically connected to the other megohmmeter terminal. The resistance measurement shall then be performed as specified in 4.9.5.4. In a similar manner, the resistance between each remaining conductor within the specimen and the short-circuited combination of all other conductors plus the overall shield (if any) shall be measured. If the specimen incorporates an overall shield, then an additional resistance measurement shall be made between the shield and the short-circuited combination of all specimen conductors. If the specimen does not incorporate an overall shield or armor over the jacket, the test shall be made after placing the reel of cable in water for a period of not less than 1 hour. (Additional measurements may be required; see d.)

c. Specimens incorporating one or more individually shielded insulated conductor transmission lines (these measurements shall be required only when shield to shield measurements are specified in the applicable specification sheet): One shield shall be selected and electrically connected to one terminal of the megohmmeter. All other specimen shields, including the overall shield (if any), shall be electrically connected to the other megohmmeter terminal, and the resistance measurement shall then be performed as specified in 4.9.5.4. In a similar manner, the resistance between each remaining shield within the specimen and the short-circuited combination of all other shields plus the overall shield (if any) shall be measured. If the specimen incorporates an overall shield, then an additional resistance measurement shall be made between the overall shield and the short-circuited combination of all specimen shields. (Additional measurement may be required; see d.)

d. Specimens incorporating double overall shields: One shield shall be electrically connected to one terminal of the megohmmeter, and the other shield shall be connected to the other terminal. The resistance measurement shall be performed as specified in 4.9.5.4. (Additional measurements may be required; see a, b, and c above.)

e. Specimens incorporating single or multiple conductors without shield or armor shall be tested to water. The specimens shall be submerged for not less than 1 hour, except for its extreme ends (which shall remain exposed to air), in the water tank. The single conductors are to be electrically connected to one terminal of the megohmmeter. The other megohmmeter terminal shall be electrically connected to the water surrounding the specimen (by means of a bare metal wire that is both connected to the megohmmeter terminal and suspended within the water surrounding the specimen). The resistance measurement shall be performed as specified in 4.9.5.4.

The lowest value of resistance measured in accordance with each of a, b, c, and d, as applicable, shall be modified to include the effects of specimen length and temperature by using the following formula:

Insulation resistance, in megohms per 1,000 feet = RLF/1,000

Where: R = Lowest value of resistance measured in accordance with a, b, c, or d, as applicable, in megohms.

L = Measured specimen length, in feet; measurement accuracy shall be within ±2 percent.

F = An ad-hoc number, which shall accurately modify the measured resistance R to that value of resistance that would be measured if the specimen were at a uniform temperature of 59.9 °F (15.5 °C). This number shall be determined by the specimen manufacturer via ad hoc testing for the specific insulation or jacketing material separating the conductors between which the resistance measurement was made and shall be normalized to unity for a material temperature of 59.9 °F (15.5 °C).

4.9.5.5 Observation. Specimen failure shall be construed if any calculated value of insulation resistance is less than that specified in the applicable specification sheet.

4.9.6 Mutual inductance. This test shall determine the ability of shielded, insulated conductor pairs within completed cable incorporating both shielded, insulated conductor pairs and unshielded insulated conductors to minimize crosstalk caused by the inductive effects of signal currents flowing in adjacent insulated conductors.

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4.9.6.1 Specimen. The specimen shall consist of a 120±0.5-inch length of completed cable, which shall have all overall components (such as armors, jackets, or overall binders) and all shields removed from each end for a distance of 1.5±0.125 inches, except that shields may be pushed back for this distance rather than removed. The insulation shall then be stripped from each end of each insulated conductor for a distance of 0.5±0.125 inch, and both insulated conductors within each shield shall be electrically connected by twisting each conductor pair tightly together at one specimen end.

4.9.6.2 Special apparatus. The apparatus shall include the following:

a. A sinusoidal signal generator, which shall produce an unchanging signal at a constant frequency of 1,000±200 Hz, which shall produce a signal current appropriate for use in the test as specified in 4.9.6.3 (a signal current of approximately 1A is typically found necessary), and the signal frequency and signal current shall be known within an accuracy of ±2 percent.

b. An RMS indicating microvoltmeter, intended for use at the signal generator frequency, exhibiting a signal input impedance not less than 10,000 ohms, shall be used for the specimen measurements as specified in 4.9.6.3 and shall exhibit a measurement error of not more than ±6 percent.

4.9.6.3 Procedure. One shielded, insulated conductor pair within the specimen shall be selected, and the microvoltmeter shall be electrically connected across this conductor pair at its unshorted end. The shield shall remain disconnected. The signal generator shall then be electrically connected across any two unshielded, insulated conductors at the specimen end opposite that of the microvoltmeter connection. These same two unshielded, insulated conductors shall be connected together electrically at the opposite specimen end (the end with the attached microvoltmeter) by twisting their two conductors tightly together. The signal current from the signal generator shall next be adjusted until a reasonable indication is noted on the microvoltmeter, and both the microvoltmeter reading and the signal current magnitude shall then be recorded. In a similar manner, additional measurements shall be taken such that microvoltmeter measurements are taken for each shielded, insulated conductor pair as the signal generator is sequentially reconnected to circulate signal current in each possible pairing of unshielded, insulated conductors. (Examples: type LSMCOS-5 cable shall require three measurements because it incorporates one shielded, insulated conductor pair plus three unshielded, insulated conductors and type LSMCOS-6 cable shall require two measurements because it incorporates two shielded, insulated conductor pairs plus two unshielded, insulated conductors.) The mutual inductance represented by each measurement shall then be calculated by using the following formula:

Mutual inductance, in microhenries = V/2 * 3.14159 fI

Where: V = Measured potential between two shielded, insulated conductors, in microvolts

f = Signal frequency, in Hz

I = Measured signal current, in A

4.9.6.4 Observation. Specimen failure shall be construed if any calculated value of mutual inductance is greater than that specified in the applicable specification sheet.

4.9.7 Pulse response time. This test shall determine the ability of completed cable that incorporates one or more shielded, insulated conductor pairs for pulse data transmission to propagate a square wave signal without causing unacceptable signal distortion.

4.9.7.1 Specimen. The specimen shall consist of a piece of completed cable, which shall have a length of not less than 200 feet. Individual transmission lines within this specimen (where each insulated conductor pair plus surrounding shield is considered to be a transmission line) shall be selected for testing as follows:

a. Specimens containing four or fewer transmission lines: every transmission line shall be tested.

b. Specimens containing more than four but fewer than 21 transmission lines: four different transmission lines shall be tested. No fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

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c. Specimens containing 21 or more transmission lines: the number of transmission lines tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of transmission lines. At least one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

4.9.7.2 Special apparatus. The apparatus shall include the following:

a. Pulse generator: a signal generator that shall produce uniform square wave voltage pulses at a constant rate of repetition and shall be used for the specimen measurements as specified in 4.9.7.3. The pulses produced by this generator shall each exhibit a 10 to 90 percent voltage rise time of not more than 50 nanoseconds, a 90 to 10 percent voltage decay time of not more than 50 nanoseconds, and a pulse width of not less than 2 microseconds.

b. Oscilloscope: a dual beam oscilloscope shall have a bandwidth of not less than 10 megahertz (MHz), and a time base capable of producing a horizontal beam sweep rate of not slower than 0.254 milliseconds (ms) per inch (0.1 ms per cm). This oscilloscope shall have provision for a high impedance, balanced or differential input on both signal channels, and be used for the specimen measurements as specified in 4.9.7.3.

c. Pulse transformers: two winding transformers, each of which shall provide essentially distortion-free transmission of the voltage pulses from the pulse generator and shall be used for the specimen measurements as specified in 4.9.7.3. One of the pulse transformers (the "input" pulse transformer, see figure 17) shall have a primary winding impedance equal to the nominal signal terminal impedance of the pulse generator; the secondary winding of this transformer shall be centertapped and shall have an impedance equal to the impedance of the specimen specified in the applicable specification sheet. The other pulse transformer (the "output" pulse transformer, see figure 17) shall have a primary winding that shall be centertapped and shall have an impedance equal to the impedance of the specimen specified in the applicable specification sheet. The secondary winding impedance of this transformer shall equal the nominal resistance of the matching resistor (see d).

d. Matching resistor: an electrical resistor shall be essentially non-inductive in construction (such as a carbon type resistor), which shall have a power rating appropriate for its intended use (see 4.9.7.3), and shall exhibit an electrical resistance appropriate for use with the output pulse transformer (see c).

4.9.7.3 Procedure. The test apparatus shall be electrically interconnected as shown on figure 17; one of the selected transmission lines shall be connected as shown. The connections between this transmission line and the pulse transformers shall be as short as practicable. In addition, both connecting cables between the transmission line and the oscilloscope shall be of identical length and construction and shall not be longer than 5 feet each in order to minimize pulse distortion. The pulse generator shall then be turned on, and both signal channels on the oscilloscope shall be adjusted such that the pulse waveform displayed by each on the oscilloscope screen shall be of the same height. The horizontal beam sweep rate of the oscilloscope shall then be increased, if necessary, and the time required for the leading edge of each signal channel pulse to rise from 10 to 90 percent of its final value shall be measured as shown on figure 18. The rise time of the transmission line shall then be calculated by using the following formula:

Rise time, in ms per 500 feet = $500 (Y_r^2 - X_r^2)^{1/2}/L$

Where: X_r = Measured rise time from signal channel number one, in microseconds

Y_r = Measured rise time from signal channel number two, in microseconds

L = Measured specimen length, in feet; measurement accuracy shall be within ±1 percent

Following this, the time required for the trailing edge of each signal channel pulse to decay from 90 to 10 percent of its initial value shall be measured as shown on figure 19. The decay time of the transmission line shall then be calculated by using the following formula:

Decay time, in ms per 500 feet = $500(Y_d^2 - X_d^2)^{1/2}/L$

Where: X_d = Measured decay time from signal channel number one, in microseconds

Y_d = Measured decay time from signal channel number two, in microseconds

L = Measured specimen length, in feet; measurement accuracy shall be within ±1 percent

In an identical manner, the remaining selected transmission lines shall also be tested.

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4.9.7.4 Observation. Specimen failure shall be construed if any transmission line exhibits either a rise time or a decay time that is greater than that specified in the applicable specification sheet.

4.9.8 Voltage withstand. This test shall determine the ability of completed cable to withstand overvoltages without exhibiting electrical discharge through insulation or jacketing material, either between conductors or between these conductors and any conducting material that may be in contact with the surface of the cable during in-service use.

4.9.8.1 Specimen. The specimen shall consist of a unit of product as specified in 6.8.4.

4.9.8.2 Special apparatus. The apparatus shall include the following:

a. An adjustable voltage source, which shall be in accordance with either of the following, as specified in the applicable specification sheet:

(1) An AC voltage source that shall produce a single-phase, approximately sinusoidal voltage under all test conditions, which shall have a frequency of not more than 100 Hz and can be adjusted to the RMS magnitude specified in the applicable specification sheet. This source shall have the capacity to provide an effective power of not less than 5 kVA and shall incorporate an overload relay that shall reduce the source voltage to zero whenever an applied load (that is, the specimen, see 4.9.8.3) attempts to draw electrical current to exceed this effective power capacity.

(2) A source of DC voltage that can be adjusted to the magnitude specified in the applicable specification sheet and shall have the capacity to provide an electrical current of not less than 0.25A. This source shall incorporate an overload relay that shall reduce the source voltage to zero whenever an applied load (that is, the specimen, see 4.9.8.3) attempts to draw electrical current to exceed this current capacity.

b. When required, a water tank, which shall be filled with water and shall be of sufficient size to contain the specimen, shall be used as specified in 4.9.8.3.f and 4.9.8.3.g (may not be required; see 4.9.8.3.f and 4.9.8.3.g).

4.9.8.3 Procedure. The voltage source shall be electrically connected to the specimen as specified (see 4.9.8.3.a, 4.9.8.3.b, 4.9.8.3.c, 4.9.8.3.d, 4.9.8.3.e, 4.9.8.3.f, and 4.9.8.3.g). For each connection, the test voltage from the voltage source shall be increased, in a uniform manner, from 0V to the voltage specified in the applicable specification sheet during a period of not more than 60 seconds or less than 10 seconds, except that specified voltages of less than 600V may be applied instantaneously. (This gradual voltage application is generally necessary to prevent a current in rush and the consequent tripping of the voltage source overload relay caused by specimen capacitance.) This specified voltage shall then be maintained for a period of 1±0.5 minute (unless the voltage source overload relay trips, see 4.9.8.4), following which the test voltage shall be reduced to zero at approximately the same rate as which it was applied.

a. Conductor to conductor (components): specimens incorporating one or more insulated conductor transmission lines (as pairs or triples) with overall shield within each transmission line. One transmission line shall be selected for measurement. One insulated conductor shall be selected and electrically connected to one terminal of the voltage source. All other conductors within that transmission line shall be electrically connected to the other voltage source terminal. The transmission line shield shall be left unconnected. The voltage withstand test shall then be performed as specified in 4.9.8.3. In a similar manner, the voltage withstand test shall be performed between each remaining conductor within the selected transmission line and the short-circuited combination of all other conductors within that transmission line. Following this, and in an identical manner, the voltage withstand test shall be performed on all remaining transmission lines within the specimen. (Additional measurements may be required; see 4.9.8.3.b, 4.9.8.3.c, 4.9.8.3.d, 4.9.8.3.e, and 4.9.8.3.f.)

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b. Conductor to conductor (cable): specimens incorporating only multiple insulated conductors without individual shields, or a combination of multiple unshielded, insulated conductors (as individual conductors or component pairs, triples, or groups) and one or more individually shielded, insulated conductor transmission lines: one unshielded, insulated conductor shall be selected and electrically connected to one terminal of the voltage source. All other specimen unshielded, insulated conductors shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified in 4.9.8.3. In a similar manner, the voltage withstand test shall be performed between each remaining unshielded, insulated conductor within the specimen and the short-circuited combination of all other unshielded, insulated conductors. If the specimen incorporates an overall shield, then the voltage withstand test shall additionally be performed between the overall shield and the short-circuited combination of all unshielded, insulated conductors that are outmost in the specimen. (Additional measurements may be required; see 4.9.8.3.a, 4.9.8.3.c, 4.9.8.3.d, 4.9.8.3.e, and 4.9.8.3.f.)

c. Conductor to component shield (shielded components): specimens incorporating one or more individually shielded, insulated conductor transmission lines. One transmission line shall be selected for measurement. For single, pair, or triple conductor(s): the short-circuited combination of all conductors within this transmission line shall be electrically connected to one terminal of the voltage source, and the transmission line shield shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified in 4.9.8.3. Following this, and in an identical manner, the voltage withstand test shall be performed on all remaining transmission lines within the specimen. (Additional measurements may be required; see 4.9.8.3.d, 4.9.8.3.e, and 4.9.8.3.f.)

d. Conductor to component shield (cable): specimens incorporating combinations of one or more individually shielded, insulated conductor transmission lines, and one or more unshielded, insulated conductors (as individual conductors or in component pairs, triples, or groups). One shield of a shielded transmission line shall be selected and electrically connected to one terminal of the voltage source. All specimen unshielded, insulated conductors shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified in 4.9.8.3. In a similar manner, the voltage withstand test shall be performed between each remaining shielded transmission line within the specimen and the short-circuited combination of all unshielded, insulated conductors. (Additional measurements may be required; see 4.9.8.3.e and 4.9.8.3.f.)

e. Component shield to shield: specimens incorporating two or more individually shielded, insulated conductor transmission lines (these measurements shall be required only when component shield to shield measurements are specified in the applicable specification sheet). One component shield shall be selected and electrically connected to one terminal of the voltage source. All other specimen component shields shall be electrically connected to the other voltage source terminal, and the voltage withstand test shall then be performed as specified in 4.9.8.3. In a similar manner, the voltage withstand test shall be performed between each remaining component shield within the specimen and the short-circuited combination of all other component shields. If the specimen incorporates an overall shield, then the voltage withstand test shall additionally be performed between the overall shield and the short-circuited combination of all component shields that are outmost in the specimen. (Additional measurements may be required; see 4.9.8.3.f.)

f. Jackets: all one-conductor shielded specimens and all multi-conductor specimens (these measurements shall be required only when measurements to water or to ground are specified in the applicable specification sheet). The specimen shall be submerged, except for its extreme ends (which shall remain exposed to the air), in the water tank for a period of not less than 1 hour. Following this, and while the specimen is still submerged, the specimen shall be tested as follows. The overall specimen shield (if any) or, if there is no overall shield, all individual shields that are outermost in the specimen core plus all unshielded conductors that are outermost in the specimen core, shall be electrically connected to one terminal of the voltage source. The other voltage source terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire that is both connected to the voltage source terminal and suspended within the water surrounding the specimen). The voltage withstand test shall then be performed as specified in 4.9.8.3.

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g. Conductor to ground: specimens incorporating a single, insulated, jacketed conductor without shield or armor shall be tested to water. The specimen shall be submerged, except for its extreme ends (which shall remain exposed to the air), in the water tank for a period of not less than 1 hour. Following this, and while the specimen is still submerged, the specimen shall be tested as follows. The single conductor shall be electrically connected to one terminal of the voltage source. The other voltage source terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire that is both connected to the voltage source terminal and suspended within the water surrounding the specimen). The voltage withstand test shall then be performed as specified in 4.9.8.3.

4.9.8.4 Observation. Specimen failure shall be construed if the voltage source overload relay trips prior to removal of the test voltage from any specimen connection. If the insulation resistance test is specified in the applicable specification sheet, the specimen shall be retained for use in that test.

4.9.9 Jacket flaws. All finished cables with overall shields shall be subject to the spark test in accordance with Clause 900 of UL 1581 at 3 kV RMS. Cables without overall shields, but with shielded components without shield insulation in outer cable layer, shall also be subject to the jacket flaws spark test.

4.9.10 Conductor and shield continuity. All finished cable shall be tested for continuity prior to shipment. To establish continuity, not more than 25 volts DC shall be applied across each conductor and shield of the cable through an appropriate indicator, such as an ohmmeter, light, or buzzer. The test voltage may be applied to conductors and shields individually or in series. Specimen failure shall be construed if the specimen exhibits an open electrical circuit.

4.9.11 Surface transfer impedance. The surface transfer impedance characteristics of the overall cable shield of finished cable shall be tested in accordance with ANSI/NEMA WC 61 and as specified in 4.9.11.1. The EMP response of the cable shield shall be determined in accordance with 4.9.11.2.

4.9.11.1 Cable sample preparation. The individual shields of shielded components, when present, shall be connected to the conductors and the connected conductors and shields shall be the conductor as specified in ANSI/NEMA WC 61.

4.9.11.2 Determination of EMP response. The surface transfer impedance determined as specified shall be known over the frequency range DC to 100 MHz. Z_t shall be assumed constant in the frequency range DC to 0.1 MHz. The EMP response in dB shall be determined as follows:

$$\text{EMP response} = (-185) - 10 \log \left[\int_0^{10^8} \frac{\left[\left[Z_t(f) \right] \right]^2}{\left(\alpha^2 + f^2 \right) \left(\beta^2 + f^2 \right)} df \right]$$

Where: $Z_t(f)$= surface transfer impedance at frequency f (Ω/m)

α = 2.39×10^5

β = 4.12×10^7

f = frequency (Hz)

df = decade of frequency

The integral expression may be approximated by a summation if at least 100 points per decade of frequency are used.

4.9.11.3 Observation. Specimen failure shall be construed if the surface transfer impedance is greater than as specified in the applicable specification sheet or if the calculated EMP response of the overall cable shield is less than as specified in the applicable specification sheet.

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4.9.12 Swept electrical parameters. Completed cable shall be tested in accordance with ANSI/TIA 568.2 for transmission performance characteristics as specified in the applicable specification sheet. This standard specifies the appropriate measurement procedures and formulas for a four-pair balanced cabling system for all transmission parameters.

4.9.12.1 Specimen. The specimen shall consist of a length of finished cable 328 feet in length. Every pair shall be tested for basic electrical and swept electrical parameters as specified in the applicable specification sheet (see 3.1).

4.9.12.2 Observation. Specimen failure shall be construed if any pair fails to meet the swept electrical parameters specified in the applicable specification sheet.

4.9.13 Stress resistance. Cable resistance to stress (see 6.8.2) imposed during shipboard installation shall meet the requirements of 3.3.22 when determined as specified in 4.9.13.1 through 4.9.13.7.

4.9.13.1 Specimen. The test specimen shall be a 328-foot (about 100-meter) length of cable. Depending on the test equipment used, such as a network analyzer or hand-held link tester, it may be necessary to terminate each end with an RJ-45 connector. If RJ-45 connectors are used, their losses shall be taken into consideration for the measurement.

4.9.13.2 Measurement. All measurements shall be in accordance with ANSI/TIA 568.2. Measurements shall be made as follows:

a. Baseline: Two baseline measurements of electrical characteristics shall be made on the specimen prior to initiating the stress resistance test. The first set of baseline measurements shall be taken while the cable is still on its original shipping reel. The second set of baseline measurements shall be taken after the cable is unspooled from its reel, but prior to performance of the stress tests.

b. Operating tensile load: Electrical characteristic measurements of the specimen shall be recorded in the stressed condition simulating operating tensile load at 24-hour intervals and in the unstressed condition at the completion of the operating tensile load test.

c. Long-term minimum bend: Electrical characteristic measurements of the specimen shall be recorded in the stressed condition simulating long-term minimum bend upon completing the initial wraps on the fixture and after a 24-hour period.

d. Cable compression: Electrical characteristic measurements of the specimen shall be recorded in the stressed condition simulating cable compression upon completing the initial banding of the cable and after a 24-hour period.

e. Cable re-spooling: Electrical characteristic measurements of the specimen shall be recorded upon completion of cable re-spooling.

4.9.13.3 Operating tensile load. This test subjects the cable specimen to tensile and bending forces. The test specimen shall be installed onto a tensile loading fixture similar to figure 20. The diameter of the sheaves shall be 12.0 inches. A minimum preload of 10 pounds per cable strand shall be placed on the cable. The load shall be increased to 112 pounds per cable strand and held for 72 hours. Electrical characteristic measurements shall be made in accordance with 4.9.13.2.b.

4.9.13.4 Long-term minimum bend. This test subjects the cable specimen to multiple bends at the long-term minimum bend radius of the cable. The test specimen shall be installed onto a bend fixture similar to figure 21. The OD of the mandrels on figure 21 shall be 6.75 inches. Twenty-six 90-degree bends shall be formed in the specimen. Each bend shall occur at approximately 3-foot intervals from the preceding bend and shall be formed to the long-term bend radius of the cable. The specimen shall be maintained on the fixture for 24 hours. Electrical characteristic measurements shall be made in accordance with 4.9.13.2.c.

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4.9.13.5 Cable compression. This test subjects the cable specimen to multiple compression loads along its length. This test shall be performed while the specimen is still installed on the bend fixture. Cable straps shall be installed at 32-inch intervals in the approximate locations as indicated on figure 22. The straps shall be installed in accordance with the following figures in MIL-STD-2003-4, "Single cableway on aluminum decks and bulkheads", "Cable clamps for single cableway on steel and aluminum decks and bulkheads", and "Single cableway for coaxial cable". Electrical characteristic measurements shall be made in accordance with 4.9.13.2.d.

4.9.13.6 Cable re-spooling. This test subjects the cable specimen to the twisting stresses that occur during re-spooling. At the completion of tests 4.9.13.3 through 4.9.13.5, the test specimen shall be un-banded and removed from the bend fixture. Using the reel on which the cable originally came from the manufacturer, the test specimen shall be coiled onto the reel in a smooth and uniform manner. While re-spooling, sufficient tension on the specimen shall be maintained to permit tight adherence to the reel. Electrical characteristic measurements shall be made in accordance with 4.9.13.2.e.

4.9.13.7 Observation. Specimen failure shall be construed if any of the measured baseline electrical characteristics change at the conclusion of each stated stress testing procedure.

4.9.14 Input impedance. This test shall determine the input impedance encountered by signals propagating along completed cable.

4.9.14.1 Specimen. The specimen shall be a 150±0.25-foot-length of completed cable and shall be the same length that will be used for determining insertion loss (see 4.9.15.1).

4.9.14.2 Special apparatus. The apparatus shall include a 4-port network analyzer that has the capability of providing input port impedance of 75 ohms and sinusoidal signal over the test frequency specified in the applicable specification sheet. The apparatus shall also include a connector assembly that allows the center conductors of the cable under test to be connected to the center pins of two connectors while the shield of the cable under test is common to the ground plane of the two connectors. An example is two SUBMiniature version A female-female panel mount connectors soldered together at the panel mount flange.

4.9.14.3 Procedure. Prior to calibration, the input ports' impedance setting shall be set to 75 ohms. This provides a differential input impedance of 150 ohms. The four-port network analyzer shall then be calibrated in accordance with the calibration procedure provided by the equipment manufacturer. The cable under test shall interface with the network analyzer's port extension cables with a connector assembly, conductors under test shall be connected to the center pins of the connector assembly, and the shield connected to the ground plane. The ends of one component shall be connected to ports 1 and 3 and the other component ends shall be connected to ports 2 and 4. The analyzer shall be powered on and the test readings shall be recorded.

4.9.14.3.1 Settings. The following settings shall be used:

a. Gating – On

b. Conversion – On

c. Response Magnitude – Linear

d. Measurement – Sdd11

4.9.14.4 Observation. Specimen failure shall be construed if the specimen exhibits an input impedance other than that specified in the applicable specification sheet at the stated test frequencies.

4.9.15 Insertion loss. This test shall determine the insertion loss of completed cable that is designed with 150-ohm differential pairs.

4.9.15.1 Specimen. The specimen shall be the same 150±0.25-foot-length of completed cable that was used for input impedance (see 4.9.14.1).

4.9.15.2 Special apparatus. The apparatus used for this test shall be in accordance with the apparatus described in 4.9.14.2.

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4.9.15.3 Procedure. The procedure used for this test shall be in accordance with 4.9.14.3, with the settings changed to those specified in 4.9.15.3.1.

4.9.15.3.1 Settings. The following settings shall be used:

a. Gating – On

b. Conversion – On

c. Response Magnitude – Linear

d. Measurement – Sdd11

4.9.15.3.2 Calculation. The insertion loss of the cable shall be calculated using the following formula:

Insertion loss in dB per 100 meters = 328 * A / L

Where: A = Measured insertion loss value in dB

L = Measured specimen length in feet

4.9.15.4 Observation. Specimen failure shall be construed if any insertion loss value per 328 feet (100 meters) is greater than that specified in the applicable specification sheet.

4.10 Fungus resistance. All nonmetallic materials shall be tested in accordance with ASTM G21 (see 3.3.16).

4.11 Fire testing provisions. See 4.1.1 for in-house testing requirements. Fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or the International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.2.5).

4.12 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3.1 as required by the qualifying activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Cables specified herein are intended for use in various applications involving naval ships, submarines, and shore stations. Existing stock of low halogen smoke cable manufactured to MIL-DTL-24643C (I1) dated 11 July 2011 and MIL-DTL-24643C dated 1 October 2009 is acceptable for use until depleted.

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6.1.1 Ampacity. Guidance for MIL-DTL-24643 cable ampacities is found in MIL-HDBK-299. The cable ampacities in the handbook are based on an upper temperature limit of +90 °C-rated service. Cables rated for upper temperature limits of +105 °C or higher are limited to +90 °C-rated service when placed with other cables that are rated for +90 °C, regardless of whether in raceways, trays, cableways, or otherwise bundled together; unless the higher temperature rated cables are physically or thermally isolated from other cables of a different (lower) upper temperature rating. (Example: +105 °C-rated cables in proximity with +90 °C-rated cables are limited to ampacity at +90 °C. +105 °C-rated cables in proximity with other +105 °C- or +150 °C-rated cables only, or physically/thermally isolated from +90 °C-rated cables, could [if deemed appropriate] operate at an ampacity rating of +105 °C.)

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Part number or identifying number (PIN) (see 1.3).

c. Type and size of cable required with reference to applicable specification sheet (see 3.1).

d. Jacket color required (see 3.3.6.1).

e. Additional fire testing provisions (see 4.11 and 6.2.5).

f. Packaging requirements (see 5.1).

g. Requirements for certified test reports (see 6.2.1).

h. Quantity required (see 6.2.2).

i. Special construction requirements (see 6.2.3).

6.2.1 Certified test reports. The contractor should prepare test reports, including those for watertightness and for the specific fire tests (flame propagation, gas flame [circuit integrity] smoke index, toxicity index, acid gas generation, and halogen content), as specified (see 6.2). The test report should contain the following information:

a. Serial number and date of NAVSEA letter of approval.

b. A statement to the effect that the product was constructed from materials listed on the manufacturer's approved details of construction sheet.

c. A statement that the product meets all of the requirements of this specification.

d. Results of all conformance tests showing actual values obtained.

e. Year and month the cable was manufactured.

f. Serial numbers of the marker tape taken from each end of each length of cable.

g. Customer's name and contract or order number.

h. A statement that the product contains no metallic mercury or mercury compounds and is free from mercury contamination.

i. The serial number of all NAVSEA letters approving deviations from the approved specification sheets.

j. Name and address of the test laboratory that conducted the tests and test report numbers when applicable.

6.2.2 Quantity. The quantity of each type and size on a contract or order should be specified (see 6.2) as an integral multiple of the unit ordering length shown by the specification sheet.

6.2.3 Manufactured product. Unless otherwise specified (see 6.2), the manufacturer has the option of construction where such options are specifically allowed in the applicable specification sheet. Typical specification sheet options could include, but are not limited to, one or more of the following: conductor material, component braid shield material, component color code identification, component braid shield single end size, component shield insulation.

6.2.3.1 Product uniformity. Each individual type and size of cable manufactured for an inspection lot (see 4.4.1) should be homogenous in material and construction with regards to manufacturer allowed options (see 6.2.3). For example, product manufactured to a detail specification sheet that allows the option of coated or uncoated copper conductors should not mix both types of copper together in the same cable.

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6.2.4 Lengths. The range of standard, random, remnant, and scrap lengths for each nominal length is shown in table XVI. In order to compensate for handling and probable loss on issue, remnant lengths are subject to the price reductions specified in table XVI. Scrap lengths are not acceptable. It is not intended that items being acquired in exact lengths for a specific job or ship will necessarily be shipped in lengths that are integral multiples of the unit ordering length or that price reductions will apply in the event that exact footages or fractional lengths are required and so indicated in acquisition of cable other than stock purposes.

TABLE XVI. Cable lengths.

Type of package	Nominal length (feet)	Standard lengths no price reduction (feet)	Random lengths no price reduction (feet)	Remnant lengths price reduction		Scrap lengths not acceptable (feet)
				5 percent (feet)	10 percent (feet)	
Coil	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0
Reel	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1,000	1,100 to 900	899 to 300	299 to 200	199 to 100	99 to 0
Reel	1,500	1,650 to 1,350	1,349 to 450	449 to 300	299 to 150	149 to 0
Reel	2,000	2,200 to 1,800	1,799 to 600	599 to 400	399 to 200	199 to 0
Reel	2,500	2,750 to 2,250	2,249 to 750	749 to 500	499 to 250	249 to 0
Reel	3,000	3,300 to 2,700	2,699 to 900	899 to 600	599 to 300	299 to 0

6.2.5 Additional fire testing provisions. NAVSEA 05P5 is the designated NAVSEA Technical Warrant Holder (TWH) for fire testing provisions. NAVSEA reserves the right to witness the tests, where such action is deemed necessary to assure compliance to the prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.3 Shipment of cables. Unless otherwise specified in the applicable specification sheet, cables covered by this specification should be shipped on reels.

6.3.1 Cable packaging. Cables shipped on reels should be packaged in accordance with the requirements of NEMA WC 26.

6.3.2 Placement of cable on the reel. The cable on each reel should be one continuous length and should have both ends readily available for testing without re-reeling.

6.3.3 Year marking. In addition to any other required marking, reels should be marked with a keyed series of colors to indicate the year of manufacture. The cycle of colors should be repeated every fifth year. The reel marking should consist of a stripe of one coat of commercial quality outside paint approximately 2 inches wide, colored to designate the particular year of manufacture as follows:

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Year of manufacture	Identifying color
2017	Green
2018	Orange
2019	Blue
2020	White
2021	Red
2022	Green
2023	Orange
2024	Blue
2025	White
2026	Red
2027	Green

And so on.

6.3.4 Location of year marking. The colored stripe on reels should be applied circumferentially over the lagging or the alternative to lagging and midway between the flanges. The stripe should consist of one coat of commercial quality outside paint of the appropriate color. In addition to the stripe, both flanges of the reel should be stenciled with 4-inch-high figures to show the year of manufacture.

6.3.5 Standard reel markings. Each reel should be plainly marked on both flanges with the following information:

a. Reel number
b. Type and size of cable
c. Footage
d. Contract Order or Number
e. Contractor's name
f. Manufacturer's name (if other than contractor)
g. Gross weight
h. One continuous length on the reel indicated in feet

6.3.6 Coils. Coils should contain one continuous length of cable (standard length, see table XVI).

6.3.7 Year marking. In addition to any other required marking, coils should be marked with a keyed series of colors to indicate the year of manufacture. This marking should consist of a stripe approximately 2 inches wide and colored for the particular year of manufacture. The cycle of colors should be the same as those used for year marking on reels (see 6.3.3).

6.3.8 Identification. Two shipping tags should be securely attached to each coil, one inside and one outside the wrapping, and marked with the following information.

a. Type and size of cable.
b. Footage.
c. Contract or order number.
d. Contractor's name.
e. Manufacturer's name (if other than contractor).
f. Gross weight.

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6.4 Special clause. Except when small quantities are to be purchased, invitations for bid and contracts or orders should specify the following:

"In order to compensate for handling and probable loss in issue, remnant lengths shall be subject to the price reductions specified in table XVI. In order that the number of lengths be kept to a minimum consistent with good manufacturing practices, for each type and size of cable on the contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, and not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is six nominal lengths or less, a footage approximately two nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is two nominal lengths or less, the total footage may be in any combination of random lengths."

6.5 Time delay. Conformance inspection will normally require 40 calendar days from the date the sample is sent to the government-designated laboratory to the date the inspection results are received by the manufacturer. This time delay should be taken into consideration by a contractor when estimating delivery time.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24643 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Definitions. For the purpose of the conformance inspection, the sampling terms specified in 6.8.1 through 6.8.5 apply.

6.8.1 Accredited independent testing laboratory (AL). ALs are those in the contiguous United States (CONUS) and which are accredited to ISO/IEC 17025 through A2LA or IAS. The Scope of Accreditation from the accrediting organization must specifically reference each of the following tests found in table V for which accreditation is given: flame propagation, gas flame (circuit integrity), smoke index, toxicity index, acid gas generation, and halogen content. Manufacturers may use more than one accredited independent testing laboratory for the required testing, but each laboratory used must be accredited for the specific test(s) they are performing.

6.8.2 Cable stress. The result of, but not limited to, tensile strain imposed by pulling; crushing forces imposed by cable clamps, straps, stuffing tubes, and multi-cable transits; bending forces imposed by routing; and twisting of the cable subcomponents due to re-spooling.

6.8.3 Sample. One unit of product (see 6.8.5) selected from an inspection lot.

6.8.4 Specimen. An individual length of cable or an individual length of a part of cable that has been taken from a sample.

6.8.5 Unit of product. The unit ordering length as specified in the specification sheet. When the unit ordering length is 500 feet, two such lengths constitute a unit of product. Random and remnant lengths or special-order lengths may be added together to equal a unit ordering length. Length tolerances as given for standard lengths apply (see 6.2.4).

For basic electrical test purposes (see table V), a unit of product is defined as an electrically continuous, splice-free length. This is to allow for longer production lengths to be tested electrically prior to preparation for shipment. Lengths longer than the unit ordering lengths specified in the applicable specification sheet may be supplied when the purchaser and manufacturer agree.

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6.9 Subject term (key word) listing.

Armor

Flame propagation

Gas flame (circuit integrity)

Toxicity index

Water-block

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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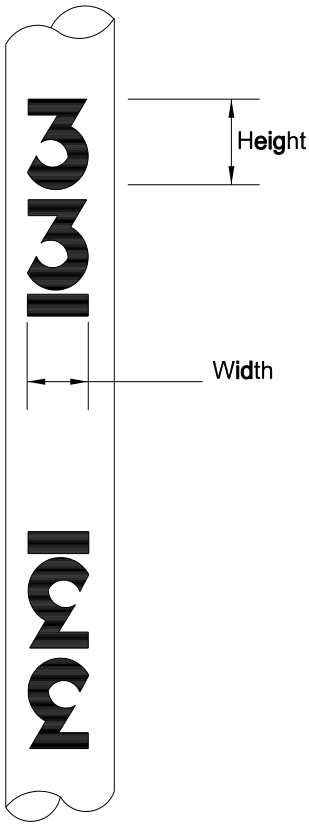


FIGURE 1. Conductor of pair number 33.

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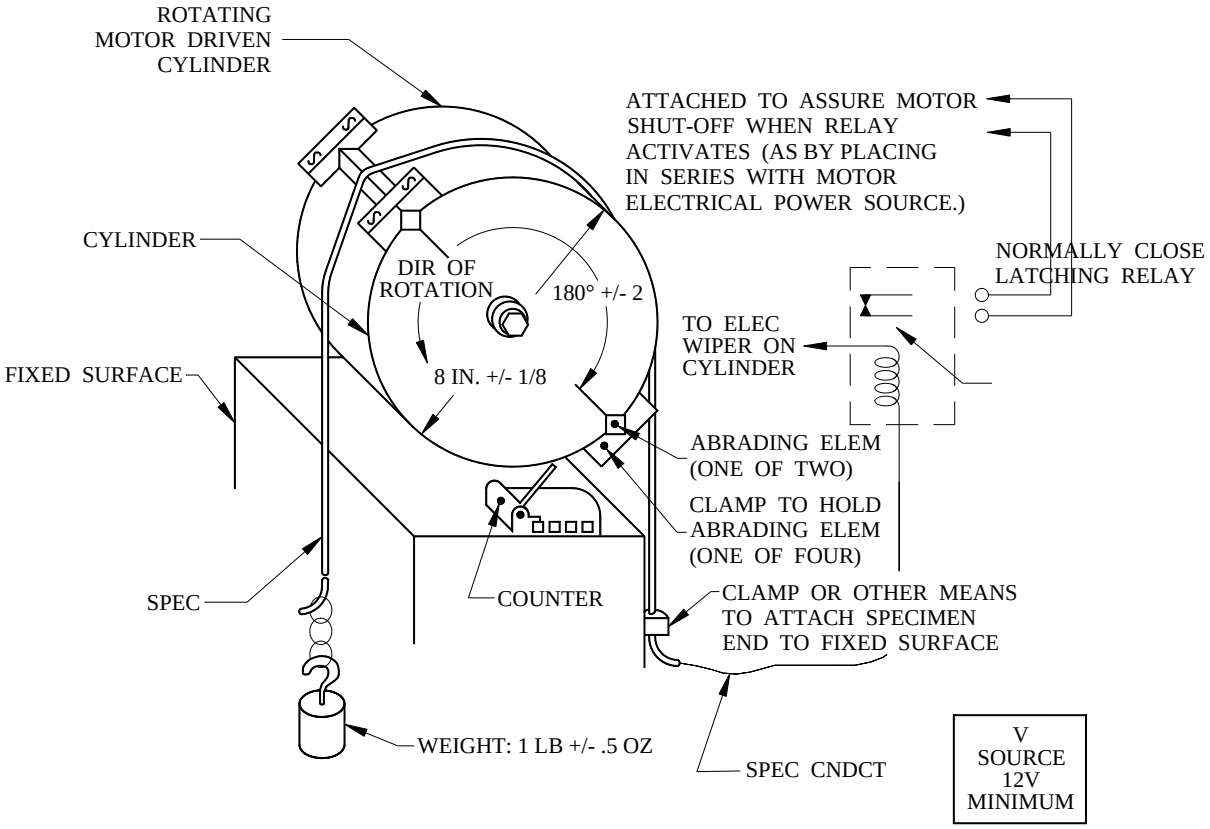


FIGURE 2. Diagram of abrading machine for the abrasion resistance test.

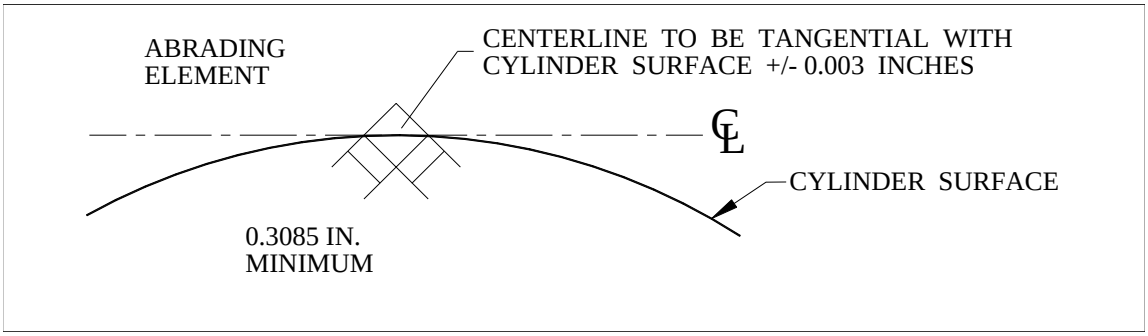


FIGURE 3. Detail diagram of abrading element fit on cylinder surface.

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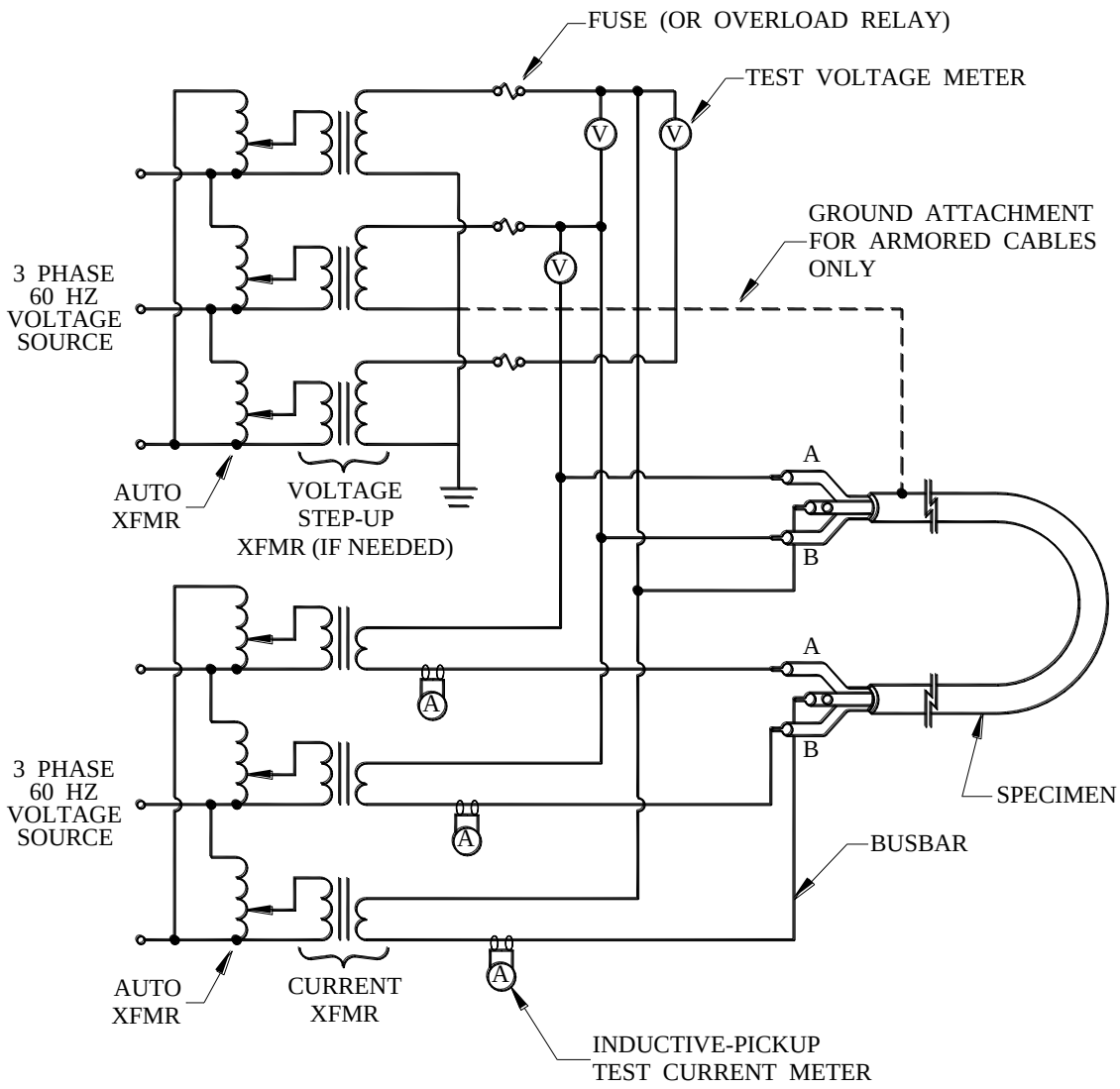


FIGURE 4. Application of test voltage and test current for accelerated service test.

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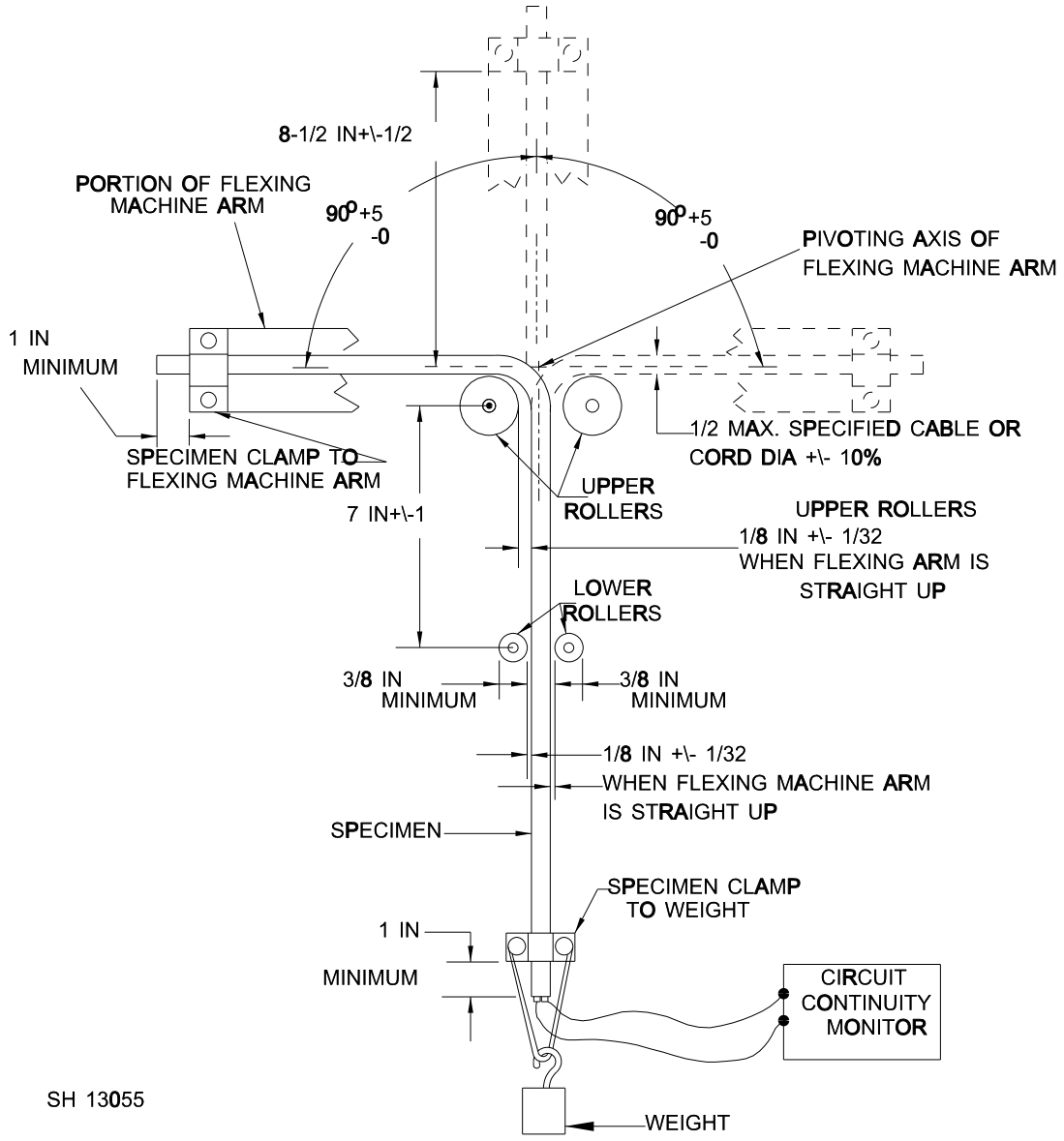


FIGURE 5. Diagram of flexing machine for the bending endurance test.

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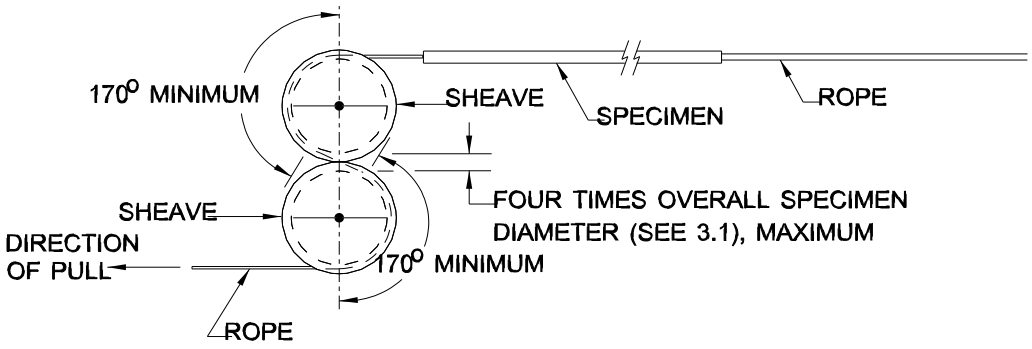


FIGURE 6. Diagram of bending apparatus for the cold working (minus 4 °F [minus 20 °C]) test.

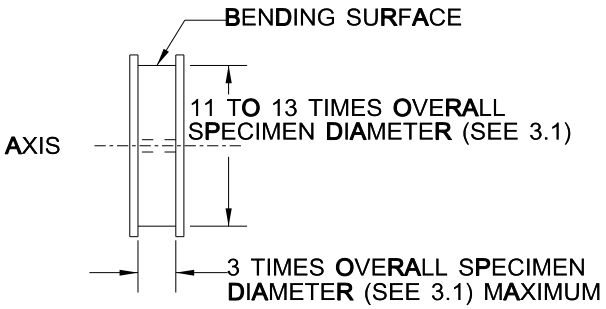


FIGURE 7. Edge view of either sheave for the bending apparatus of the cold working (minus 4 °F [minus 20 °C]) test.

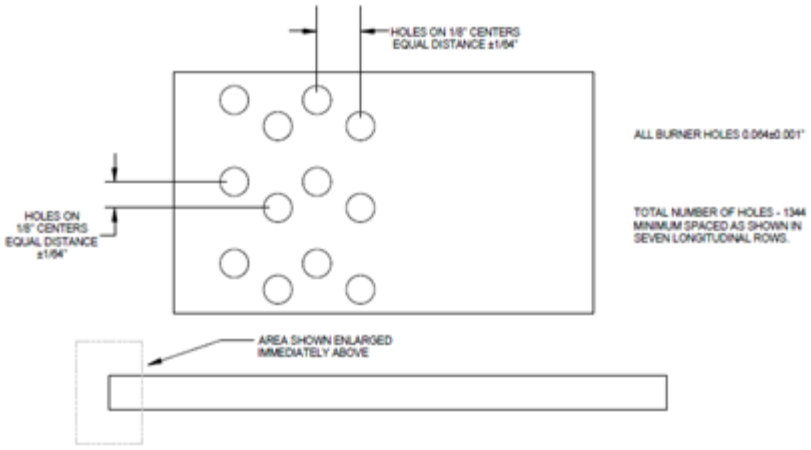


FIGURE 8. Top view of burner for gas flame (circuit integrity) test.

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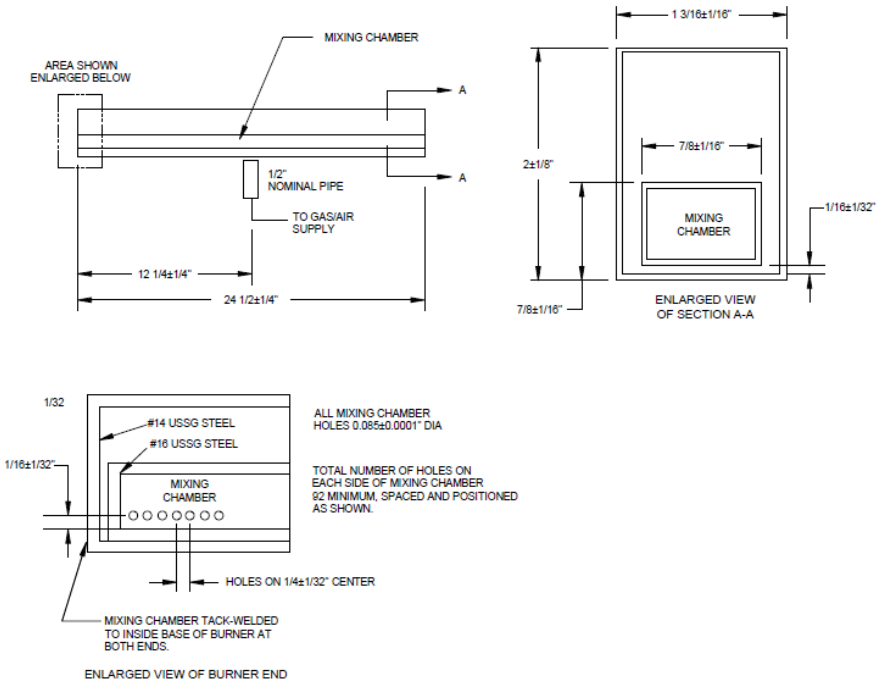


FIGURE 9. Side view of burner showing internal mixing chamber for the gas flame (circuit integrity) test.

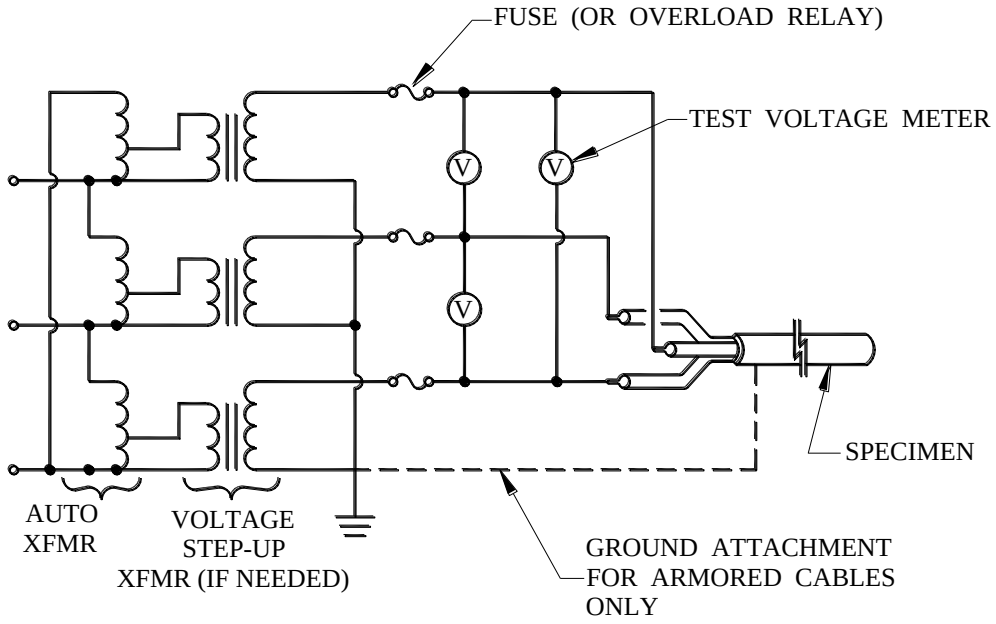


FIGURE 10. Application of test voltages for the gas flame (circuit integrity) test.

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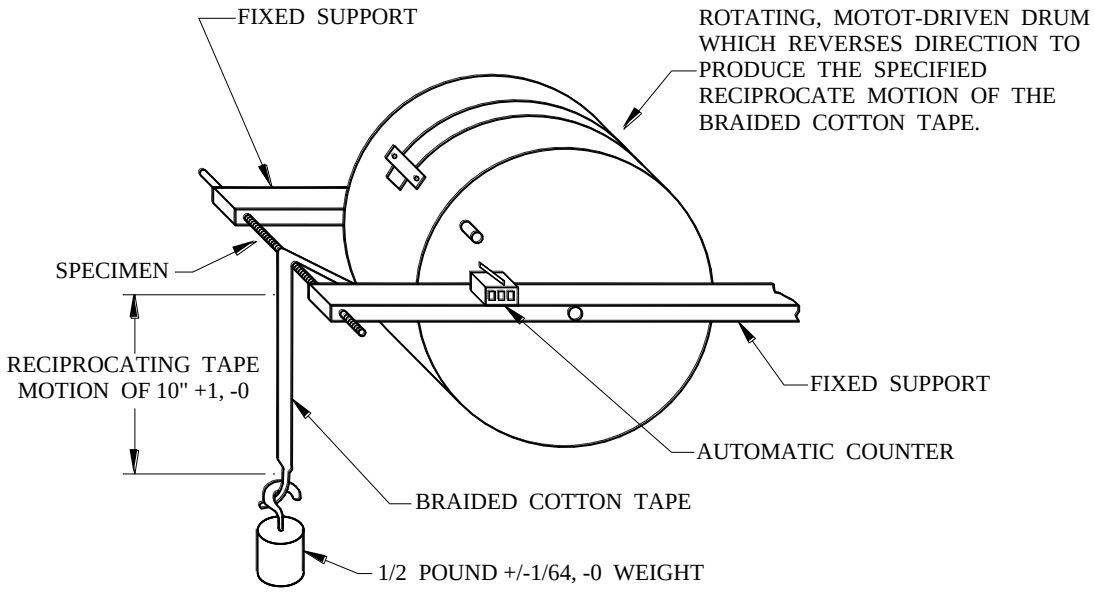


FIGURE 11. Diagram of typical abrading machine for the permanence of printing (conductor insulation) test.

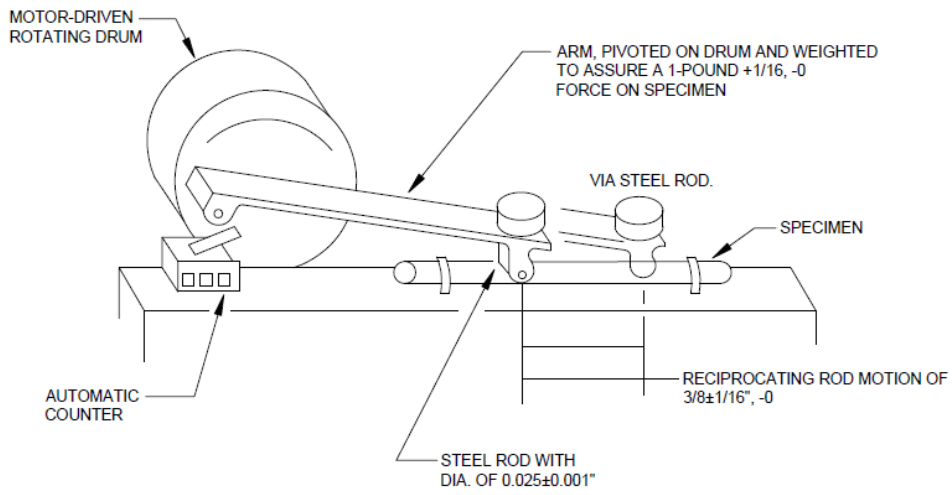


FIGURE 12. Diagram of abrading machine for the permanence of printing (jacket) test.

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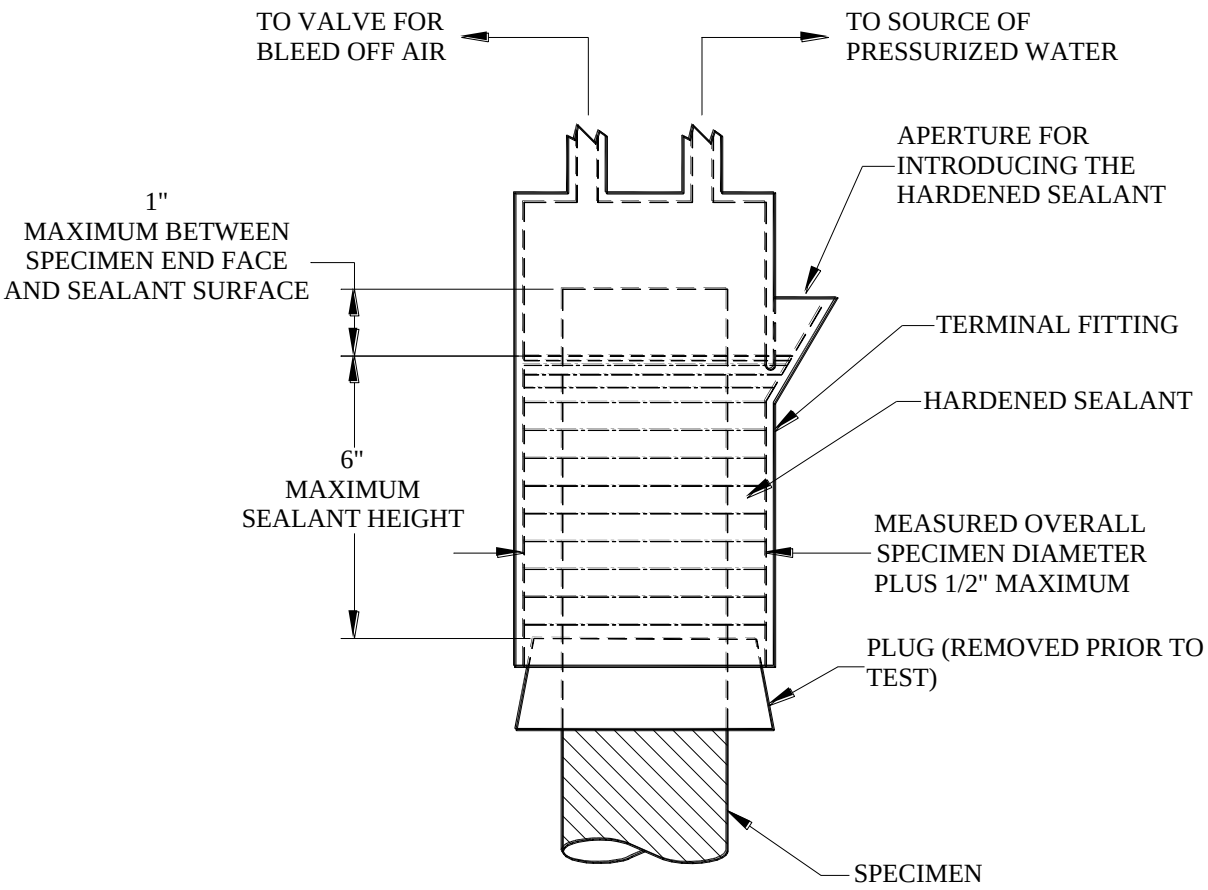


FIGURE 13. Diagram of terminal fitting, showing internal details, for the watertightness test.

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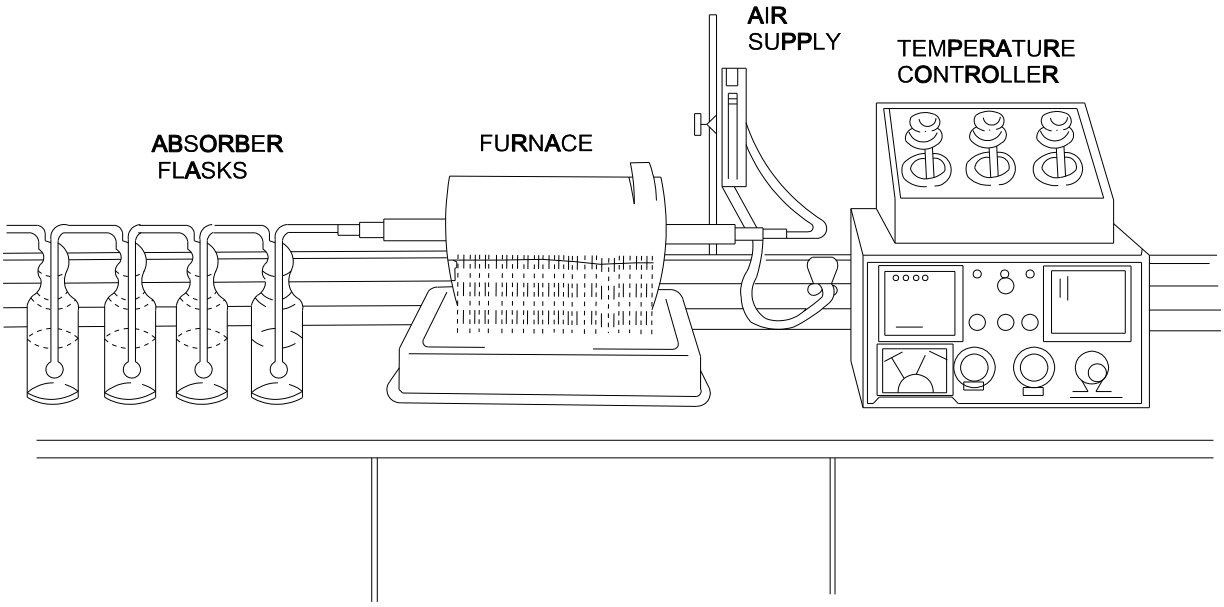


FIGURE 14. Acid gas test apparatus.

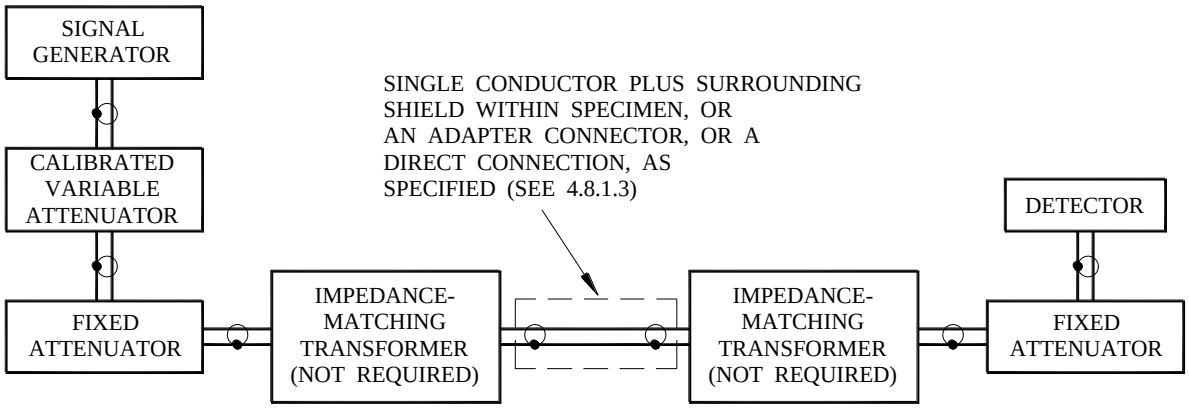


FIGURE 15. Block diagram of electrical connections for the attenuation test when testing a single conductor plus surrounding shield.

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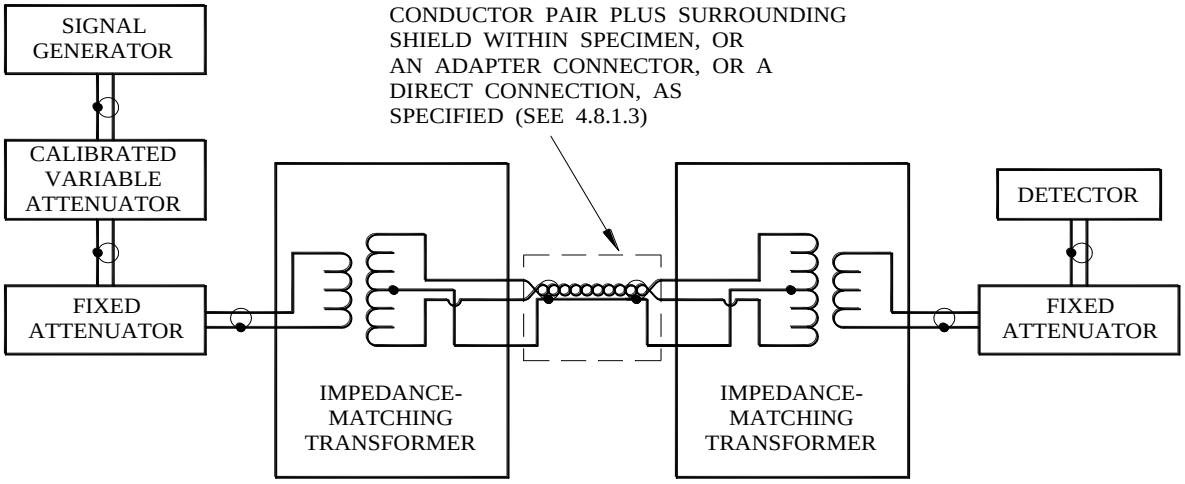


FIGURE 16. Block diagram of electrical connections for the attenuation test when testing a conductor pair plus surrounding shield.

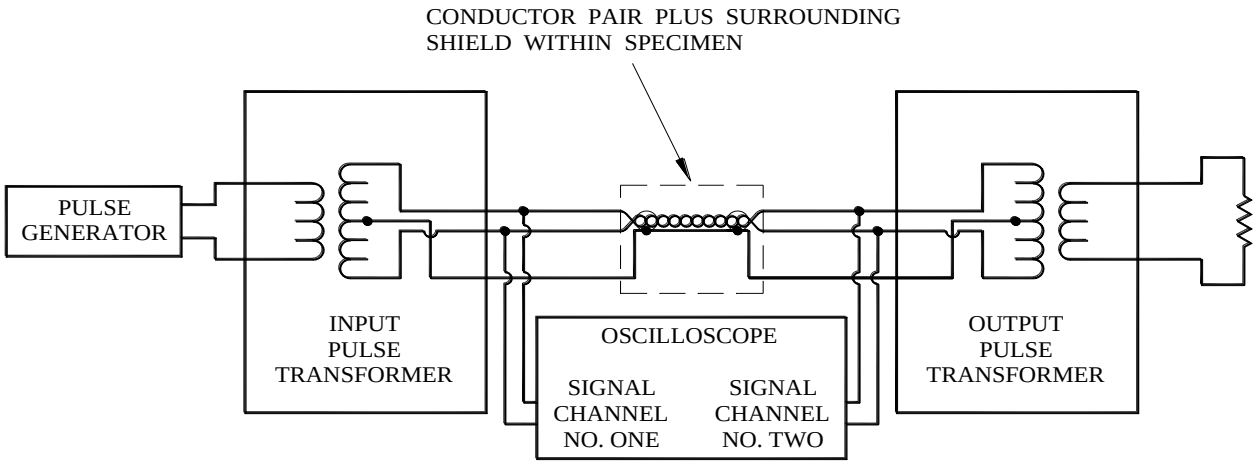


FIGURE 17. Block diagram of electrical connections for the pulse response test.

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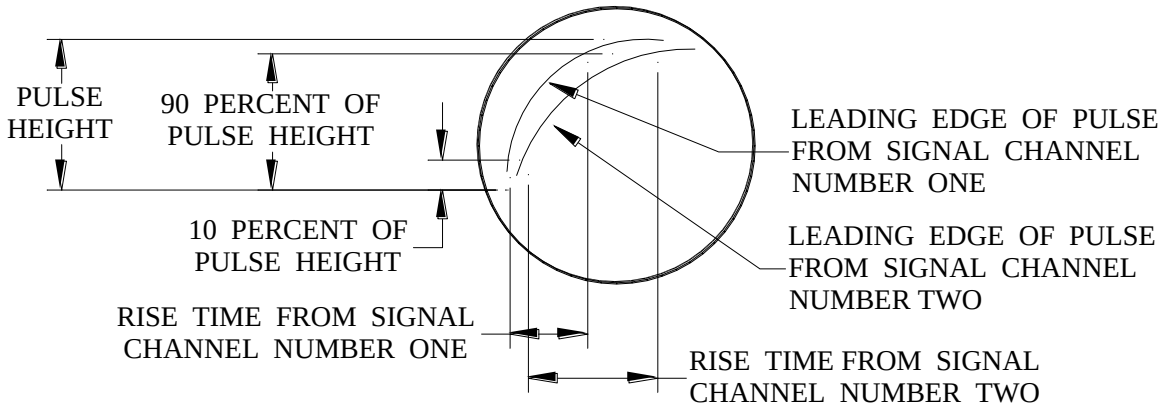


FIGURE 18. Oscilloscope display showing rise time measurements for the pulse response time test.

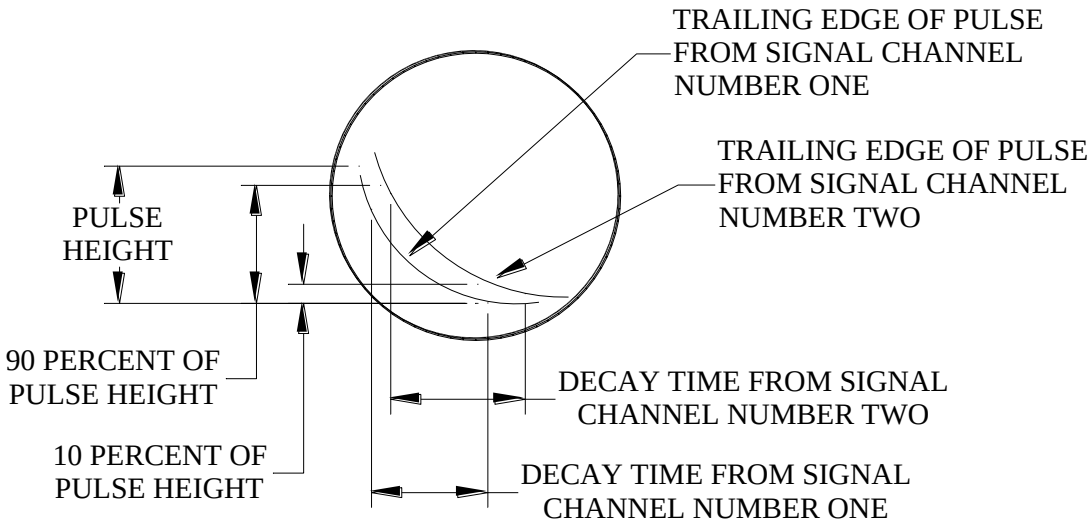


FIGURE 19. Oscilloscope display showing decay time measurements for the pulse response time test.

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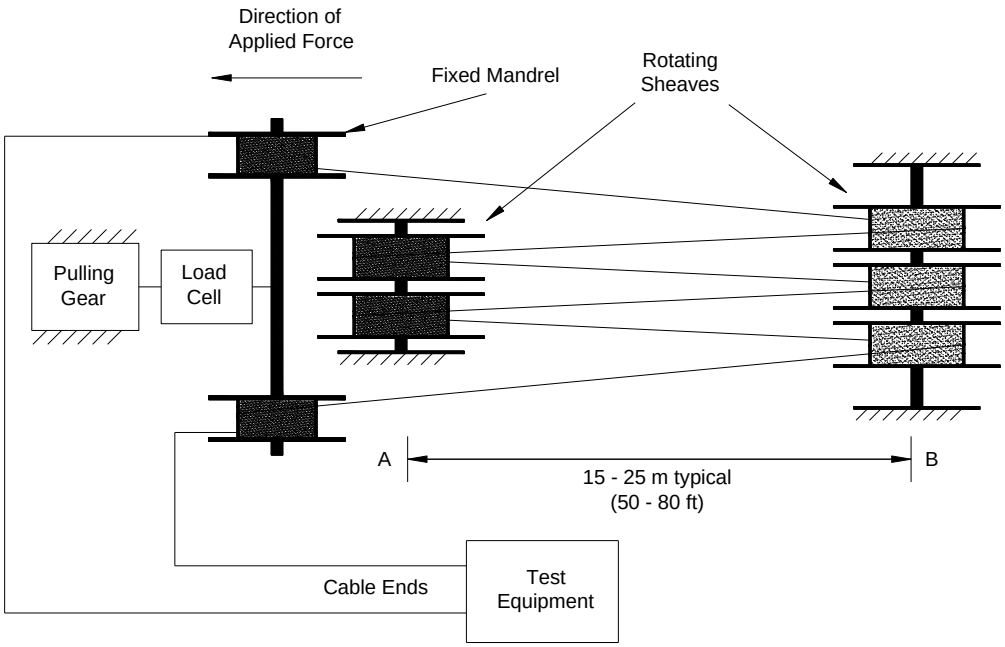


FIGURE 20. Tensile loading fixture.

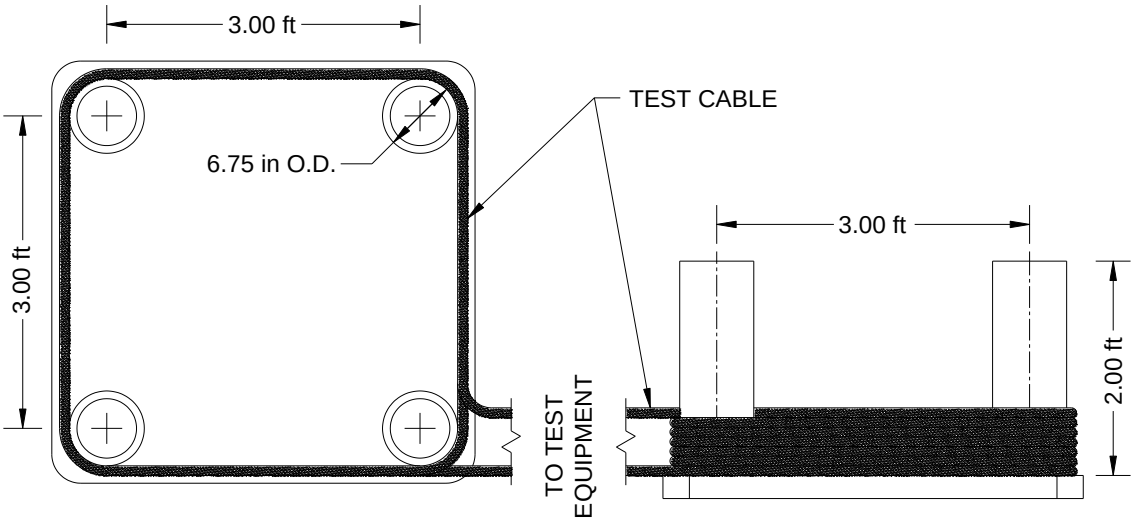


FIGURE 21. Bend fixture.

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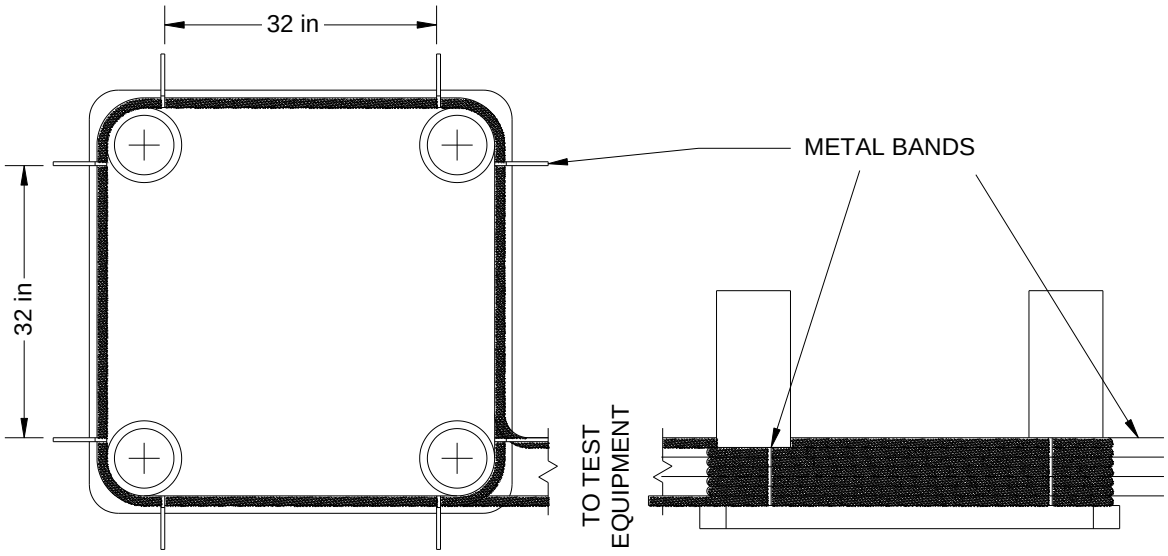


FIGURE 22. Cable banding.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-018)

Review activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

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